



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

SECP2523: DATABASE

SEM I 2025/ 2026

DATABASE DESIGN DESCRIPTIONS: LOGICAL DATABASE DESIGN

FOR <Hasta Vehicle Rental Booking System>

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Stakeholder: Hasta Travel & Tours Sdn Bhd

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TABLE OF CONTENTS

CHAPTER	TOPIC	PAGE
1	Introduction	
1.1	Overview about the company	5
1.2	The Existing System	6 - 24
1.3	The Proposed System	25 - 28
2	Global Conceptual Data Model (Global ERD)	
2.1	Global Conceptual Database Design (Conceptual ERD)	30
2.2	Data Dictionary for the global conceptual ERD	31 - 39
3	Logical Data Model (Logical ERD)	
3.1	Logical Database Design (Logical ERD)	41
3.2	Data Dictionary for the Logical ERD	42 - 53
3.3	Relational Database Schema(s)	54 - 56
3.4	Normalization	57 - 61
4	Interface Design and SQL Implementation	
4.1	User Management Module	62

4.1.1	Interface Design	62 - 70
4.1.2	SQL Implementation	71 - 75
4.2	Car Rental Booking Module	76
4.2.1	Interface Design	76 - 83
4.2.2	SQL Implementation	84 - 89
4.3	Fleet Management Module	90
4.3.1	Interface Design	90 - 91
4.3.2	SQL Implementation	92 - 93
4.4	Reporting Module	94
4.4.1	Interface Design	94 - 96
4.4.2	SQL Implementation	97 - 98
4.5	Loyalty Card and Rewards Module	99
4.5.1	Interface Design	99 - 101
4.5.2	SQL Implementation	102 - 103

5	Summary	104
6	Appendices	105
6.1	Use Case Diagram for the system.	105
6.2	Activity Diagrams for every use case identified	106 - 121
6.3	Meeting Log	122 - 127

NOTE: A module can consist of one or multiple (related) use case. A team member in a group must be responsible to at least one module.

1.0 Introduction

1.1 Overview about the company

Hasta Travel & Tours Sdn. Bhd. is a car rental company that operates as a car rental service provider, offering short-term vehicle rental service specifically for Universiti Teknologi Malaysia (UTM) students and staff. As a car rental operator, its daily operation relies heavily on accurate vehicle availability tracking, precise rental scheduling, timely payment verification and well-maintained customer records.

In addition to handling daily bookings, Hasta Travel manages a high volume of transactional data, including customer profiles, vehicle details, rental schedules, payment records, deposits, penalties, and loyalty rewards. These data elements are highly interrelated and require consistent updates throughout the rental lifecycle.

However, the current rental operation relies on a largely manual and semi-digital booking system, supported by basic tools such as WhatsApp, WhatsForm, Google Sheets , physical documents, whiteboards and the Drive Wahdah system. The company adopts a file-based system approach, where information is stored in separate files on computers rather than within a centralized database. These tools are used independently rather than as part of an integrated platform, resulting in fragmented workflows and duplicated data handling. The Drive Wahdah system functions mainly as a client-side rental record system, accessible only by staff and administrators, while customers interact with the company primarily through manual communication channels.

As a result, as the volume and complexity of rental-related data increase over time, an organized and structured data management approach becomes essential to support efficient operations and long-term scalability.

1.2 The Existing System

1.2.1 Purpose and What the Existing System Does

The existing booking system used by Hasta Travel operates through manual processes and basic digital tools, resulting in several limitations in daily operations. The daily operations such as customer booking, payment verification and record keeping are handled using physical documents, digital forms and simple spreadsheets files. The current system for booking cars that is used in Hasta Travel is DRIVE WAHDAH. The system is only for client-side and only accessible by staff and admin of HASTA Travel & Tours Sdn Bhd.

1.2.2 Type of Existing System (Manual / Automated)

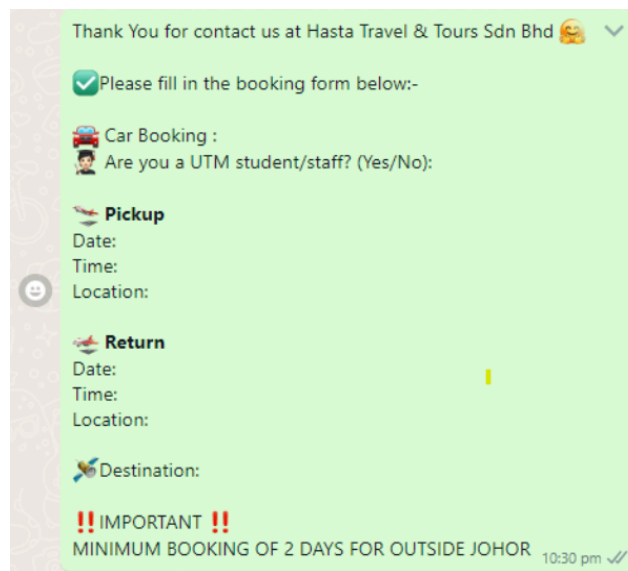
The existing system is mainly manual with limited digital support. The car rental operations rely on Whatsapp, WhatsForm, physical documents, manual boards, and spreadsheets, while Drive Wahdah is used by staff/admin to store booking and payment information.

1.2.3 Existing System Workflow / Processes

1.2.3.1 Car Rental Booking Process

1.2.3.1.1 Getting Information

The car rental booking process at Hasta Travel is handled through Whatsapp or Whatsform. Currently, customers send a booking request message to rent a car via Whatsapp. The staff will immediately provide a booking template that includes booking information such as a confirmation of whether the customer is a UTM student or staff, pickup date, pickup time, pickup location, return date, return time, return location and destination to the customers and customers must fill in before proceeding with the reservation. The template strictly informs the customer that the bookings require a minimum of two days for trips outside Johor.



Thank You for contact us at Hasta Travel & Tours Sdn Bhd 🙏

✅ Please fill in the booking form below:-

🚗 Car Booking :
👤 Are you a UTM student/staff? (Yes/No):

✈️ **Pickup**
Date:
Time:
Location:

✈️ **Return**
Date:
Time:
Location:

📍 Destination:

!! IMPORTANT !!
MINIMUM BOOKING OF 2 DAYS FOR OUTSIDE JOHOR

10:30 pm ✓✓

Figure 1.2.1: Booking Template

The staff of Hasta Travel will review the customer's reply to ensure that all booking details meet the requirements. They will verify that the customer is a UTM student or staff member, confirm that the pickup and return locations are at UTM Student Mall, and ensure that the pickup and return times fall within office hours. The booking is strictly limited to UTM students and staff. Any booking made by a non-UTM individual will be rejected. Hasta Travel currently uses a manual record system, such as a sales board, to display car availability for booking on specific dates .

SALES BOARD	1	2	3	4
MYVI JPN 1416 YEAR 2017	16/04/18 9am-12pm	17/04/18 9am-12pm	X	▶
AXIA VC 6493 YEAR 2016	X	X	X	X
AXIA CEF 6048 YEAR 2010	X	X	X	▶
MYVI VC 6522 YEAR 2016	X	X	18/04/18 12pm-3pm	19/04/18 12pm-3pm

Figure 1.2.2: Sales Board Hasta Travel

If the certain dates marked with an “X”, indicating the car are unavailable, which other slots remain open. This system allows staff to inform customers about available cars and confirm their preferred choice. The staff will inform customers that the price is different for different types of car.

The staff will provide the booking form to customers after confirming their chosen car .The booking forms will include customer details, pickup and return details, payment details , destination, details for deposit refund. The booking will inform the user that the deposit has a minimum RM50 for booking, deposit will be burned for cancellation less than 24 hours and Payment Reference ID.

BOOKING FORM
Please read to the end before filling out the form, thank you

CAR MODEL (depends on availability) : MYVI JPN

CUSTOMER DETAILS :
Name :
I/C @ Passport :
Phone number :
Matric No :
Race :
College :
Faculty :
Email :

PICKUP
Date :
Time :

RETURN
Date :
Time :

PAYMENT
Deposit (RM) : 50
Rental (RM) :
Duration Rental (hrs) :

Your Destination :
Pick up Location :
P.O. :

DETAILS FOR DEPOSIT REFUND
Account Bank No :
Bank :
Name :

Please make a PAYMENT as DEPOSIT minimum RM50 for booking

!!! IMPORTANT !!!
DEPOSIT will be BURNED for CANCELLATION LESS THAN 24 HOURS

Payment to 012-7623282

MAYBANK
HASTA TRAVEL TOURS SDN. BHD.
5513 0554 1568

Please put your name on REFERENCE and send the proof payment to SECURE your booking thru whatsapp or email to hastatraveltours@gmail.com

Figure 1.2.3: Booking Form

If the customer is not booking the car during staff operating hours and the location is not UTM Student Mall or after the customer pays the deposit, the staff will ask the customer to provide their pictures of IC or passport, driving license and matric card in a single PDF.

Another way for customers to book a car is filling in WhatsForm that was prepared by Hasta Travel. The WhatsForm allows customers to book a car by displaying the rental price per hour for each car type and available promotions on weekdays from Monday to Thursday for 9 hour bookings. It collects essential information for booking from customers, including desired car model, name, email, student ID or staff ID, IC number or passport number, phone number, address and faculty. It also collects pickup and return details such as pickup date, pickup time, pickup location, return date, return time and return location, rental duration, rental fee, destination and additional remarks such as car preferences, Bluetooth, promotions, promo codes or vouchers. Once completed, the WhatsForm automatically converts the information into a booking template and sends it directly to Hasta Travel's Whatsapp for processing.

**PRICE WEEKDAY PROMO (MONDAY TO THURSDAY)
FOR 9 HOURS ONLY**

HARGA "*WEEKDAY PROMO 20% DISCOUNT*"

MODEL	PERODUA AXIA		PERODUA MYVI	PERODUA BEZZA		MOTOR HONDA	
	1ST GEN	2ND GEN	2ND GEN	1ST GEN	2ND GEN	DASH 125	BEAT SKUTER
HOOR	PRICE (RM)						
9	85	90	90	90	95	36	35
PROMO	68	72	72	72	76	28	28

Depends on availability

Car Rental Booking Form

* Car Model

If car not in option, please mention in remarks section.

* Name

Please fill in your Full Name

* Enter your email

We send all our best offers and discounts to our customers via email

Figure 1.2.4: WhatsForm for Booking Car

1.2.3.1.2 Deposit

The staff will remind the customer to fill in the Booking Form and pay the deposit. Then, the staff will ask for the receipt from the customer and verify that the RM50 deposit amount is correct. They will also ensure that the deposit time matches the time of the payment was received. If the customer forgets to pay or is late in paying the deposit, the staff of Hasta Travel will remind the customer to make the payment. If the customer asks about the deposit, the staff will explain that the RM50 deposit will be returned within 10 working days if the car has no damage, the oil level is the same as before, there are no summonses, the car is clean and it is returned on time. These checks allow the staff to refund the deposit without the deductions. After receiving the deposit from the customer, the staff of Hasta Travel will download the deposit receipt and save it in the specific folder for that year and specific file for that month. The file name format is <year, month, day> <RM50> <name>.

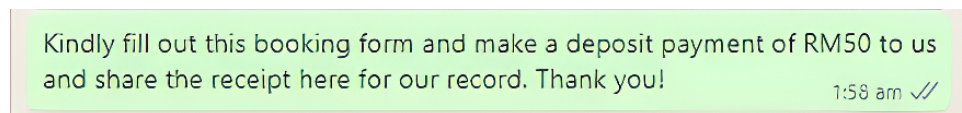


Figure : Reminder for customer to pay the deposit

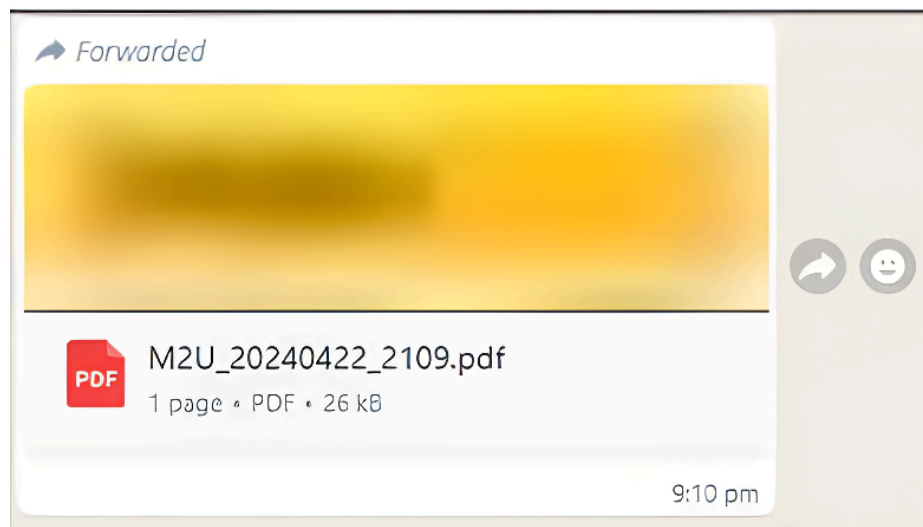


Figure 1.2.5: Receipt of Deposit

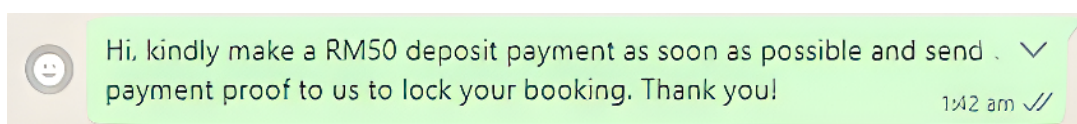


Figure 1.2.6: Reminder for Late Customers to Pay Deposit

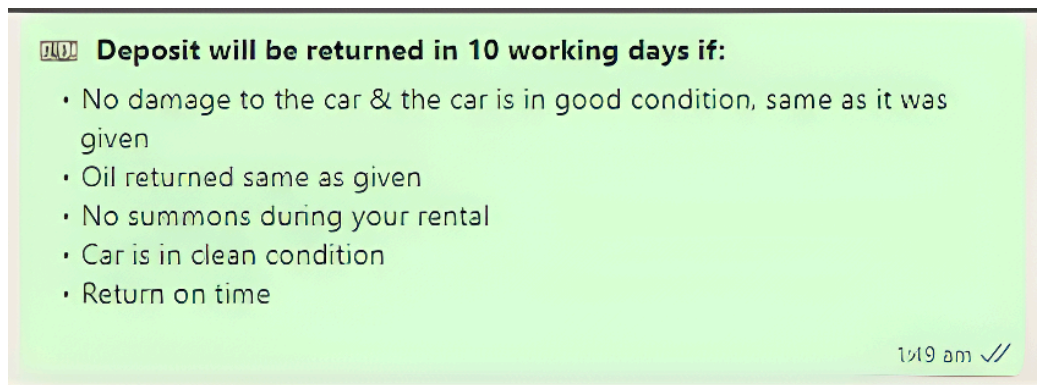


Figure 1.2.7: Reply Message of Deposit

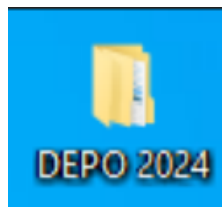


Figure 1.2.8: Folder for Storing Deposit Receipt by Year

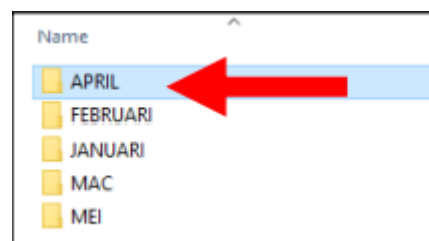


Figure 1.2.9: File for Storing Deposit by Month

Name	Date modified
 20240401 RM50 Irdina	27/4/2024 1:26 AM

Figure 1.2.10: Example format file name that follows <year, month, day> <RM50> <name>.

1.2.3.1.3 Booking Records

The staff will manually record the booking forms on the booking whiteboard to ensure that all bookings are immediately visible and can be easily tracked. They will also use the Drive Wahdah website and a spreadsheet to maintain daily sales reports and booking records, which helps to organize the data systematically, prevent errors, and provide a reliable reference for future verification and reporting purposes.

For the booking whiteboard, each successful booking is recorded in a clearly defined space on the board. The customer's information is entered following a specific format in which the customer's name is written first, followed by the pickup time of the car, the return time and the rental data range. Different colours are used on the board to distinguish types of bookings and users. Black represents staff or UTM students, blue indicates external bookings, green is for Hasta staff and red marks cars that are unavailable for rent. This system allows staff to quickly view which cars are booked, by whom, and during what times, helping to organize daily operations efficiently and prevent scheduling conflicts.

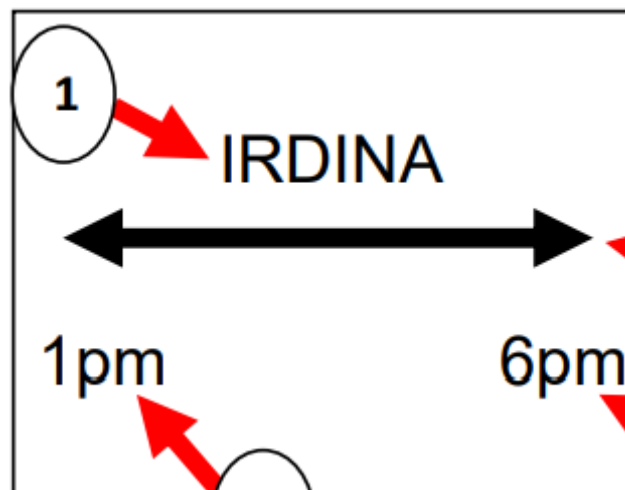


Figure 1.2.11: The format for writing on the whiteboard

For the Drive Wahdah website, staff are responsible for adding the rental form and filling in all the required information using the booking form and the receipt provided by the customer. This includes recording the customer's personal details such as name, email address, identification number, residential address and mobile number. In addition, the staff will enter the pickup and return information, vehicle details and deposit details written on the note. The pricing section of the form clearly indicates the payment status, specifying whether the payment has been made or remains unpaid. If the customer pays the full rental amount, staff will record the payment date and method of payment, ensuring that all financial transactions are properly documented. Once all the information is completed and verified, the staff will submit the form, thereby creating a complete and accurate record of the rental transaction for future reference, reporting and audit purposes.

Dashboard / Bookings / Add Sale Rental

Add Sale Rental

CUSTOMER INFO

Name:

Email: ID/Passport Number:

Address:

Mobile Number: +60 12-345 6789

VEHICLE INFO

Vehicle Name: Please Select Vehi...

Plate Number: Please Select Plate #

Staff: Please Select Staff

Currency: MYR

RENTAL INFO

Pickup Location:

Return Location:

DATE & TIME

Add-on Details:

Figure 1.2.12: Rental Form of Drive Wahdah Website

PRICING

Payment Status

PAID

Payment Date **Payment Method**

24 Apr 2024 Online Banking

Figure 1.2.13: Pricing section when the payment status is paid

For the spreadsheet that contains daily sales reports, staff are responsible to fill in the customer's information into the spreadsheet. The process begins by selecting the specific sheet tab at the bottom that corresponds to the actual date of rental. For example, if a customer books a car on 1/4/2025 for the date 2/4/2025, the staff needs to select the tab labeled "2" to record the booking correctly. Then staff must transfer information from the customer's booking form into the "Customer Details" section, including name, bank account and destination. In the "Rental details" section, the staff need to record the destination, the car plate of the rental car and the payment in either the "Cash" or "Transfer" column depending on the method used. Staff must record the deposit amount in the "IN" section and place the value under either the "Cash" or "Transfer" column based on the customer's payment method. They must also include the exact date the payment was received. The "OUT" column is for the deduction part such as deductions for fuel, late returns, or other charges as these are only filled in after the customer returns the vehicle. Lastly , the staff needs to enter

their name in the “Key In Data” column. If the person keying in the data is different from the person handing over the keys, both names should be written using a slash (/).

DSR APRIL 2024

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Figure 1.2.14: Spreadsheet Daily Sales Reports

IN		
CASH (RM)	TRANSFER (RM)	DATE IN
	50	29/03/2024
	50	01/04/2024

Figure 1.2.15: “IN” Section of Deposit Details

DEPOSIT DETAILS							PIC (NAME)	
OUT (DEDUCTION)					TOTAL	REFUND (RM)	GIVE CAR/KEY IN DATA	CHECK CAR RETURN
EXTEND PAID (RM)	EXTEND UNPAID (RM)	FUEL (RM)	LATE RETURN (RM)	DATE OUT				
					0	50	AQIL / ADZIM	NAZIFAH
					0	50	AQIL / ADZIM	NAZIFAH

Figure 1.2.16: “Out” Section of Deposit Details

1.2.3.2 Pick-up Process

The customer can choose to arrange their car pickup either online or offline, depending on their preference and convenience. Both methods ensure that the customer receives the car and car keys once all necessary steps have been completed.

1.2.3.2.1 Online Car Pick-Up Process

Before the pick-up process, the staff must ensure that the customer has made the full payment. If the full payment is made by the customer, the staff needs to update the payment status to “Paid” on the Drive Wahdah system. Then, the staff must download both the agreement form and the inspection form in PDF format from the Drive Wahdah system. These forms must be sent to the customer at least 30 minutes before the pick-up time. The staff should also send the procedure of picking up and returning the car to the customer.

For the pick-up process, the customer must sign the agreement form and inspection form and send it back to the Hasta staff. The customer must also ensure the full payment is completed and provide the payment receipt to the staff. After that, the customer can collect the car key, which is placed under the driver’s seat. The customer must take photos of the mileage, oil bar, and all four sides of the car.

15 minutes before the pick-up time, the staff will remind the customer about the required documents which consist of the signed agreement form and inspection form, the photos of the car, and the full payment. After receiving all required documents, the staff will then print all the signed agreement and inspection forms, and the PDF copies of the customer’s driving licence, IC, and matric card, and store them as hardcopy in the file. Then, the staff must send the customer the car return conditions, which include the rules for oil level, cleanliness, damage, return time, and key return procedure and the place to return the keys.

Lastly, the staff will check the inspection form submitted by the customer. Once the customer sends the four-side photos of the car, the staff must upload the pickup date and time into the spreadsheets.

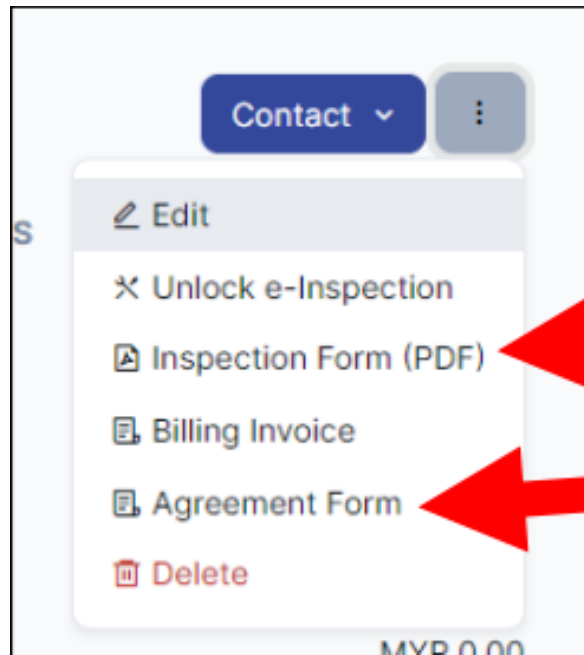


Figure 1.2.17: The Agreement Form and Inspection Form in the Drive Wahdah System

The image shows a screenshot of the 'Inspection Form' in the Drive Wahdah System. The form is titled 'INSPECTION FORM PICK-UP / RETURN'. It includes the following fields: 'Model : Perodua Myvi (3rd Gen)', 'Reg No : JSM6249', 'Vehicle Type : Compact', and 'Invoice Number :'. Below these fields, there is a grid of car diagrams for inspection. The grid is divided into two main sections: 'PICK-UP' and 'RETURN'. Each section contains three rows of diagrams. The first two rows show side views of the car, and the third row shows front and rear views. The 'PICK-UP' section has four diagrams, and the 'RETURN' section has four diagrams. The form is displayed on a tablet or smartphone screen, with a navigation bar at the top showing '1 / 2' and '100%'. On the left side of the screen, there is a sidebar with a menu icon and a list of items, including '1' and '2'.

Figure 1.2.18: Inspection Form

AGREEMENT FORM

HASTA
HASTA TRAVEL & TOURS SDN. BHD.
(1359376-T)

HASTA TRAVEL & TOURS SDN. BHD. (1359376-T)
7A, JALAN KEBUDAYAAN 1A,
TAMAN UNIVERSITI,
81310 SKUDAI
JOHOR BAHRU
JOHOR
Office : +6011-10900700

INVOICE NUMBER
2024-04-HASTA/INV004827

Rental Details	
Vehicle	PERODUA Myvi 1.3 (3rd Gen)
Pick Up Location	
Return Location	
Pick Up Date	05-05-2024
Return Date	06-05-2024
Pick Up Time	09:00 AM
Return Time	12:00 PM
Duration	1 day 3 hours 0 minute
Note	job under bus palli

Figure 1.2.19: Agreement Form

Hi, Please follow this procedure correctly or **you will be penalized.**

CAR: JPN1416

How to pickup the car:

1. Sign the agreement form and send it back to us in pdf format for our reference
2. Pay the FULL RENTAL AMOUNT & send the receipt to us
3. Take the car key below the driver's seat
4. Snap the 4 sides pictures of the car (front, back, left and right side) & the mileage and fuel of the car

How to return the car:

1. Park the car at student mall utm in front of our office
2. Snap the 4 sides pictures of the car (front, back, left and right side) & the mileage and fuel of the car
3. Put the car key below the driver's seat

THANK YOU FOR YOUR COOPERATION, HAVE A SAFE DRIVE 😊

Edited 12:09 pm ✓✓

Figure 1.2.20: Procedure of Pick-up and Return Car

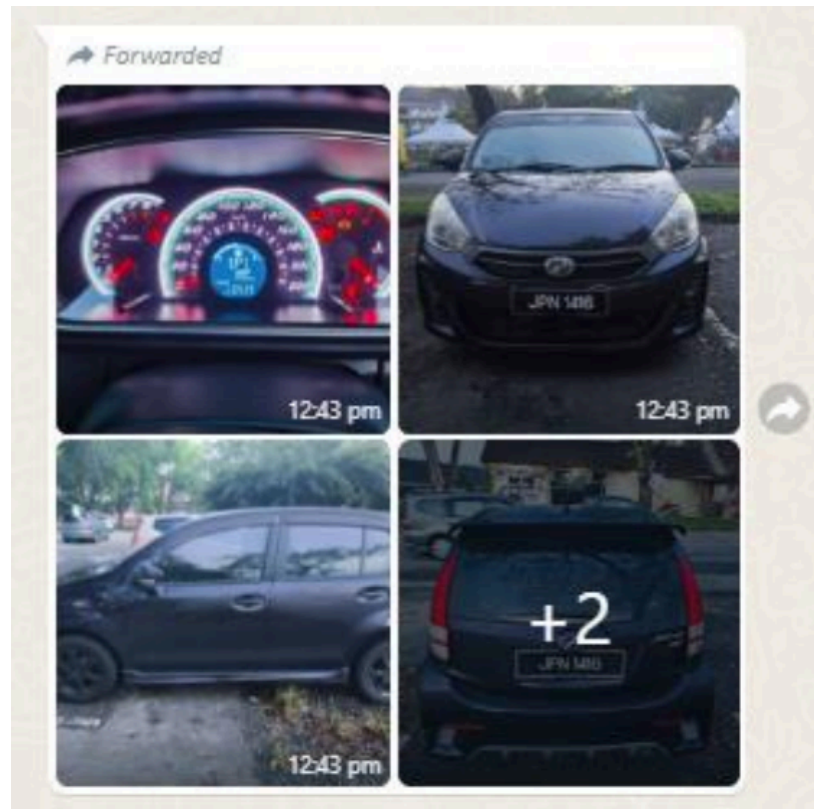


Figure 1.2.21: The Four-Side Photos of Car that Sent by Customer

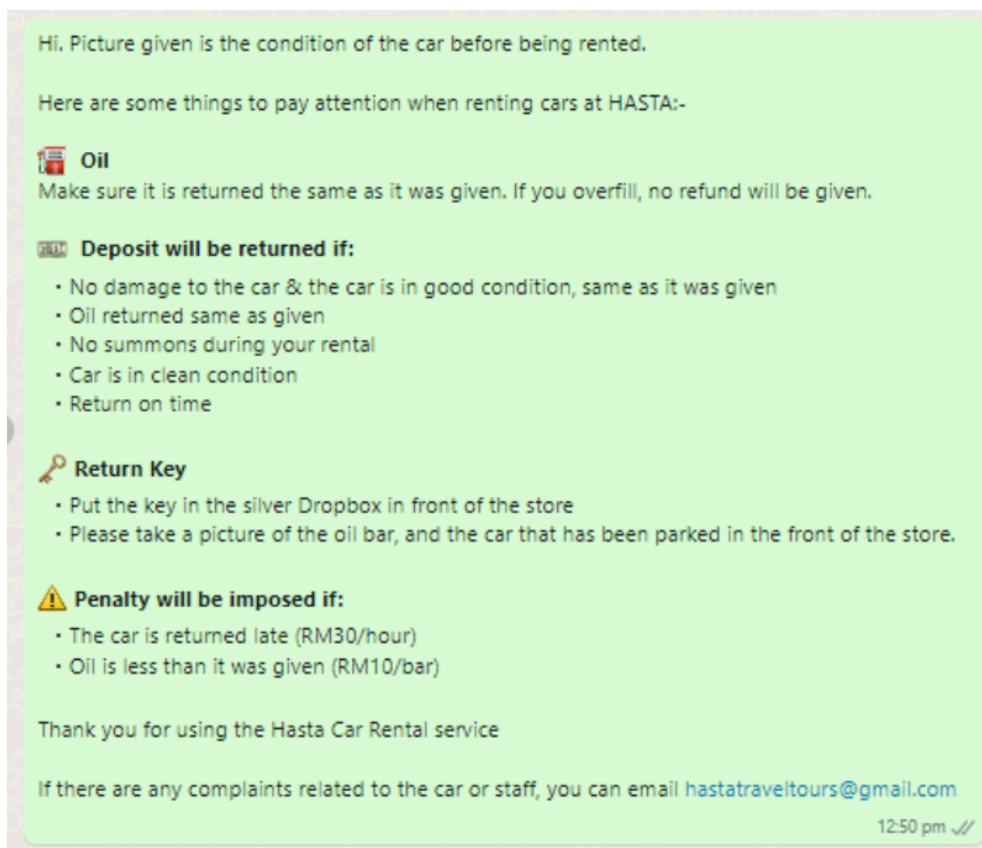


Figure 1.2.22: Car Returned Condition

1.2.3.2.2 Physical Car Pick-up Process

Just like the online pick-up process, the staff must first ensure that the customer has made the full payment before the pick-up. If the full payment is made by the customer, the staff needs to update the payment status to “Paid” on the Drive Wahdah system. Next, the staff needs to print the agreement form, the inspection form and the PDF documents that contain the customer’s driving licence, IC, and matric card as hardcopies.

Then, the staff will provide these forms to the customer and ask them to sign the agreement form and the inspection form physically. After signing, the customer must take photos of the mileage, oil bar, and all four sides of the car. Once the customer has taken and submitted all the required photos, the staff may hand over the car key to the customer.

After giving the key, the staff must send the customer the car return conditions, which include the rules for oil level, cleanliness, damage, return time, and key return procedure. Finally, the staff must upload the pickup date and time into the spreadsheet immediately after handing the key to the customer.

The image shows the front page of an 'Agreement Form' for HASTA Travel & Tours Sdn. Bhd. The form is divided into several sections: Rental Details, Customer Details, Car Information, and a Vehicle Price table. The Rental Details section includes fields for Vehicle (Perodua Axia 1.0 (A)), Pick Up Location (STUDENT MALL, UTM JB), Return Location (STUDENT MALL, UTM JB), Pick Up Date (01-08-2023), Return Date (02-08-2023), Pick Up Time (01:00 PM), Return Time (01:00 PM), Duration (1 day 0 hour 0 minute), and Note (DEPOSIT RM10.00 (PAID - 1/8)). The Customer Details section includes Name (NUR SYAFIQAH BT ALI SABRI) and Mobile Number (+601139770747). The Car Information section includes Plate Number (VC 6493) and Color (SILVER). The Vehicle Price table shows the price for Perodua Axia 1.0 (A) as 90.00, with a discount of -0.00, resulting in a Total Amount Before Tax of 90.00. The Total Amount is MYR 90.00, and the Rounding is MYR 0.00.

Rental Details	
Vehicle	Perodua Axia 1.0 (A)
Pick Up Location	STUDENT MALL, UTM JB
Return Location	STUDENT MALL, UTM JB
Pick Up Date	01-08-2023
Return Date	02-08-2023
Pick Up Time	01:00 PM
Return Time	01:00 PM
Duration	1 day 0 hour 0 minute
Note	DEPOSIT RM10.00 (PAID - 1/8)

Customer Details	
Name	NUR SYAFIQAH BT ALI SABRI
Mobile Number	+601139770747

Car Information	
Plate Number	VC 6493
Color	SILVER

Vehicle	Price
Perodua Axia 1.0 (A)	90.00
Discount	-0.00
Total Amount Before Tax	90.00
Total Amount: MYR 90.00	
Rounding: MYR 0.00	

Figure 1.2.23: Front Page of Agreement Form

1.2.3.3 Return Process

When returning the car, the customer must ensure it is returned on time, in a clean condition, with the oil level the same as when it was given, and no summons issued during the rental period. The customer should place the key in the silver Dropbox in front of the store and take a picture of the oil bar and the car that has been parked in front of the store which is on all four sides of the car, along with the mileage and fuel bar. If the car is returned late or the oil level is lower than given, penalties will apply. The customer will be fined RM30 per hour if returned late and RM10 per bar if the oil is less than it was given. Customers should record the date, time, mileage, and fuel level of the car, and fill in their name in the PIC section of the inspection form. The staff should enter 0 in the Extend Paid, Extend Unpaid, Fuel, and Late Return fields if the car is returned in the same condition as rented. After completing the return, staff should notify the customer that the deposit will be returned within 10 days.

If the customer late returns the car, staff must record the delay duration in the REMARKS section and note the lateness on the front page of the Agreement Form. The delay is counted after 30 minutes past the scheduled return time. If the oil level is not the same as when given, the staff should record the amount of oil missing in REMARKS and make a note on the front page of the Agreement Form. In the case of car damage, such as dents, multiple scratches, or broken parts, staff must inform the customer by comparing the car with the pre-rental photos, mark the damaged areas on the photos, label the type of damage, and also record the details on the front page of the Agreement Form.

If a customer requests to extend the car rental, staff should record the extension fee per hour according to the type of car in the “Extend” column. For example, RM35 per 1 hour extension for a Myvi. Extensions can be marked as “Extend Unpaid” if the fee is deducted from the customer’s deposit, or “Extend Paid” if the customer has paid the fee directly. If the customer returns the car late, staff should record the late fee per hour according to the car type in the “Late Return” column. For example, RM30 per 1 hour delay for an Axia. If the car is returned with less fuel than given, staff should record RM10 per missing fuel bar in the “Fuel” column. For example, RM10 for 1 missing fuel bar.

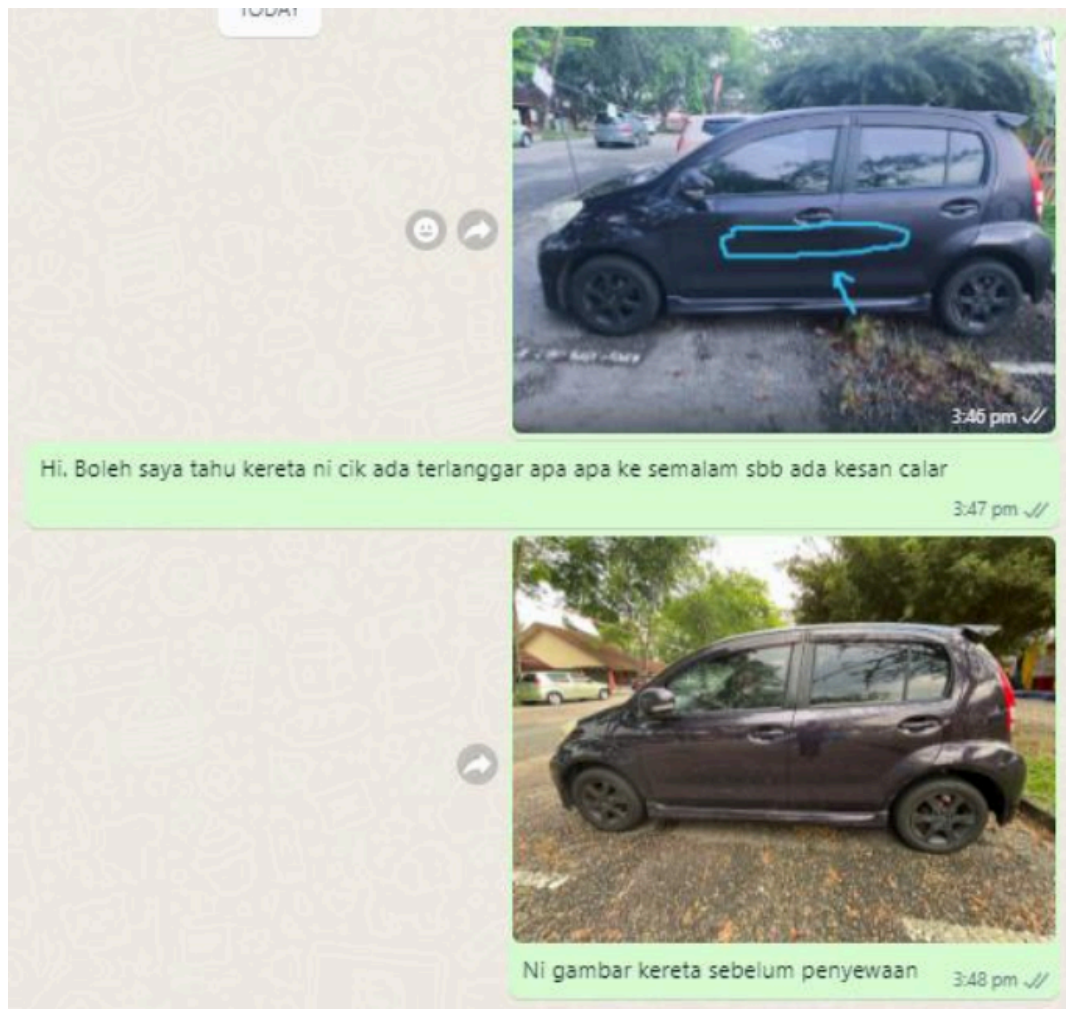


Figure 1.2.26: Comparison with Pre-Rental Photos of Car

1.2.3.4 Loyalty Card and Reward

The customer's loyalty card and rewards will be recorded in Excel. To get a loyalty card, customers must rent a car for 12 hours, regardless of car type, which earns them the first stamp. To get a stamp, the customer must rent a car for 9 hours. After issuing the loyalty card, staff should record the customer's name in Excel according to the serial number on the card, mark the first box with a (/) and note the date and the name of the person-in-charge (PIC). If the customer rents for 9 hours again and reaches a box that qualifies for a discount, staff should mark the "Claimed" button to indicate the reward has been used.

CARD NO	CUSTOMER DETAILS
	NAME
001	

Figure 1.2.27: Loyalty Card that Need Serial Number and Name of Customer

BOX 1	TARIKH	PIC
☐		
☐		

Figure 1.2.28: Loyalty Card

BOX 3	TARIKH	PIC
30% Claimed		
30% Claimed		
Not yet Claimed		

Figure 1.2.29: Loyalty Card that Can Claim

1.2.4 Problems Faced by Client (Users) in the Existing System

Hasta Travel's manual system has a number of serious problems that impact the accuracy, scalability, and operational efficiency of the business. The system is unsustainable as booking volume rises due to its significant reliance on manual Driveo entries, WhatsForm submissions, and WhatsApp messaging.

First of all, the booking procedure is inefficient and incredibly prone to mistakes. The staff must manage several requests at once because all correspondence and confirmations are completed manually via WhatsApp, which raises the possibility of duplicate bookings and miscommunication. An effective synchronization of hourly and daily rental schedules is impossible in the absence of a real-time vehicle tracking system.

Second, the customer experience is adversely affected by the absence of automation and a digital user interface design. In contrast to Wahdah, SOCAR and GoCar systems, Hasta's website does not let users make booking or payment directly online. Instead, it only shows basic text information. Customers are forced to use manual communication, which reduces satisfaction and delays response times.

Third, the current booking system requires customers to provide numerous documents, such as IC, driver's license, and matric card, which makes the process lengthy and complicated. As a result, customers have to go through many steps to complete a booking, leading to potential frustration and dissatisfaction with the service.

Fourth, scheduling and geographic limitations make services even less accessible. The current booking system restricts the pickup location to only Student Mall and the drop-off location to areas around UTM. This limitation prevents students who are outside these locations from making a booking, which reduces the number of potential customers.

Additionally, the current system enforces a "cool-down" period following every rental, meaning that if a car is returned the next morning, the system blocks the entire day from new bookings. This limitation requires staff to manually check car availability, which slows down the booking process and can lead to reduced efficiency and customer dissatisfaction.

Obviously, these limitations make it abundantly evident that Hasta's manual workflow is unable to meet customer expectations or operational demands in the modern digital world.

1.3 The Proposed System

This section describes the proposed To-Be system for the Hasta Vehicle Rental Booking System. It outlines the purpose and goals of the system, explains how the system will operate compared to the current manual process, and defines its scope and boundaries. The section also presents the system's mission statement, key objectives, main functionalities, involved user roles, and the technologies proposed for designing and developing the system.

1.3.1 Mission Statement

The mission of the Hasta Vehicle Rental Booking System is to provide a centralized and efficient web-based platform that automates the vehicle rental booking process for Hasta Travel & Tours. The system is intended to replace the existing manual workflow by improving operational efficiency, reducing human errors, and enhancing transparency in booking activities, fleet management, and customer interactions.

1.3.2 Overview of the To-Be System

The proposed Hasta Vehicle Rental Booking System is a role-based web application designed to manage the full vehicle rental lifecycle in a structured and organized manner. The system supports two main user groups: customers, which include UTM students and staff, and clients, consisting of Hasta staff and administrators. Customers can register and log in to the system, browse available vehicles, make rental bookings, upload payment receipts, submit vehicle inspection images, and provide reviews after completing the rental period. Meanwhile, Hasta staff and administrators use the system to manage fleet information, review and approve bookings, verify customer submissions, monitor rewards usage, and generate administrative reports. By digitalizing these processes, the system ensures accurate record-keeping, improves traceability, and increases overall operational efficiency.

1.3.3 Mission Objectives

The proposed database system has been designed with a set of clearly defined objectives to address the shortcomings of the current car rental booking process and fulfill the requirements of both clients and staff.

1. To automate vehicle rental booking operations, reducing reliance on manual processes.
2. To minimize errors associated with manual record-keeping and data entry.
3. To improve overall service efficiency for both customers and staff.
4. To provide real-time vehicle availability information to support faster decision-making.
5. To establish a structured booking approval workflow that ensures consistency and accountability.

6. To organize and manage rental documents in a centralized and accessible manner.
7. To support administrative decision-making through comprehensive reporting features.
8. To enhance customer satisfaction by offering a clear and guided rental process.

1.3.4 Scope

Our proposed system is a usable and easy to maintain system which is implemented and designed to manage the car rental operations of Hasta Travels. There are three types of system users that are categorized into customer, staff and admin. In the Hasta Vehicle Rental Booking System, there are several modules including user management, car management, booking management, payment and penalty handling, loyalty rewards and admin dashboard modules.

The system can be accessed through the general website via any search engine. Users can either log into their existing account or sign up for a new account. As for staff and administrators, they are only permitted to log into the system. Staff accounts can only be created through a page that is exclusively accessible by administrators.

1.3.4.1 User Management

As for customers, upon logging into the system, they will be redirected to the customer interface. The customer interface consists of a main dashboard which redirects to multiple different pages including the home, vehicle, booking, vouchers and help pages. Customers also have their own profile page which contains all of their personal information, including documents such as their identity card or passport, matriculation card and driving licence pictures. Before proceeding to make a booking, customers are required to complete their profile and have their documents manually verified by staff.

If a user is blacklisted because of outstanding payments, they will still be able to log in and view the car models and details of the car. However, they would be restricted from accessing other system features. Once the blacklisted user wants to rent a car, they will be redirected to a payment page to clear their outstanding payments.

1.3.4.2 Car Management

In the vehicles page, customers are able to view all car models handled under the company along with their respective details such as price, colour, engine, power and number of seats. Real-time availability is also shown via an interactive calendar that highlights when a car is booked and enforces a 1-hour cooling period between bookings. Customers may proceed to make a booking directly from this page.

1.3.4.3 Booking Management

Other than that, customers can perform a rental booking which involves selecting a vehicle and a preferred pick-up and return location, as well as the date and time. If a customer chooses a location other than Student Mall for pick-up or return, an extra fee will be charged. Customers are also given the option to input a voucher code obtained from the vouchers page. Following that, the booking fee is automatically calculated based on the rental period, pick-up point, a deposit of RM50 per booking and any voucher applied. A full payment breakdown will be displayed and customers may proceed to pay the total rental amount via QR payment. Once payment has been made, the customer will need to wait for staff verification and approval in order to activate the booking, after which a confirmation notification will be sent through their email and on the website.

Before picking up the car, customers are required to take photos of the car, fill in the current fuel level and upload the documents along with the agreement form. The system will then automatically start the countdown once the uploading process has been completed. A few minutes before the rental period ends, customers are required to upload after-rental photos along with the inspection form. Customers are also given the option to leave feedback and ratings regarding their overall rental experience, whether about the service or the car. If a customer wishes to cancel the booking, it must be done at least 24 hours before the pick-up time in order to receive a full refund including the deposit. Cancellations made within 24 hours of the pick-up time will not be eligible for a refund.

1.3.4.4 Payment and Penalty Handling

The system will also handle payment and penalty tracking. The system supports QR payment, and payment receipts must be uploaded for verification. Late returns are subject to penalties, as users are given a 30-minute grace period after the return time. After this period, an additional fine of RM50 will be charged for each hour of overdue return, regardless of whether the vehicle is a car or a motorcycle.

1.3.4.5 Rewards

The vouchers page will display all available vouchers that can be redeemed by customers based on their loyalty stamps. Once a voucher has been redeemed, it must be used within a specified time limit before it expires. Vouchers such as a 12-hour free booking can be applied during the booking process to reduce the total rental cost.

1.3.4.6 Admin Dashboard

The admin dashboard displays important activity and information needed to be reviewed, such as total revenue, active rentals, pending tasks, quick links and an action center. The manager dashboard contains more visualised graphs that show performance trends, top rented models, customer analytics and fleet and booking status. Both managers and administrators are able to verify user documents, approve or reject customer bookings, update car availability and manage blacklists while tracking payments.

The fleet page allows staff to add or edit vehicle information, including rental prices, vehicle documentation and other relevant details. The bookings page displays all bookings recorded in the system along with all relevant information compiled throughout the booking process, including receipts, photos, inspection forms and reviews.

The reporting page displays revenue and performance data based on preferences set by the administrator. The data can be filtered by yearly, monthly or weekly intervals and adjusted by categories. This page also offers a print feature for documentation purposes.

1.3.4.7 Out of Scope

The system does not handle vehicle insurance claims or policy renewals. All insurance-related matters would remain under Hasta Travels' external insurance provider to avoid integration with third-party insurance APIs and confidential legal data. In addition, the system does not include real-time vehicle tracking using GPS. Vehicle location monitoring and live tracking are not implemented to avoid additional hardware requirements and system complexity. Lastly, although the system accepts payment receipts, it does not connect directly to bank systems. All payment verifications are done manually by staff to avoid the complexity of financial API compliance.

2.0 Global Conceptual Data Model (Global ERD)

The Global Conceptual Data Model (Global ERD) refines and integrates the local conceptual data models defined in the previous chapter, Database System Requirement, covering the User Management, Fleet Management, Car Rental Booking, Rewards and Reporting modules. The Global ERD consists of 6 entity types and 8 relationships, providing a detailed snapshot of how the data in Hasta Travel Vehicle Rental Booking System is stored and interrelated in the system. By combining the individual module requirements, this model resolves the potential conflicts and redundancies serving as the blueprint for physical database implementation. It ensures that all critical data flows ranging from user verification and fleet management to complex booking transactions and loyalty rewards are effectively supported while maintaining the strict data integrity.

2.1 Global Conceptual Database Design (Lucidchart)

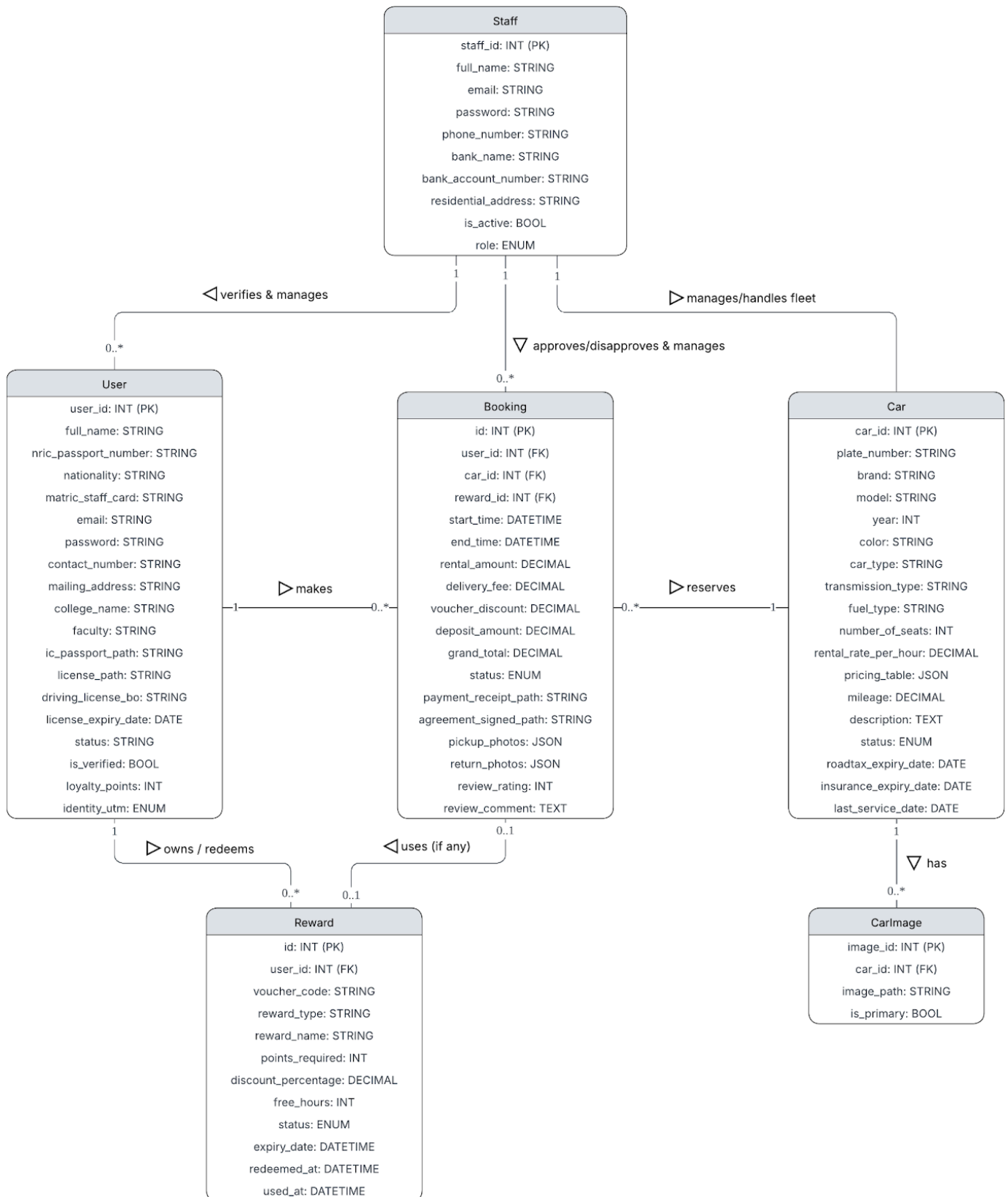


Figure 2.1 : Global Conceptual Entity Relationship Diagram

2.2 Data Dictionary for the global conceptual ERD

2.2.1 Description of Entities

Entity Name	Description	Aliases	Occurrence
User	Represents the customers either UTM students or UTM staff who register and undergo the verification process in order to book a vehicle rental.	Customer	Each occurrence represents a registered customer account containing the personal data either being verified, not yet verified or blacklisted. The entity can serve as a data source for the Reporting Module by allowing the system to group the user details like faculty, college and identity type to create reports and insights about the customer base.
Staff	Represents the administrative and operational personnel who are responsible for managing the fleet, verifying the users and approving as well as handling the bookings.	Admin, Employee/Staff	Each occurrence represents a staff with specific access control or rights. The Admin defined within this entity are authorized to access the Reporting Module dashboard to view financial reports, system analytics and alter the details of a staff.
Car	Represents the specifications, status or availability of the vehicles in the system.	Vehicle	Each occurrence represents a vehicle in the fleet inventory. The records of fleet allow the system to track the usage of a vehicle and can be used to help the admins analyze the performance and vehicle availability trends.
Booking	Represents rental records and transactions which links a user to a specific vehicle/car for a specific duration of time.	Reservation	Each occurrence represents a booking record from the initial request to completion. The entity provides the Reporting Module with key details like the total payment and timestamps that are used to calculate the monthly

			sales and generate the income reports.
Reward	Represents loyalty vouchers or coupons that are owned by users and can be redeemed using loyalty points.	Voucher	Each occurrence represents a specific redeemable voucher owned by a user, The records track the discount applied to bookings, ensuring that the value of each booking reflects the applied discount and accurately shows the user's savings.
CarImage	Represents the images associated with a specific vehicle for display purposes.	Vehicle Photo	Each occurrence represents an image file path linked to a specific car, allowing the system to display multiple angles of a vehicle in the fleet without cluttering the main Car table

2.2.2 Description of Relationships

Entity 1	Multiplicity	Relationship	Multiplicity	Entity 2	Description
User	1	makes	0..*	Booking	A single user can make multiple bookings.
Staff	1	verifies and manages	0..*	User	A staff can verify documents of multiple users and manage multiple user accounts.
Staff	1	approves or disapproves and manages	0..*	Booking	A staff can process multiple bookings either approving or rejecting booking requests sent by the users and manage the booking process.
Staff	1	manages or handles fleet	0..*	Car	A staff can add, update and change the availability status of multiple vehicles in the fleet inventory.
Booking	0..*	reserves	1	Car	Each booking reserves exactly a specific car from the fleet ensuring that the vehicle is booked for the selected time duration.
User	1	owns or redeems	0..*	Reward	A user can redeem and own multiple loyalty vouchers or rewards.
Booking	0..*	Uses (if any)	0..1	Reward	A booking may apply a voucher or reward for discount
Car	1	has	0..*	CarImage	A vehicle or car can be associated with multiple images.

2.2.3 Description of Attributes

Entity Name	Attributes	Data Type	Constraint	Description
User	user_id	INT	Primary Key (PK), Auto Increment	Unique identifier for the user
	full_name	STRING	Not Null	Full name of the user
	nric_passport_number	STRING	Unique, Not Null	Unique NRIC or Passport number
	nationality	STRING	Not Null	Nationality of user
	matric_staff_card	STRING	Unique, Not Null	Unique Matric ID of student or Staff ID of staff
	email	STRING	Unique, Not Null	Registered email address (login credential)
	password	STRING	Unique, Not Null	Hashed login password (hashed for security)
	contact_number	STRING	Not Null	User mobile phone number
	mailing_address	STRING	Not Null	Permanent residential or mailing address
	college_name	STRING	Not Null	Name of residential college inside or outside UTM (if applicable)
	faculty	STRING	Nullable	Name of the faculty (if applicable)
	ic_passport_path	STRING	Nullable	File path to the uploaded NRIC/Passport document image
	license_path	STRING	Nullable	File path to the uploaded driving license image
	driving_license_no	STRING	Nullable	Unique driving license number
	license_expiry_date	DATE	Nullable	Expiry date of the driving license

	status	STRING	Enum: 'Active', 'Pending', 'Blacklisted' (Default: 'Active')	Account status (e.g. Active, Blacklisted)
	is_verified	BOOL	Default: False	Flag indicating if the documents of user have been verified
	loyalty_points	INT	Default: 0	Current loyalty points earned by user
	identity_utm	ENUM	Enum: 'Student', 'Staff'	User type: "Student" or "Staff"
Staff	staff_id	INT	Primary Key (PK), Auto Increment	Unique identifier for the staff
	full_name	STRING	Not Null	Full name of staff
	email	STRING	Unique, Not Null	Staff login email address
	password	STRING	Unique, Not Null	Hashed login password (hashed for security)
	phone_number	STRING	Not Null	Staff mobile phone number
	bank_name	STRING	Not Null	Staff's bank name
	bank_account_number	STRING	Not Null	Staff's bank account number
	residential_address	STRING	Not Null	Staff's home or residential address
	is_active	BOOL	Default: True	Flag to grant or revoke the access control to the system
	role	ENUM	Enum: 'Admin', 'Staff' (Default: 'Staff')	Access level: "Admin" or "Staff"
Car	car_id	INT	Primary key (PK), Auto Increment	Unique identifier for the vehicle
	plate_number	STRING	Unique, Not Null	Unique vehicle registration number

	brand	STRING	Not Null	Vehicle manufacturer
	model	STRING	Not Null	Vehicle model name
	year	INT	Not Null	Manufacturing year of vehicle
	color	STRING	Not Null	Vehicle color
	car_type	STRING	Not Null	Category of vehicle (e.g. Sedan, Hatchback)
	transmission_type	STRING	Not Null	Gear type (e.g. Auto, Manual)
	fuel_type	STRING	Not Null	Fuel type (e.g. Petrol, Diesel)
	number_of_seats	INT	Not Null	Seating capacity of the vehicle
	rental_rate_per_hour	DECIMAL	Not Null	Hourly rental rate/cost
	pricing_table	JSON	JSON, Nullable	Complex pricing rules
	mileage	DECIMAL	Not Null	Current mileage or odometer reading
	description	TEXT	Nullable	Description of vehicle
	status	ENUM	Enum: 'Available', 'In Use', 'Maintenance', 'Unavailable' (Default: 'Available')	Availability status of vehicle: "Available", "In Use" "Maintenance" or "Unavailable"
	roadtax_expiry_date	DATE	Nullable	Expiry date of vehicle roadtax
	insurance_expiry_date	DATE	Nullable	Expiry date of vehicle insurance
	last_service_date	DATE	Nullable	Most recent service/maintenance date
CarImage	image_id	INT	Primary Key (PK), Auto Increment	Unique identifier for the image

	car_id	INT	Foreign Key (FK) referencing Car (car_id), On Delete Cascade	Reference to the associated Car
	image_path	STRING	Not Null	File path where the image is stored
	is_primary	BOOL	Default: False	Flag indicating if the image is the main image used for display purpose
Reward	id	INT	Primary Key (PK), Auto Increment	Unique identifier for the reward
	user_id	INT	Foreign Key (FK) referencing User (user_id), On Delete Cascade	Reference to the User who owns the reward
	voucher_code	STRING	Unique, Not Null	Unique code used for voucher redemption to get the discount
	reward_type	STRING	Not Null	Type of the reward (e.g. 'discount_10' stands for 10% discount)
	reward_name	STRING	Not Null	Display name of the reward/voucher
	points_required	INT	Not Null	Cost in loyalty points to redeem the reward/voucher
	discount_percentage	DECIMAL	Nullable	Percentage of discount for voucher
	free_hours	INT	Nullable	Number of free hours granted (if applicable)
	status	ENUM	Enum: 'Available', 'Used', 'Expired' (Default: 'Available')	Validity status of voucher: "Available", "Used" or "Expired"
	expiry_date	DATETIME	Not Null	Date when the voucher becomes invalid
	redeemed_at	DATETIME	Not Null	Timestamp when loyalty

				points are exchanged to redeem new voucher/reward
	used_at	DATETIME	Nullable	Timestamp when voucher is applied to booking
Booking	id	INT	Primary Key (PK), Auto Increment	Unique identifier for the booking record
	user_id	INT	Foreign Key (FK) referencing User (user_id)	Reference to User who made the booking
	car_id	INT	Foreign Key (FK) referencing Car (car_id)	Reference to Car being rented
	reward_id	INT	Foreign Key (FK) referencing Reward (id), Nullable	Reference to Reward used (if any)
	start_time	DATETIME	Not Null	Scheduled vehicle pickup date and time
	end_time	DATETIME	Not Null	Schedule vehicle dropoff date and time
	rental_amount	DECIMAL	Not Null	Rental amount for selected vehicle and time
	delivery_fee	DECIMAL	Default: 0.00	Additional fee for pickup or return delivery for location other than Student Mall
	voucher_discount	DECIMAL	Default: 0.00	Amount deducted by the applied voucher (if any)
	deposit_amount	DECIMAL	Not Null	Deposit that will be refunded after completing a booking if there is no issue or damage on vehicle rented
	grand_total	DECIMAL	Not Null	Final total amount to be paid
	status	ENUM	Enum: 'Pending', 'Approved',	State of booking: "Pending", "Approved",

			‘Active’, ‘Completed’, ‘Cancelled’, ‘Rejected’ (Default: ‘Pending’)	“Active”, “Completed”, “Cancelled” or “Rejected”
	payment_receipt_path	STRING	Nullable	File path to the payment receipt image
	agreement_signed_path	STRING	Nullable	File path to the signed rental agreement
	pickup_photos	JSON	JSON, Nullable	Array of image paths taken at pickup location
	return_photos	JSON	JSON, Nullable	Array of image paths taken at return location
	review_rating	INT	Nullable, Integer (1-5)	Customer rating in stars
	review_comment	TEXT	Nullable	Customer feedback text

3.0 Logical Data Model

A Logical Data Model (Logical ERD) provides an abstract representation of a system's data and the relationships among data elements, independent of any specific database technology or physical implementation. It identifies the entities within the system, their attributes, and the relationships between entities, such as one-to-one, one-to-many, and many-to-many relationships, in a structured and business-oriented manner.

Compared to a conceptual data model, which offers a high-level view of the system, the logical ERD introduces greater detail by specifying attributes, keys, data types, and normalization rules while remaining technology-neutral. As such, it acts as an intermediate model that bridges business requirements and physical database design, helping to ensure that the data structure is consistent, non-redundant, and aligned with organizational data usage.

3.1 Logical Database Design (Logical EDR)

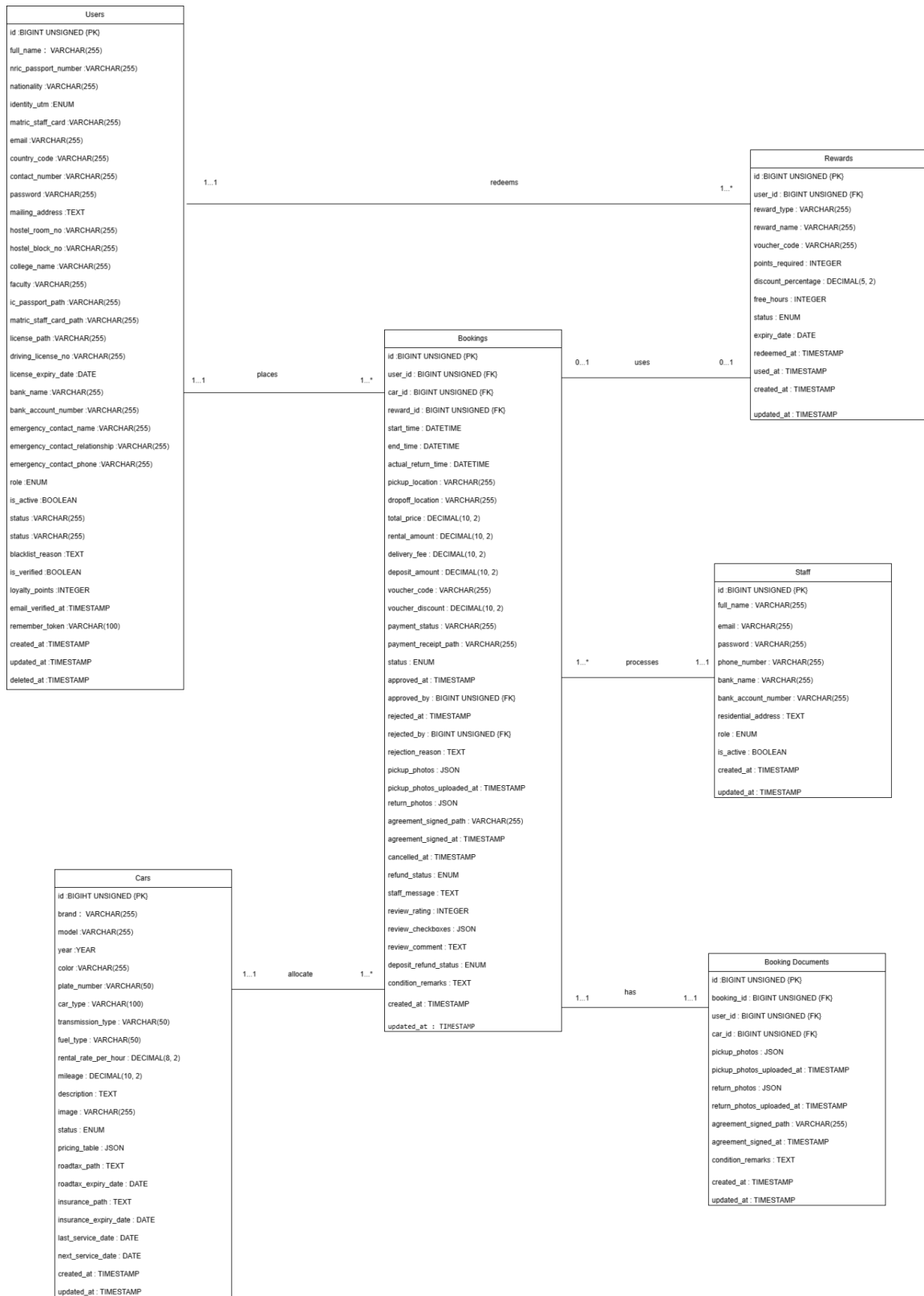


Figure :Logical ERD

3.2 Data Dictionary for Logical ERD

The data dictionary provided a detailed reference guide for the Logical ERD. It provides clear definitions for each entity and includes all the information needed to give a better understanding for readers. There shows the description of the attribute with its data type, length, key and constraint.

3.2.1 Entity 1: Users

Store all the information about the users, including customers and admin.

Attribute Name	Description	Data Type	Length	Key	Constraint
id	Unique identifier for the user	BIGINT UNSIGNED	-	PK	Auto-increment, Not Null
full_name	Full name of the user same with the identical card	VARCHAR	255		Not Null
nric_passport_number	User's NRIC or Passport number.	VARCHAR	255		Unique, Not Null
nationality	Nationality of the user.	VARCHAR	255		Not Null
identity_utm	Categorizes users as UTM Students or Staff.	ENUM	-		“Student”, ”Staff”, Not Null
matric_staff_card	Matric card ID or Staff ID number.	VARCHAR	255		UNIQUE, Nullable
email	Registered email address for login and contact.	VARCHAR	255		UNIQUE, Not Null
country_code	Country dialing code for phone number.	VARCHAR	255		Default: '+60', Not Null
contact_number	Primary contact phone number.	VARCHAR	255		Not Null
password	Hashed password for authentication.	VARCHAR	255		Not Null
mailing_address	User's permanent or mailing address.	TEXT	-		Not Null
hostel_room_no	Room number if	VARCHAR	255		Nullable

	staying in a hostel.				
hostel_block_no	Block number if staying in hostel.	VARCHAR	255		Nullable
college_name	Name of the residential college.	VARCHAR	255		Not Null
faculty	Faculty the student/staff belongs to.	VARCHAR	255		Nullable
ic_passport_path	File path to uploaded IC/Passport image.	VARCHAR	255		Nullable
matric_staff_card_path	File path to uploaded Matric/Staff card image.	VARCHAR	255		Nullable
license_path	File path to uploaded driving license image	VARCHAR	255		Nullable
driving_license_no	Driving license alphanumeric ID.	VARCHAR	255		Nullable
license_expiry_date	Expiry date of the driving license.	DATE	-		Nullable
bank_name	Name of the bank for refunds.	VARCHAR	255		Nullable
bank_account_number	Bank account number for refunds.	VARCHAR	255		Nullable
emergency_contact_name	Name of emergency contact person	VARCHAR	255		Nullable
emergency_contact_relationship	Relationship with emergency contact.	VARCHAR	255		Nullable
emergency_contact_phone	Phone number of emergency contact.	VARCHAR	255		Nullable
role	User's system role.	ENUM	-		'Customer','admin' Default : 'customer'
is_active	Flag to indicate if the account is active.	BOOLEAN	-		Default: TRUE
status	Current status of user	VARCHAR	255		Default: 'active'
blacklist_reason	Reason provided if the	TEXT	-		Nullable

	user is blacklisted.				
is_verified	Flag indicating if the document is verified.	BOOLEAN	-		Default: FALSE
loyalty_points	Accumulated reward points	INTEGER	-		Default: 0
email_verified_at	Timestamp when email was verified.	TIMESTAMP	-		Nullable
remember_token	Token for exist user to login again in fast way	VARCHAR	100		Nullable
created_at	Record creation timestamp	TIMESTAMP			Nullable
updated_at	Record last update timestamp	TIMESTAMP			Nullable
deleted_at	Timestamp for deletes.	TIMESTAMP			Nullable

3.2.2 Entity 2: Cars

Store all the information about the rental car ,including the brand, price and the availability.

Attribute Name	Description	Data type	Length	Key	Constraint
id	Unique identifier for the vehicle	BIGINT UNSIGNED	-	PK	Auto-increment, Not Null
brand	Car brand	VARCHAR	255		Not Null
model	Car model	VARCHAR	255		Not Null
year	Manufacturing year of the car	YEAR	-		Not Null
color	Exterior color of car	VARCHAR	255		Not Null
plate_number	Plate number of car	VARCHAR	50		UNIQUE, Not Null
car_type	Type of car, example: suv	VARCHAR	100		Not Null
transmission_type	Transmission type,example:a uto/manual	VARCHAR	50		Not Null
fuel_type	Fuel type,example:p etrol	VARCHAR	50		Not Null
number_of_seats	Seating capacity	INTEGER	-		Not Null
rental_rate_per_hour	Cost of renting in one hour	DECIMAL	8,2		Not Null
mileage	Current mileage of the car	DECIMAL	10,2		Not Null
description	Additional details about the car	TEXT	-		Nullable
image	File path to car's display image	VARCHAR	255		Nullable
status	Current	ENUM	-		'available',

	availability status				'unavailable', Default: 'available'
pricing_table	Store complex pricing rules	JSON	-		Nullable
roadtax_path	File path store roadtax document	TEXT	-		Nullable
roadtax_expiry_date	Expire date of roadtax	DATE	-		Nullable
insurance_path	File path to insurance document	TEXT	-		Nullable
insurance_expiry_date	Expiry date of the insurance policy	DATE	-		Nullable
last_service_date	Date of last maintenance service	DATE	-		Nullable
next_service_date	Date of next maintenance service	DATE	-		Nullable
created_at	Record creation timestamp	TIMESTAMP	-		Nullable
updated_at	Record last update timestamp	TIMESTAMP	-		Nullable

3.2.3 Entity 3: Bookings

Store all the details about the booking process, transaction records, related users, cars details and payments.

Attribute Name	Description	Data type	Length	Key	Constraint
id	Unique identifier for the booking.	BIGINT UNSIGNED	-	PK	Auto-increment, Not Null
user_id	Reference to the user table.	BIGINT UNSIGNED	-	FK	Not Null
car_id	Reference to the cars table.	BIGINT UNSIGNED	-	FK	Not Null
reward_id	Reference to the rewards table.	BIGINT UNSIGNED	-	FK	Not Null
start_time	Scheduled pickup date and time.	DATETIME	-		Not Null
end_time	Scheduled return date and time.	DATETIME	-		Not Null
actual_return_time	Actual return date and time.	DATETIME	-		Nullable
pickup_location	Location where the car is picked up.	VARCHAR	255		Not Null
dropoff_location	Location where car is returned.	VARCHAR	255		Not Null
total_price	Final total price to be paid.	DECIMAL	10,2		Not Null
rental_amount	Base rental cost before fees.	DECIMAL	10,2		Default:0
delivery_fee	Additional fee for delivery.	DECIMAL	10,2		Default:0
deposit_amount	Security deposit amount.	DECIMAL	10,2		Not Null
voucher_code	Code of voucher applied.	VARCHAR	255		Nullable
voucher_discount	Discount amount from voucher.	DECIMAL	10,2		Default:0

payment_status	Status of payment transaction.	VARCHAR	255		Default:"Pending"
payment_receipt_path	Path to payment receipt image.	VARCHAR	255		Nullable
status	Current status of booking workflow.	ENUM	-		'pending', 'approved', 'active', 'completed', 'cancelled', 'rejected', Default: 'pending'
approved_at	Timestamp when booking was approved.	TIMESTAMP	-		Nullable
approved_by	Reference to staff who approved.	BIGINT UNSIGNED	-	FK	Nullable
rejected_at	Timestamp when booking was rejected.	TIMESTAMP	-		Nullable
rejected_by	Reference to STAFF who rejected	BIGINT UNSIGNED	-	FK	Nullable
rejection_reason	Reason for rejection.	TEXT	-		Nullable
pickup_photos	Photos taken during pickup.	JSON	-		Nullable
pickup_photos_uploaded_at	Timestamp of pickup photo upload.	TIMESTAMP	-		Nullable
return_photos	Photos taken during return.	JSON	-		Nullable
agreement_signed_path	Path to signed rental agreement.	VARCHAR	255		Nullable
agreement_signed_at	Timestamp of agreement signing.	TIMESTAMP	-		Nullable
cancelled_at	Timestamp if booking was cancelled.	TIMESTAMP	-		Nullable

refund_status	Status of deposit/payment refund.	ENUM	-		'pending', 'refunded', 'charged_and_refunded', 'no_refund', Nullable
staff_message	Notes from staff.	TEXT	-		Nullable
review_rating	Customer rating (1-5 stars).	INTEGER	-		Nullable
review_checkboxes	Selected feedback tags.	JSON	-		Nullable
review_comment	Comment from customer	TEXT	-		Nullable
deposit_refund_status	Specific status for deposit refund.	ENUM	-		'pending', 'refunded', 'charged', Nullable
condition_remarks	Remarks on vehicle condition	TEXT	-		Nullable
created_at	Record creation timestamp.	TIMESTAMP	-		Nullable
updated_at	Record last update timestamp.	TIMESTAMP	-		Nullable

3.2.4 Entity 4: Rewards

Store all the information about the reward system ,it shows the input data when a booking is completed and the output data when the rewards are redeemed. There store all the information that needed to run the mechanism of the reward system.

Attribute Name	Description	Data type	Length	Key	Constraint
id	Unique identifier for the reward.	BIGINT UNSIGNED	-	PK	Auto-increment , Not Null
user_id	Reference to the users table	BIGINT UNSIGNED	-	FK	Not Null
reward_type	Type of reward,example: 10% discount	VARCHAR	255		Not Null
reward_name	Display name of the reward.	VARCHAR	255		Not Null
voucher_code	Unique code to redeem the reward.	VARCHAR	255		UNIQUE, Not Null
points_required	Points cost to redeem this reward.	INTEGER	-		Not Null
discount_percentage	Percentage value for discount rewards.	DECIMAL	5,2		Nullable
free_hours	Number of free hours granted.	INTEGER	-		Nullable
status	Current state of the reward voucher.	ENUM	-		'available', 'Unavailable', Default: 'available'
expiry_date	Last valid date for the voucher.	DATE	-		Not Null
redeemed_at	Timestamp when user redeemed points for this.	TIMESTAMP	-		Not Null
used_at	Timestamp	TIMESTAMP	-		Nullable

	when voucher was applied to a booking.				
created_at	Record creation timestamp.	TIMESTAMP	-		Nullable
updated_at	Record last update timestamp.	TIMESTAMP	-		Nullable

3.2.5 Entity 5: Staff

Store all the information about the staff and administrators. Administrators have full access to advanced features such as staff management and report generation

Attribute Name	Description	Data type	Length	Key	Constraint
id	Unique identifier for the staff.	BIGINT UNSIGNED	-	PK	Auto-increment, Not Null
full_name	Full legal name of the staff.	VARCHAR	255		Not Null
email	Staff login email address.	VARCHAR	255		UNIQUE, Not Null
password	Hashed login password.	VARCHAR	255		Not Null
phone_number	Contact phone number.	VARCHAR	255		Not Null
bank_name	Bank name for salary/claims.	VARCHAR	255		Not Null
bank_account_number	Bank account number.	VARCHAR	255		Not Null
residential_address	Home address of the staff member.	TEXT	-		Not Null
role	Permission level in the system.	ENUM	-		'admin', 'staff', Default: 'staff'
is_active	Flag to indicate if the account is active.	BOOLEAN	-		Default: TRUE
created_at	Record creation timestamp.	TIMESTAMP	-		Nullable
updated_at	Record last update timestamp.	TIMESTAMP	-		Nullable

3.2.6 Entity 6: Booking_documents

Store all the information about the booking process ,including the details of other entities such as users and cars.This entity also produces the booking document as an entity.

Attribute Name	Description	Data type	Length	Key	Constraint
id	Unique identifier for the document record.	BIGINT UNSIGNED	-	PK	Auto-increment , Not Null
booking_id	Reference to the bookings table	BIGINT UNSIGNED	-	FK	Not Null
user_id	Reference to users	BIGINT UNSIGNED	-	FK	Not Null
car_id	Reference to cars	BIGINT UNSIGNED	-	FK	Not Null
pickup_photos	Collection of image paths taken at pickup.	JSON	-		Nullable
pickup_photos_uploaded_at	Timestamp when pickup photos were uploaded.	TIMESTAMP	-		Nullable
return_photos	Collection of image paths taken at return.	JSON	-		Nullable
return_photos_uploaded_at	Timestamp when return photos were uploaded.	TIMESTAMP	-		Nullable
agreement_signed_path	Path to the signed PDF agreement.	VARCHAR	255		Nullable
agreement_signed_at	Timestamp when agreement was signed.	TIMESTAMP	-		Nullable
condition_remarks	Notes on vehicle condition	TEXT	-		Nullable
created_at	Record creation timestamp	TIMESTAMP	-		Nullable
updated_at	Record last update timestamp.	TIMESTAMP	-		Nullable

3.3 Relational Database Schema

Here is the Relational Database Schema before normalization (UNF) :

1. User (id , full_name, nric_passport_number, nationality ,identity_utm ,
matric_staff_card, email, country_code , contact_number, password,
mailing_address , hostel_room_no , hostel_block_no, college_name,
faculty , ic_passport_path , matric_staff_card_path , license_path ,
driving_license_no , license_expiry_date , bank_name ,
bank_account_number , emergency_contact_name ,
emergency_contact_relationship , emergency_contact_phone , role ,
is_active , status , blacklist_reason , is_verified , loyalty_points
,email_verified_at ,remember_token , created_at , updated_at , deleted_at
)

Primary Key : id

2. CARS (id,brand , model , year , color , plate_number , car_type ,
transmission_type , fuel_type, number_of_seats , rental_rate_per_hour ,
mileage , description , image , status , pricing_table , roadtax_path ,
insurance_path , insurance_expiry_date , last_service_date ,
next_service_date , created_at , updated_at)

Primary Key : id

3. BOOKINGS (id, user_id , car_id ,reward_id , start_time , end_time ,
actual_return_time , pickup_location , dropoff_location , total_price ,

rental_amount , delivery_fee , deposit_amount , voucher_code ,
voucher_discount , payment_status , payment_receipt_path , status ,
approved_at , approved_by , rejected_at , rejected_by , rejection_reason ,
pickup_photos , pickup_photos_uploaded_at , return_photos ,
agreement_signed_path , agreement_signed_at , cancelled_at ,
refund_status , staff_message , review_rating , review_checkboxes ,
review_comment , deposit_refund_status , condition_remarks ,
created_at, updated_at)

Primary Key : id

FK : user_id references users(id)

FK : car_id references cars(id)

FK : reward_id references rewards(id)

FK : approved_by references staff.id

FK : rejected_by references staff.id

4. REWARDS (id , user_id , reward_type , reward_name , voucher_code ,
points_required , discount_percentage , free_hours , status , expiry_date ,
redeemed_at ,used_at, created_at , updated_at)

Primary Key : id

FK : user_id references users(id)

5. STAFF (id, full_name , email , phone_number , bank_name ,
bank_account_number , residential_address , role , is_active , created_at ,
updated_at)

Primary Key : id

6. BOOKING_DOCUMENTS (id , booking_id , user_id , car_id ,
pickup_photos , pickup_photos_uploaded_at , return_photos ,
return_photos_uploaded_at , agreement_signed_path ,
agreement_signed_at , condition_remarks , created_at, updated_at)

Primary Key : id

FK : user_id references users(id)

FK : booking_id references bookings(id)

FK : car_id references cars(id)

3.4 Normalization

3.4.1 First Normal Form (1NF)

Anomalies Identified:

- CARS: The attribute pricing_table contains a JSON object with multiple values.
- BOOKINGS: The attributes pickup_photos and return_photos contain JSON arrays with multiple file paths and review_checkboxes contains multiple tags in a single column

Resulting Relations (1NF):

- CARS (removed the attribute pricing_table)
- CAR_PRICING (created new relation)
 - Attributes: id, car_id, hours, price
- BOOKINGS (removed pickup_photos, return_photos, review_checkboxes)
- BOOKING_PICKUP (created new relation)
 - Attributes: id, booking_id, pickup_frontview_path, pickup_leftview_path, pickup_rightview_path, pickup_backview_path, pickup_dashboardview_path
- BOOKING_DROPOFF (created new relation)
 - Attributes: id, booking_id, return_frontview_path, return_leftview_path, return_rightview_path, return_backview_path, return_dashboardview_path, return_keyview_path
- BOOKING_REVIEW_TAGS (created new relation)
 - Attributes: id, booking_id, tag_name

3.4.2 Second Normal Form (2NF)

All relations in the 1NF schema utilize id as the Primary Key. Since Primary Key consists of a single attribute, no partial dependencies exist as it is impossible for a non-key attribute to depend on only a “part” of the Primary Key.

3.4.3 Third Normal Form (3NF)

Transitive Dependencies Identified:

- BOOKINGS: booking_id $\rightarrow$ rental_amount, delivery_fee, voucher_discount $\rightarrow$ total_price
- BOOKINGS: document_id $\rightarrow$ review_rating, review_comment
- BOOKING_DOCUMENTS: document_id $\rightarrow$ booking_id $\rightarrow$ user_id, car_id

Resulting Relations (3NF):

- BOOKINGS (removed the attribute total_price)
- BOOKING_DOCUMENTS (removed the attributes user_id and car_id)
- REVIEWS (created new relation)
 - Attributes: id, booking_id, rating, comment, created_at

3.4.4 Boyce-Codd Normal Form (BCNF)

Determinants Identified:

- USERS:
 - fd1: nric_passport_number $\rightarrow$ user_id
 - fd2: email $\rightarrow$ user_id
- CARS:
 - fd1: plate_number $\rightarrow$ car_id
- REWARDS:
 - fd1: voucher_code $\rightarrow$ reward_id
- STAFF:
 - fd1: email $\rightarrow$ staff_id

All relations in the 3NF satisfy the Boyce-Codd Normal Form (BCNF) requirements. For every functional dependency identified, the determinant is defined as a Candidate Key via Unique constraints. Since every determinant is a Candidate Key, no violations exist and hence the schema remains unchanged from the 3NF.

3.4.4 Database Schema after Normalization

3.4.4.1 Primary Key (PK) and Foreign Key (FK)

1. USERS
 - PK: id
 - FK: -
2. CARS
 - PK: id
 - FK: -
3. CAR_PRICING
 - PK: id
 - FK: car_id references CARS(id)
4. STAFF
 - PK: id
 - FK: -
5. REWARDS
 - PK: id
 - FK: user_id references USERS(id)
6. BOOKINGS
 - PK: id
 - FK: user_id references USERS(id), car_id references CARS(id), reward_id references REWARDS(id), approved_by references STAFF(id), rejected_by references STAFF(id)
7. BOOKING_DOCUMENTS
 - PK: id
 - FK: booking_id references BOOKINGS(id)
8. BOOKING_PICKUP
 - PK: id
 - FK: booking_id references BOOKINGS(id)
9. BOOKING_DROPOFF
 - PK: id
 - FK: booking_id references BOOKINGS(id)
10. REVIEWS
 - PK: id

- FK: booking_id references BOOKINGS(id)

11. BOOKING_REVIEW_TAGS

- PK: id
- FK: booking_id references BOOKINGS(id)

3.4.4.2 Final Normalized Schema

1. USERS (id PK, full_name, nric_passport_number, nationality, identity_utm, matric_staff_card, email, country_code, contact_number, password, mailing_address, hostel_room_no, hostel_block_no, college_name, faculty, ic_passport_path, matric_staff_card_path, license_path, driving_license_no, license_expiry_date, bank_name, bank_account_number, emergency_contact_name, emergency_contact_relationship, emergency_contact_phone, role, is_active, status, blacklist_reason, is_verified, loyalty_points, email_verified_at, remember_token, created_at, updated_at, deleted_at)
2. CARS (id PK, brand, model, year, color, plate_number, car_type, transmission_type, fuel_type, number_of_seats, rental_rate_per_hour, mileage, description, image, status, roadtax_path, roadtax_expiry_date, insurance_path, insurance_expiry_date, last_service_date, next_service_date, created_at, updated_at)
3. CAR_PRICING (id PK, car_id FK, hours, price, created_at, updated_at)
4. STAFF (id PK, full_name, email, password, phone_number, bank_name, bank_account_number, residential_address, role, is_active, created_at, updated_at)
5. REWARDS (id PK, user_id FK, reward_type, reward_name, voucher_code, points_required, discount_percentage, free_hours, status, expiry_date, redeemed_at, used_at, created_at, updated_at)
6. BOOKINGS (id PK, user_id FK, car_id FK, reward_id FK, start_time, end_time, actual_return_time, pickup_location, dropoff_location, rental_amount, delivery_fee, deposit_amount, voucher_code, voucher_discount, payment_status, payment_receipt_path, status, approved_at, approved_by FK, rejected_at, rejected_by)

FK, rejection_reason, cancelled_at, refund_status, staff_message,
deposit_refund_status, condition_remarks, created_at, updated_at)

7. BOOKING_DOCUMENTS (id PK, booking_id FK, agreement_signed_path,
agreement_signed_at, condition_remarks, created_at, updated_at)
8. BOOKING_PICKUP (id PK, booking_id FK, pickup_frontview_path,
pickup_leftview_path, pickup_rightview_path, pickup_backview_path,
pickup_dashboardview_path, created_at)
9. BOOKING_DROPOFF (id PK, booking_id FK, return_frontview_path,
return_leftview_path, return_rightview_path, return_backview_path,
return_dashboardview_path, return_keyview_path, created_at)
10. REVIEWS (id PK, booking_id FK, rating, comment, created_at, updated_at)
11. BOOKING_REVIEW_TAGS (id PK, booking_id FK, tag_name, created_at)

4.0 Interface Design and SQL Implementation

This section presents the complete interface design mapped to SQL implementation for the Hasta Vehicle Rental Booking System. Each module is organized with interface descriptions, screenshots, and corresponding DDL (Data Definition Language) and DML (Data Manipulation Language) statements that demonstrate how the database supports system functionality.

4.1 User Management Module

4.1.1 Interface Design

Public / Customer Interfaces

- Homepage (before login)

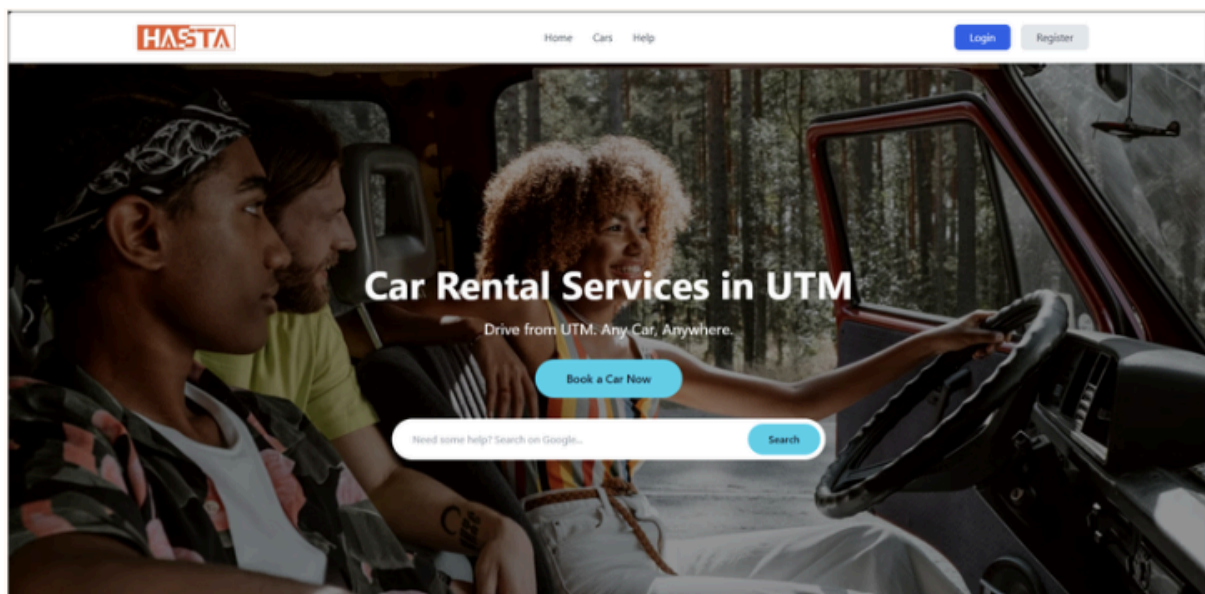


Figure 4.1: Interface for homepage before login

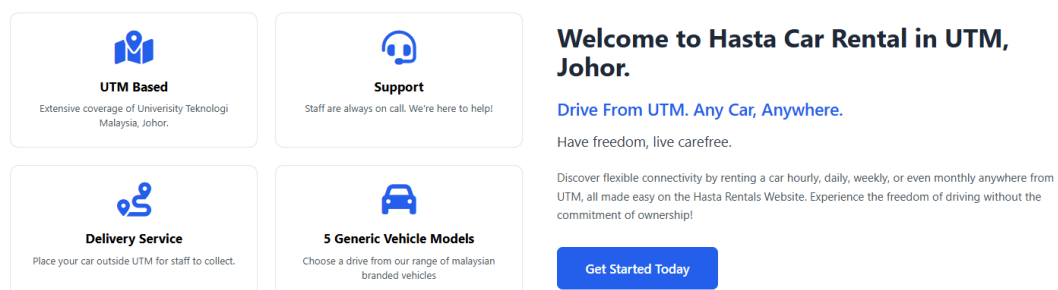


Figure 4.2: Interface for below part of homepage before login

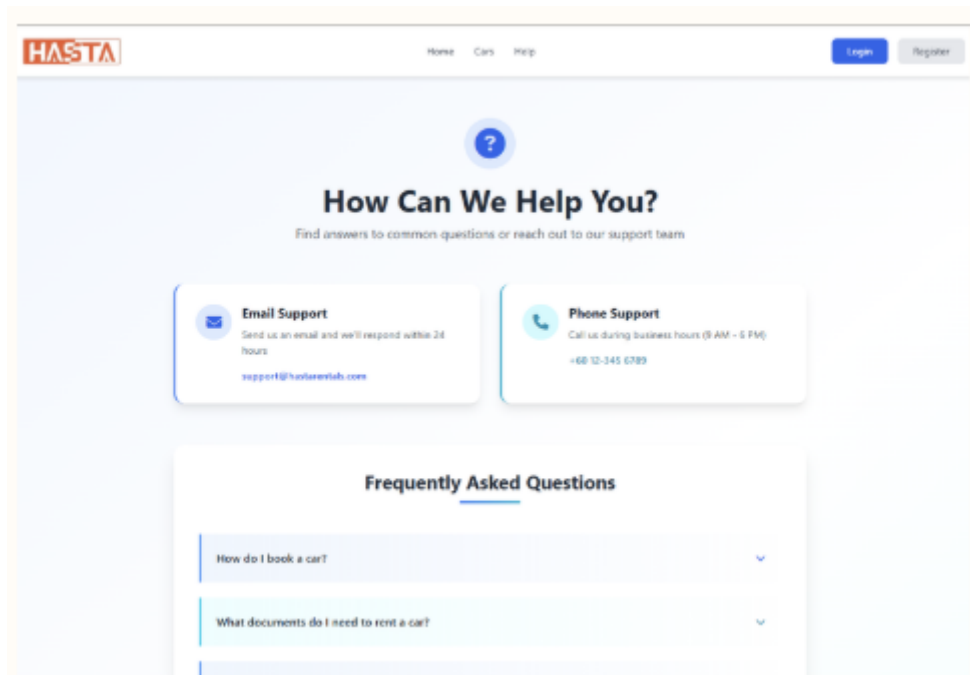


Figure 4.3: Interface for below part of homepage before login

- Browse Fleet & Car Details

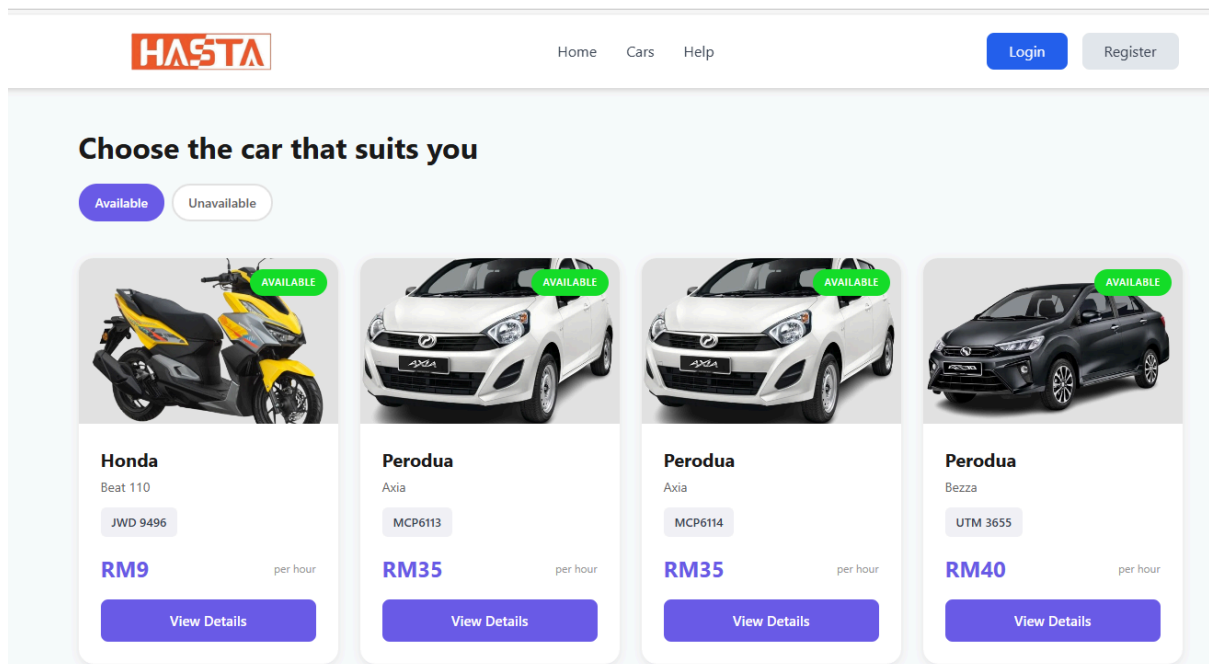


Figure 4.4: Interface for rental car display

HASTA[Home](#)[Bookings](#)[Cars](#)[Vouchers](#)[Help](#)

[Back to Car List](#)

AVAILABLE

Perodua Axia

Hatchback • White

MCP6113

Rental Rate

RM35.00/hour

[Rent Now](#)

Figure 4.5: Fleet interface

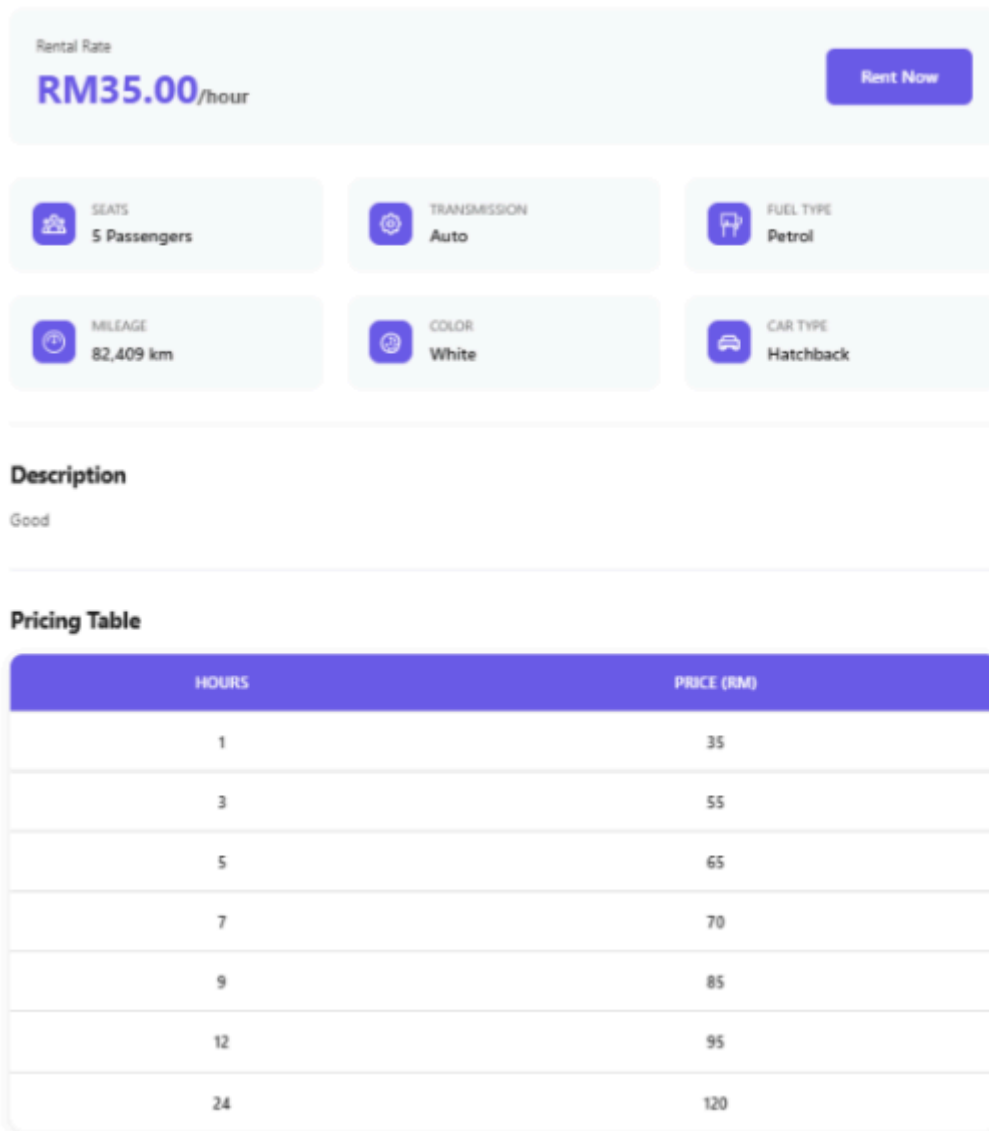


Figure 4.6: Interface for display the details of car and the pricing table

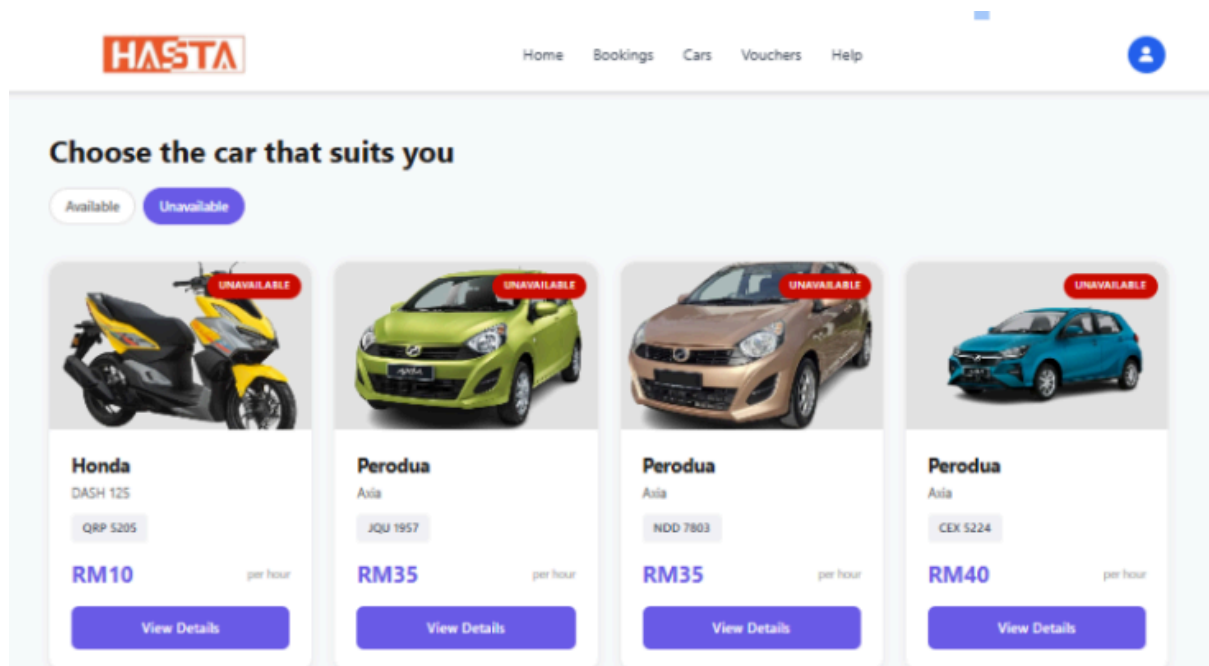


Figure 4.7: Interface for display unavailable cars

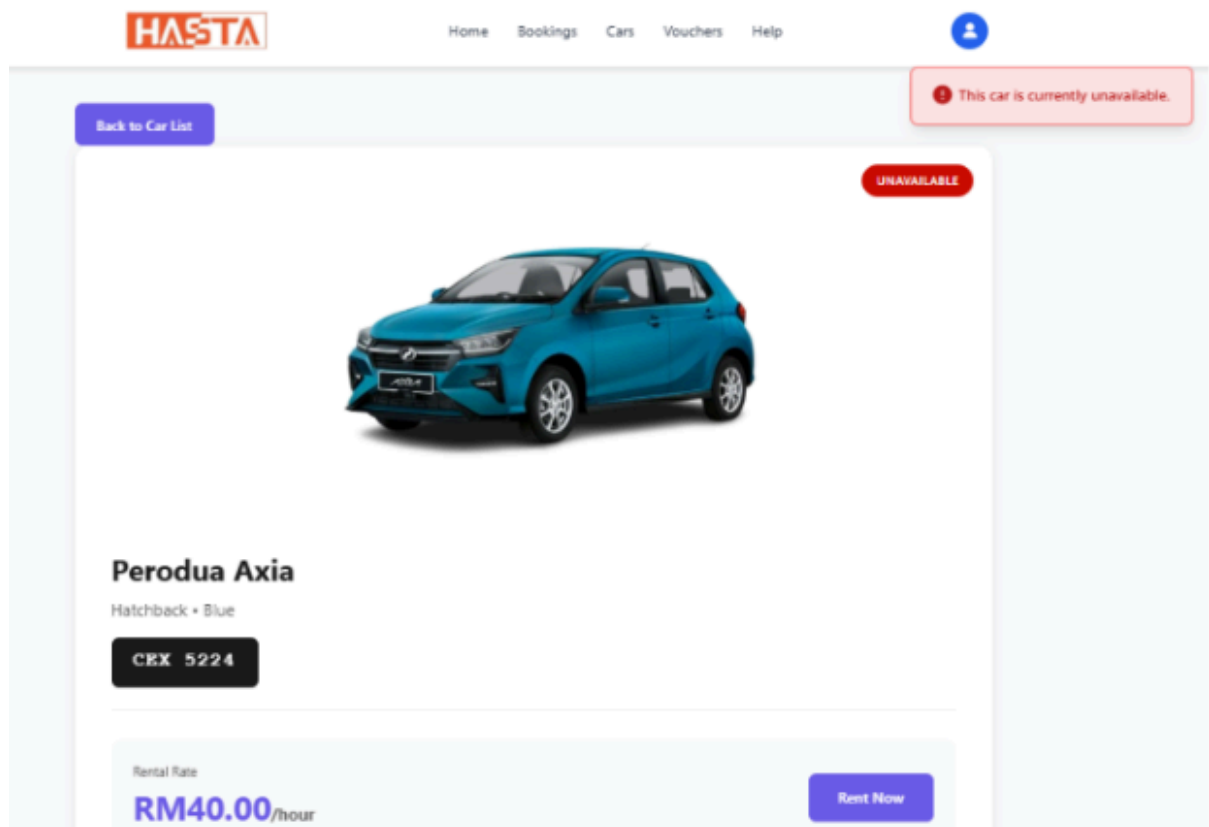


Figure 4.8: Vehicle details (unavailable) + notification message

- Authentication

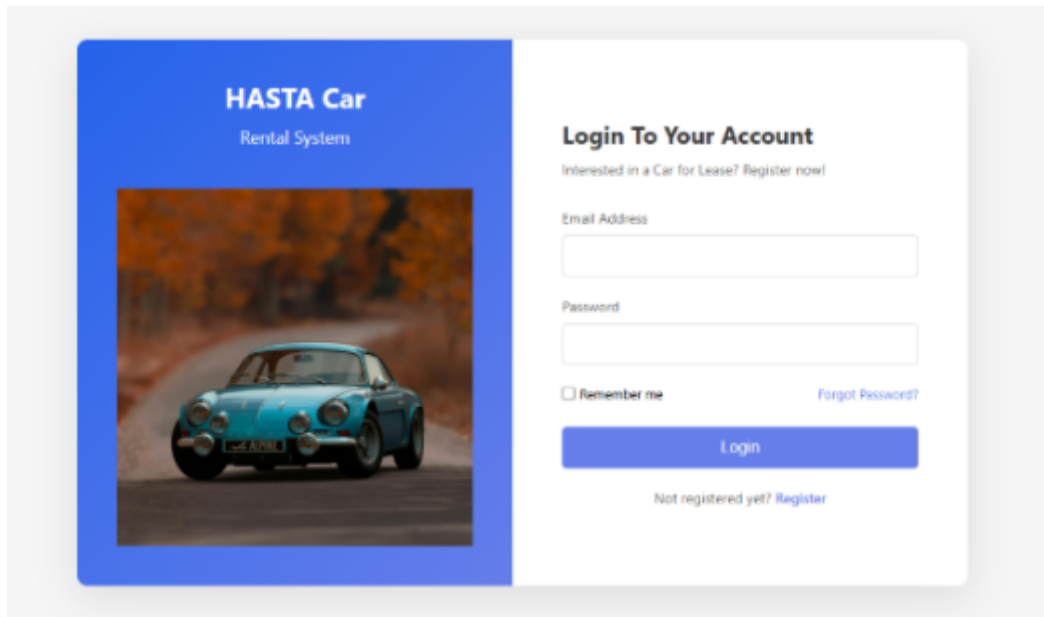


Figure 4.9: Interface for user login page

Figure 4.10: Interface for register new user's account

- Homepage (after login) + Profile



Figure 4.11: Interface for homepage after user login

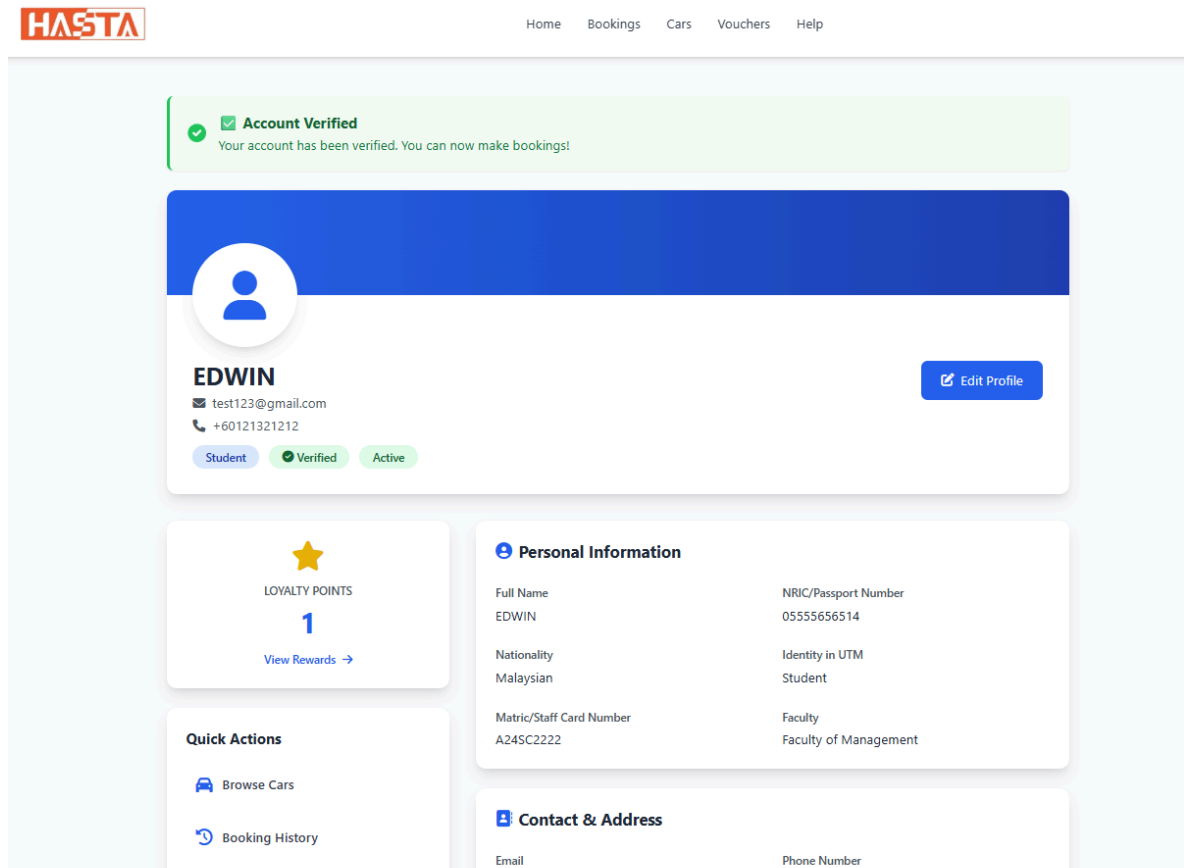


Figure 4.12: Interface for profile of the user

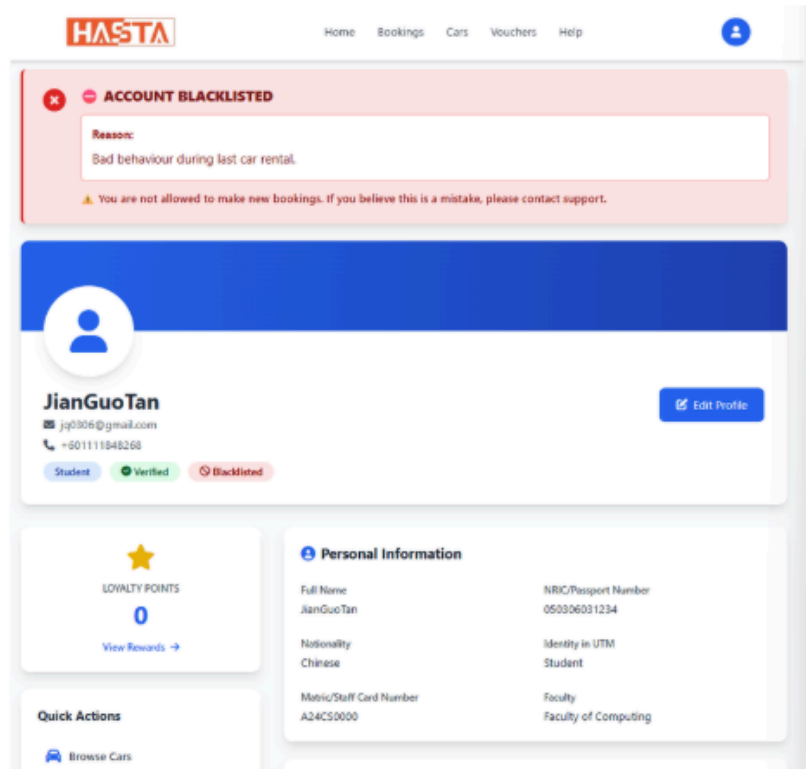


Figure 4.13: Interface for profile page of the blacklist user

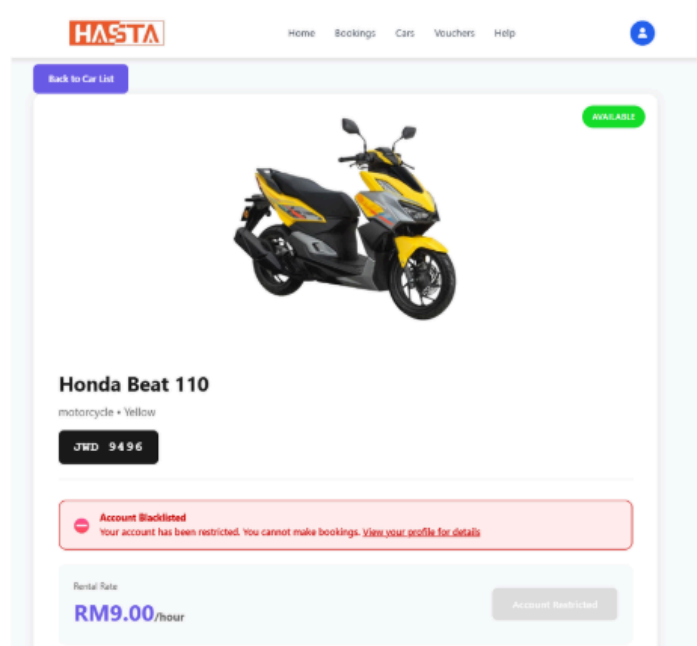


Figure 4.14: Interface for booking page of the blacklist user (disabled booking button)

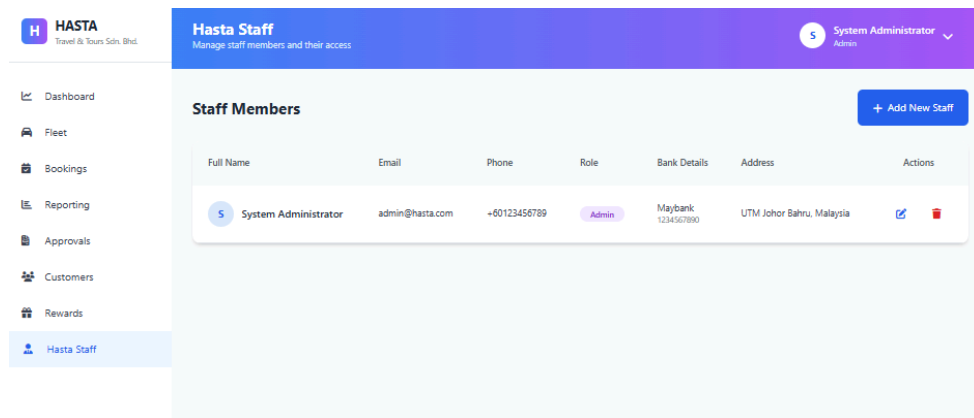


Figure 4.15: Staff Management Interface

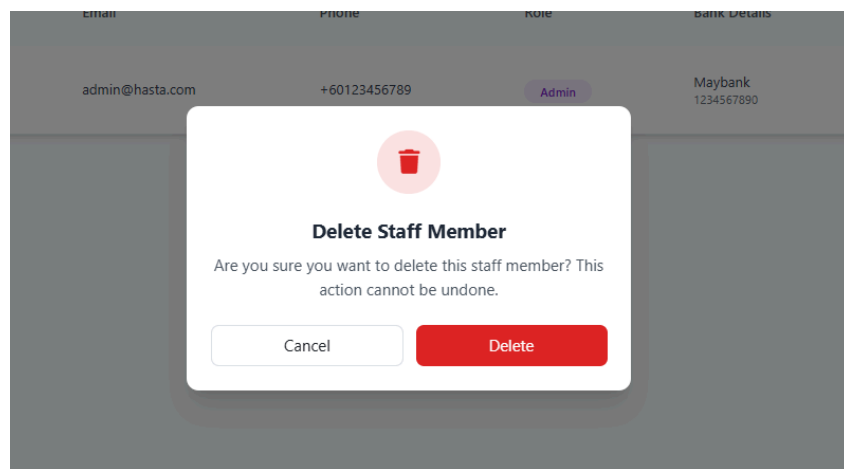


Figure 4.16: Delete Staff Confirmation

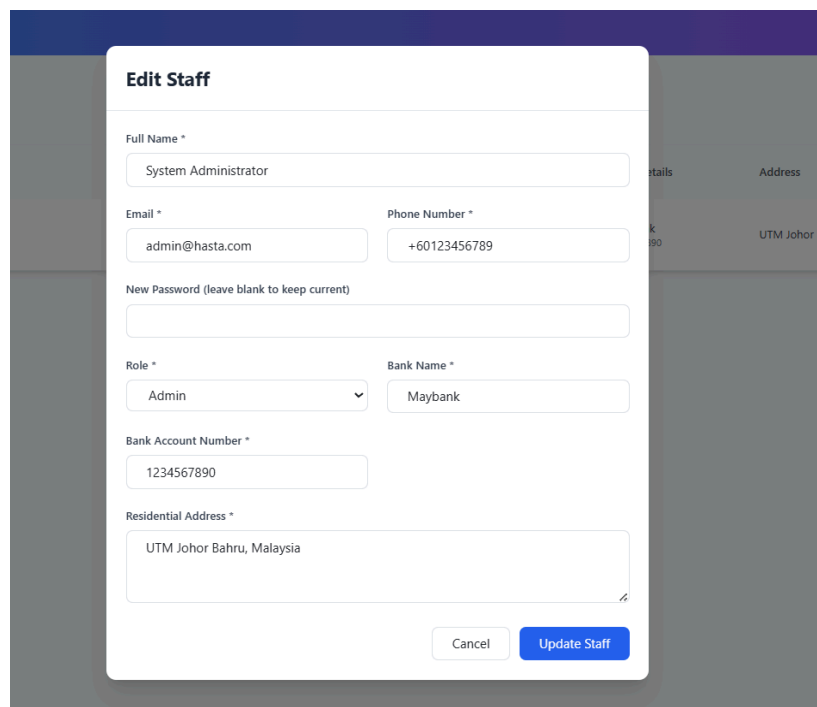


Figure 4.17: Edit Staff Information

4.1.2 SQL Implementation

4.1.2.1 DDL

USERS

```
CREATE TABLE users (  
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    full_name VARCHAR(255) NOT NULL,  
    nric_passport_number VARCHAR(255) NOT NULL UNIQUE,  
    nationality VARCHAR(255) NOT NULL,  
    identity_utm ENUM('Student','Staff') NOT NULL,  
    matric_staff_card VARCHAR(255) UNIQUE NULL,  
    email VARCHAR(255) NOT NULL UNIQUE,  
    country_code VARCHAR(255) NOT NULL DEFAULT '+60',  
    contact_number VARCHAR(255) NOT NULL,  
    password VARCHAR(255) NOT NULL,  
    mailing_address TEXT NOT NULL,  
    hostel_room_no VARCHAR(255) NULL,  
    hostel_block_no VARCHAR(255) NULL,  
    college_name VARCHAR(255) NOT NULL,  
    faculty VARCHAR(255) NULL,  
    ic_passport_path VARCHAR(255) NULL,  
    matric_staff_card_path VARCHAR(255) NULL,  
    license_path VARCHAR(255) NULL,  
    driving_license_no VARCHAR(255) NULL,  
    license_expiry_date DATE NULL,  
    bank_name VARCHAR(255) NULL,  
    bank_account_number VARCHAR(255) NULL,  
    emergency_contact_name VARCHAR(255) NULL,  
    emergency_contact_relationship VARCHAR(255) NULL,  
    emergency_contact_phone VARCHAR(255) NULL,  
    role ENUM('customer','admin') NOT NULL DEFAULT 'customer',  
    is_active BOOLEAN NOT NULL DEFAULT TRUE,  
    status VARCHAR(255) NOT NULL DEFAULT 'active',  
    blacklist_reason TEXT NULL,  
    is_verified BOOLEAN NOT NULL DEFAULT FALSE,  
    loyalty_points INT NOT NULL DEFAULT 0,  
    email_verified_at TIMESTAMP NULL,
```

```

remember_token VARCHAR(100) NULL,
created_at TIMESTAMP NULL,
updated_at TIMESTAMP NULL,
deleted_at TIMESTAMP NULL
);

```

CARS

```

CREATE TABLE cars (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  brand VARCHAR(255) NOT NULL,
  model VARCHAR(255) NOT NULL,
  year YEAR NOT NULL,
  color VARCHAR(255) NOT NULL,
  plate_number VARCHAR(50) NOT NULL UNIQUE,
  car_type VARCHAR(100) NOT NULL,
  transmission_type VARCHAR(50) NOT NULL,
  fuel_type VARCHAR(50) NOT NULL,
  number_of_seats INT NOT NULL,
  rental_rate_per_hour DECIMAL(8,2) NOT NULL,
  mileage DECIMAL(10,2) NOT NULL,
  description TEXT NULL,
  image VARCHAR(255) NULL,
  status ENUM('available','in_use','unavailable','maintenance')
NOT NULL DEFAULT 'available',
  pricing_table JSON NULL,
  roadtax_path TEXT NULL,
  roadtax_expiry_date DATE NULL,
  insurance_path TEXT NULL,
  insurance_expiry_date DATE NULL,
  last_service_date DATE NULL,
  next_service_date DATE NULL,
  created_at TIMESTAMP NULL,
  updated_at TIMESTAMP NULL
);

```

4.1.2.2 DML

UC01 — User Register

```
-- Uniqueness checks
SELECT COUNT(*) FROM users WHERE email = ?;
SELECT COUNT(*) FROM users WHERE nric_passport_number = ?;
SELECT COUNT(*) FROM users WHERE matric_staff_card = ?;

-- Create account
INSERT INTO users
(full_name, nric_passport_number, nationality, identity_utm,
matric_staff_card,
email, country_code, contact_number, password, mailing_address,
hostel_room_no, hostel_block_no, college_name, faculty,
role, status, is_active, is_verified, loyalty_points, created_at,
updated_at)
VALUES
(?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, 'customer', 'active',
TRUE, FALSE, 0, NOW(), NOW());
```

UC02 — User Log In

```
SELECT id, full_name, password, role, status, is_active,
blacklist_reason, is_verified
FROM users
WHERE email = ? AND deleted_at IS NULL;
```

```
-- Optional: block login if inactive
-- (handled in app logic using is_active/status)
```

UC03 — Browse Fleet (Public + Logged-in)

```
-- List cars (with filter by status)
SELECT id, brand, model, year, car_type, number_of_seats,
rental_rate_per_hour, status, image
FROM cars
WHERE status IN ('available','in_use','unavailable','maintenance')
ORDER BY created_at DESC;

-- Vehicle details (including pricing table)
```

```
SELECT *
FROM cars
WHERE id = ?;
```

UC04 — Upload Documents

```
UPDATE users
SET ic_passport_path = ?,
    matric_staff_card_path = ?,
    license_path = ?,
    driving_license_no = ?,
    license_expiry_date = ?,
    updated_at = NOW()
WHERE id = ?;
```

UC05 — Verify Documents

```
UPDATE users
SET is_verified = TRUE,
    status = 'active',
    updated_at = NOW()
WHERE id = ?;
```

-- If rejected / incomplete documents:

```
UPDATE users
SET is_verified = FALSE,
    status = 'pending',
    updated_at = NOW()
WHERE id = ?;
```

UC06 — Edit Customer Profile

```
UPDATE users
SET full_name = ?,
    contact_number = ?,
    mailing_address = ?,
    hostel_room_no = ?,
    hostel_block_no = ?,
    college_name = ?;
```

```
faculty = ?,  
bank_name = ?,  
bank_account_number = ?,  
emergency_contact_name = ?,  
emergency_contact_relationship = ?,  
emergency_contact_phone = ?,  
updated_at = NOW()  
WHERE id = ?;
```

4.2 Car Rental Booking Module

4.2.1 Interface Design

Booking & Checkout Flow

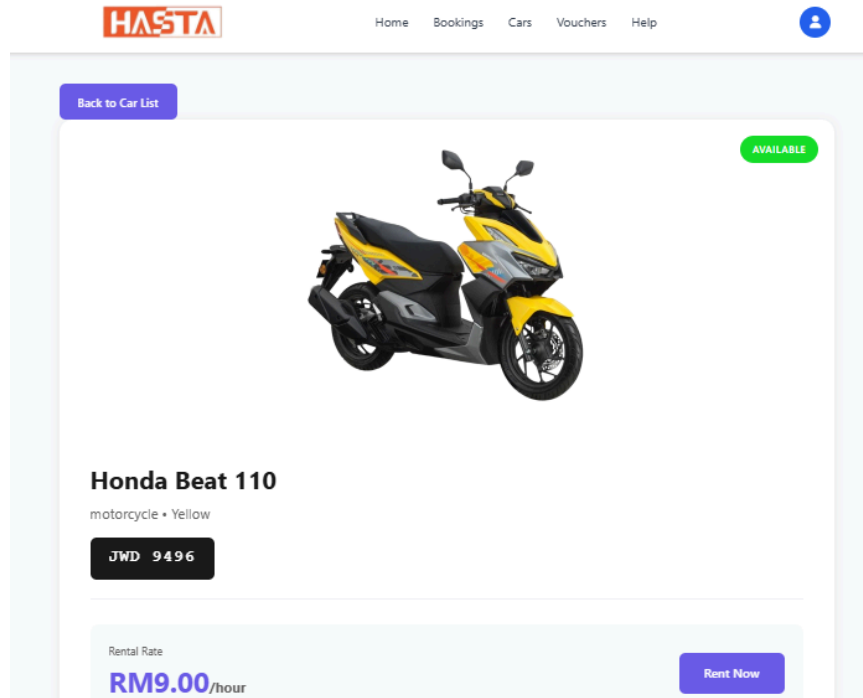


Figure 4.18: Booking vehicle page

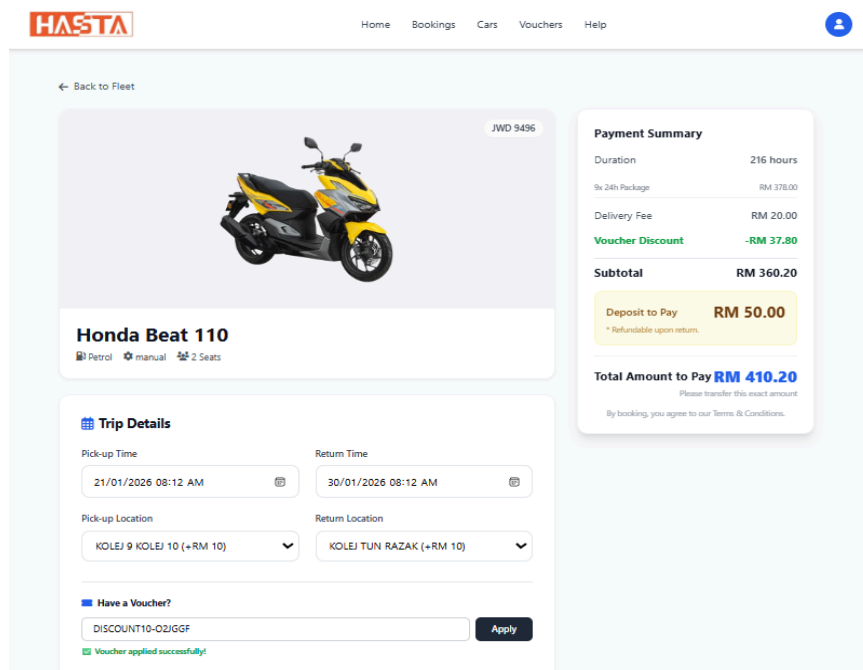


Figure 4.19: Booking Checkout Interface

[Home](#)
[Bookings](#)
[Cars](#)
[Vouchers](#)
[Help](#)

[← Back to Fleet](#)

JWD 9496

Honda Beat 110

Petrol
 manual
 2 Seats

Trip Details

Pick-up Time

21/01/2026 08:12 AM

Return Time

30/01/2026 08:12 AM

Pick-up Location

KOLEJ 9 KOLEJ 10 (+RM 10)

Return Location

KOLEJ TUN RAZAK (+RM 10)

Have a Voucher?

DISCOUNT10-O2/IGGF

Apply

✓ Voucher applied successfully!

Payment Summary

Duration	216 hours
9x 24h Package	RM 378.00
Delivery Fee	RM 20.00
Voucher Discount	-RM 37.80
Subtotal	RM 360.20
Deposit to Pay	RM 50.00
* Refundable upon return.	
Total Amount to Pay	RM 410.20
Please transfer this exact amount	
By booking, you agree to our Terms & Conditions.	

Figure 4.20: Booking Request Confirmation

[Home](#)
[Bookings](#)
[Cars](#)
[Vouchers](#)
[Help](#)

[← Back to Fleet](#)

JWD 9496

Honda Beat 110

Petrol
 manual
 2 Seats

Trip Details

Pick-up Time

21/01/2026 08:12 AM

Return Time

30/01/2026 08:12 AM

Pick-up Location

KOLEJ 9 KOLEJ 10 (+RM 10)

Return Location

KOLEJ TUN RAZAK (+RM 10)

Have a Voucher?

DISCOUNT10-O2/IGGF

Apply

✓ Voucher applied successfully!

Payment Summary

Duration	216 hours
9x 24h Package	RM 378.00
Delivery Fee	RM 20.00
Voucher Discount	-RM 37.80
Subtotal	RM 360.20
Deposit to Pay	RM 50.00
* Refundable upon return.	
Total Amount to Pay	RM 410.20
Please transfer this exact amount	
By booking, you agree to our Terms & Conditions.	

Figure 4.21: User Booking Management Dashboard

Pickup / Return Evidence Flow (Customer)



[Home](#) [Bookings](#) [Cars](#) [Vouchers](#) [Help](#)



[← Back to My Bookings](#)

Sign Rental Agreement

Please download, sign, and upload the rental agreement



Perodua Myvi
JPN 1416
18 Feb 2026, 08:02 AM - 16 Apr 2026, 08:01 AM

PHOTOS UPLOADED

Rental Agreement Process

1

Download Agreement Form
Download the rental agreement PDF template

Download Agreement Form

2

Sign the Agreement
Print the form, read carefully, and sign where indicated. You may also digitally sign the PDF.

3

Upload Signed Agreement
Upload the signed agreement form below (PDF only, max 5MB)


Click to upload signed agreement
PDF only, max 5MB

 **Important:**

- By submitting this agreement, you accept all terms and conditions
- Make sure all information is correct before uploading
- Keep a copy of the signed agreement for your records

Cancel

Submit & Activate Booking

?

Need Help?

If you're having trouble signing or uploading the agreement, please contact our support team.
[Contact Support →](#)

Figure 4.23: Sign Rental Agreement interface

[← Back to My Bookings](#)

Upload Return Photos

Please take clear photos of the car when returning it



Honda Beat 110
JWD 9496
21 Jan 2026, 08:01 AM - 24 Jan 2026, 08:01 AM

ACTIVE

Required Photos (6)

1. Front View *


Click to upload
Max 2MB

2. Left Side View *


Click to upload
Max 2MB

3. Right Side View *


Click to upload
Max 2MB

4. Back View *


Click to upload
Max 2MB

5. Dashboard *


Click to upload
Max 2MB

6. Key Location *


Click to upload
Max 2MB

Figure 4.24: Upload car photos after finish rental



Photos Uploaded Successfully!

Now please rate your rental experience



Honda Beat 110

JWD 9496

12 Mar 2026 - 01 Apr 2026

How would you rate this rental? *



Excellent

How was the car condition?

☐ Air conditioning quality is good?

☐ Car condition is clean?

☒ Car utilities are good (seatbelt, pedal, steering wheel, etc.)?

☐ Car is in overall good condition?

Additional Comments (Optional)

Tell us more about your experience...

 Important:

- By submitting, your rental will be marked as **Completed**
- Your deposit refund status will be set to **Pending**
- You will earn **+1 Loyalty Point** for this booking
- Staff will review and process your deposit refund

Submit Review & Complete Rental

Figure 4.25: Booking review interface

Blacklist restriction

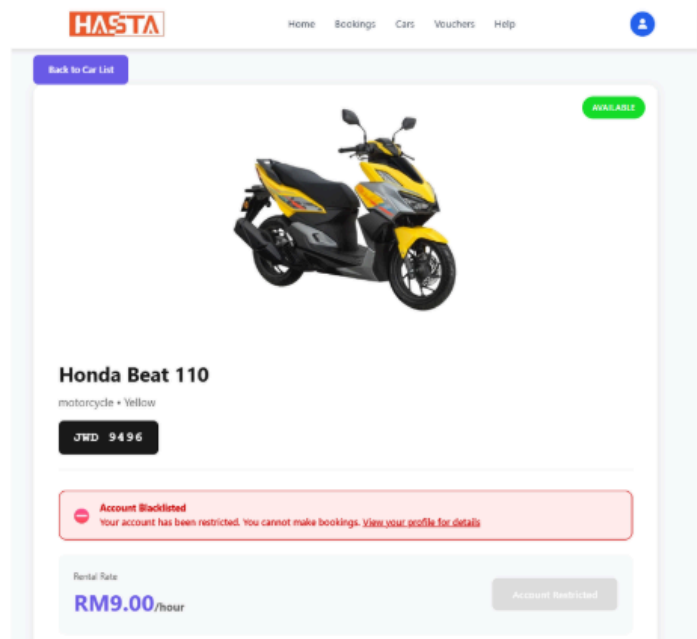


Figure 4.26: Booking page of blacklist user

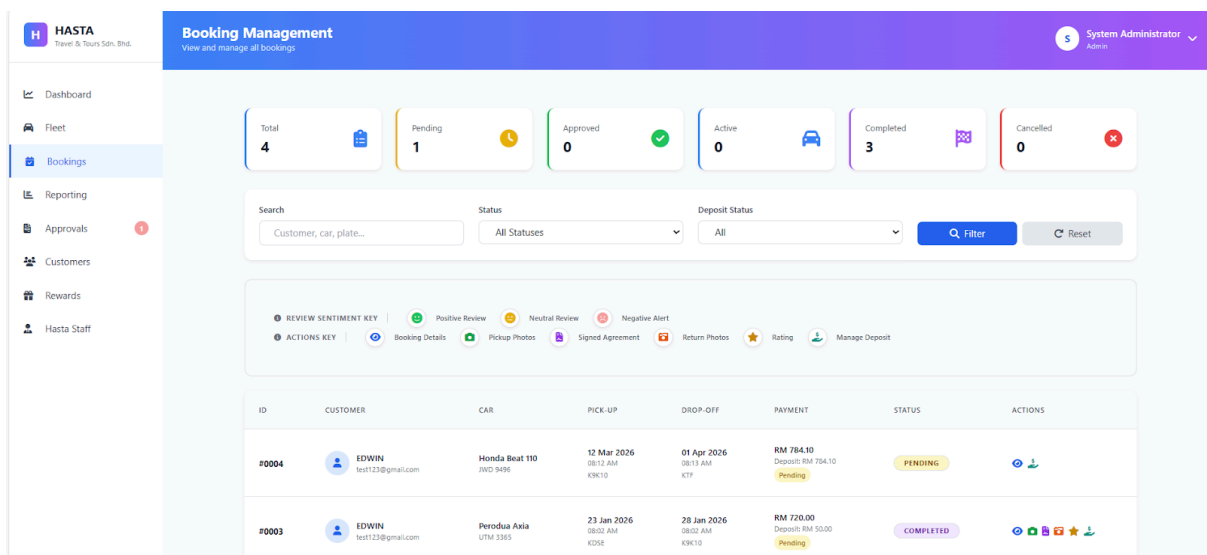


Figure 4.27: Booking Management Interface

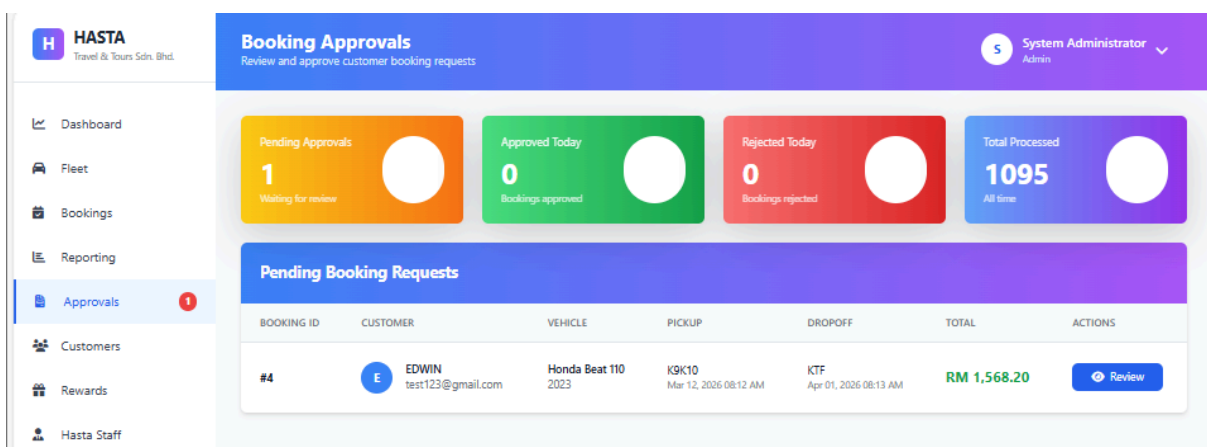


Figure 4.28: Booking Approvals Dashboard

← Back to Approvals

Customer Information

Name

EDWIN

Email

test123@gmail.com

Phone

121321212

Customer Since

Jan 2026

Vehicle Information

Vehicle

Perodua Axia

License Plate

MCP6113

Hourly Rate

RM 35.00/hour

Transmission

Auto

Booking Details

Rental Duration

360 hours

Pickup

KDSE

February 05, 2026

09:20 AM

Dropoff / Return

Payment Summary

Rental Amount

RM 1,800.00

Delivery Fee

RM 20.00

Deposit (Included)

RM 1,820.00

Included in total - Refundable upon return

Total Paid

RM 3,640.00

Exact amount paid by customer

Approve Booking

Reject Booking

Figure 4.29: Booking Request Review

EDWIN

test123@gmail.com

Honda Beat 110

JWD 9496

12 Mar 2026

08:12 AM

K9K10

01 Apr 2026

08:13 AM

KTF

RM 784.10

Deposit: RM 784.10

Pending

COMPLETED

EDWIN

test123@gmail.com

RM 720.00

Deposit: RM 50.00

Pending

COMPLETED

EDWIN

test123@gmail.com

RM 7,430.00

Deposit: RM 7,430.00

Pending

COMPLETED

EDWIN

test123@gmail.com

RM 146.00

Deposit: RM 50.00

Pending

COMPLETED

Dr. Micheal Mann

dpacocha@example.org

Perodua Axia

CEX 5224

19 Jan 2026

06:00 PM

UTM

19 Jan 2026

09:00 PM

UTM

RM 135.00

Deposit: RM 50.00

Refunded

COMPLETED

Manage Deposit

Update Status

Pending

Pending

Refunded (Green)

Charged/Forfeited (Red)

Figure 4.30: Manage Deposit Dialog

83

4.2.2 SQL Implementation

4.2.2.1 DDL

BOOKINGS

```
CREATE TABLE bookings (  
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  user_id BIGINT UNSIGNED NOT NULL,  
  car_id BIGINT UNSIGNED NOT NULL,  
  reward_id BIGINT UNSIGNED NULL,  
  start_time DATETIME NOT NULL,  
  end_time DATETIME NOT NULL,  
  actual_return_time DATETIME NULL,  
  pickup_location VARCHAR(255) NOT NULL,  
  dropoff_location VARCHAR(255) NOT NULL,  
  total_price DECIMAL(10,2) NOT NULL,  
  rental_amount DECIMAL(10,2) NOT NULL DEFAULT 0,  
  delivery_fee DECIMAL(10,2) NOT NULL DEFAULT 0,  
  deposit_amount DECIMAL(10,2) NOT NULL,  
  voucher_code VARCHAR(255) NULL,  
  voucher_discount DECIMAL(10,2) NOT NULL DEFAULT 0,  
  payment_status VARCHAR(255) NOT NULL DEFAULT 'pending',  
  payment_receipt_path VARCHAR(255) NULL,  
  status  
  ENUM('pending','approved','active','completed','cancelled','reject  
ed') NOT NULL DEFAULT 'pending',  
  approved_at TIMESTAMP NULL,  
  approved_by BIGINT UNSIGNED NULL,  
  rejected_at TIMESTAMP NULL,  
  rejected_by BIGINT UNSIGNED NULL,  
  rejection_reason TEXT NULL,  
  pickup_photos JSON NULL,  
  pickup_photos_uploaded_at TIMESTAMP NULL,  
  return_photos JSON NULL,  
  agreement_signed_path VARCHAR(255) NULL,  
  agreement_signed_at TIMESTAMP NULL,  
  cancelled_at TIMESTAMP NULL,
```



```

    refund_status
ENUM('pending','refunded','charged_and_refunded','no_refund')
NULL,
    staff_message TEXT NULL,
    review_rating INT NULL,
    review_checkboxes JSON NULL,
    review_comment TEXT NULL,
    deposit_refund_status ENUM('pending','refunded','charged') NULL,
    condition_remarks TEXT NULL,
    created_at TIMESTAMP NULL,
    updated_at TIMESTAMP NULL,
    CONSTRAINT fk_booking_user FOREIGN KEY (user_id) REFERENCES
users(id) ON DELETE CASCADE,
    CONSTRAINT fk_booking_car FOREIGN KEY (car_id) REFERENCES
cars(id) ON DELETE CASCADE,
    CONSTRAINT fk_booking_reward FOREIGN KEY (reward_id) REFERENCES
rewards(id) ON DELETE SET NULL,
    CONSTRAINT fk_booking_approved_by FOREIGN KEY (approved_by)
REFERENCES staff(id) ON DELETE SET NULL,
    CONSTRAINT fk_booking_rejected_by FOREIGN KEY (rejected_by)
REFERENCES staff(id) ON DELETE SET NULL
);

```

BOOKING_DOCUMENTS

```

CREATE TABLE booking_documents (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    booking_id BIGINT UNSIGNED NOT NULL UNIQUE,
    user_id BIGINT UNSIGNED NOT NULL,
    car_id BIGINT UNSIGNED NOT NULL,
    pickup_photos JSON NULL,
    pickup_photos_uploaded_at TIMESTAMP NULL,
    return_photos JSON NULL,
    return_photos_uploaded_at TIMESTAMP NULL,
    agreement_signed_path VARCHAR(255) NULL,
    agreement_signed_at TIMESTAMP NULL,
    condition_remarks TEXT NULL,
    created_at TIMESTAMP NULL,

```

```

    updated_at TIMESTAMP NULL,
    CONSTRAINT fk_doc_booking FOREIGN KEY (booking_id) REFERENCES
bookings(id) ON DELETE CASCADE,
    CONSTRAINT fk_doc_user FOREIGN KEY (user_id) REFERENCES
users(id) ON DELETE CASCADE,
    CONSTRAINT fk_doc_car FOREIGN KEY (car_id) REFERENCES cars(id)
ON DELETE CASCADE
);

```

(Rewards table is in Module 5, but bookings references it.)

4.2.2.2 DML

UC11 — Booking Vehicle

```

-- Check blacklist / active
SELECT status, is_active, is_verified
FROM users
WHERE id = ?;

-- Check car availability
SELECT status
FROM cars
WHERE id = ?;

-- Create booking (pending approval)
INSERT INTO bookings
(user_id, car_id, start_time, end_time, pickup_location,
dropoff_location,
    rental_amount, delivery_fee, deposit_amount, voucher_code,
voucher_discount,
    total_price, payment_status, status, created_at, updated_at)
VALUES
(?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, 'pending', 'pending', NOW(),
NOW());

-- Create 1:1 booking document row immediately (recommended)
INSERT INTO booking_documents

```

```
(booking_id, user_id, car_id, created_at, updated_at)
VALUES
(LAST_INSERT_ID(), ?, ?, NOW(), NOW());
```

UC12 — Payment (Upload receipt + update payment status)

```
UPDATE bookings
SET payment_receipt_path = ?,
    payment_status = 'pending',
    updated_at = NOW()
WHERE id = ? AND user_id = ?;
```

UC09 — View Booking History

```
SELECT
    b.id, b.start_time, b.end_time, b.pickup_location,
    b.dropoff_location,
    b.total_price, b.deposit_amount, b.voucher_discount,
    b.payment_status, b.status, b.deposit_refund_status,
    c.brand, c.model, c.plate_number, c.image
FROM bookings b
JOIN cars c ON c.id = b.car_id
WHERE b.user_id = ?
ORDER BY b.created_at DESC;
```

UC10 — Cancel Booking

```
UPDATE bookings
SET status = 'cancelled',
    cancelled_at = NOW(),
    updated_at = NOW()
WHERE id = ? AND user_id = ?
    AND status IN ('pending', 'approved');
```

UC07 — Pickup Vehicle (Upload pickup photos + sign agreement)

```
-- Upload pickup photos (JSON list of paths)
UPDATE booking_documents
SET pickup_photos = ?,
    pickup_photos_uploaded_at = NOW(),
    updated_at = NOW()
WHERE booking_id = ? AND user_id = ?;

-- Upload signed agreement
UPDATE booking_documents
SET agreement_signed_path = ?,
    agreement_signed_at = NOW(),
    updated_at = NOW()
WHERE booking_id = ? AND user_id = ?;

-- Optionally mark booking active after pickup requirements
completed
UPDATE bookings
SET status = 'active',
    updated_at = NOW()
WHERE id = ? AND user_id = ?
    AND status = 'approved';
```

UC08 — Dropoff Vehicle (Upload return photos + return time)

```
UPDATE booking_documents
SET return_photos = ?,
    return_photos_uploaded_at = NOW(),
    updated_at = NOW()
WHERE booking_id = ? AND user_id = ?;

UPDATE bookings
SET actual_return_time = ?,
    updated_at = NOW()
WHERE id = ? AND user_id = ?
    AND status = 'active';
```

Post-return Review

```
UPDATE bookings
SET review_rating = ?,
    review_checkboxes = ?,
    review_comment = ?,
    status = 'completed',
    updated_at = NOW()
WHERE id = ? AND user_id = ?;
```

```
UPDATE users
SET loyalty_points = loyalty_points + 1,
    updated_at = NOW()
WHERE id = ?;
```

4.3 Fleet Management Module

4.3.1 Interface Design

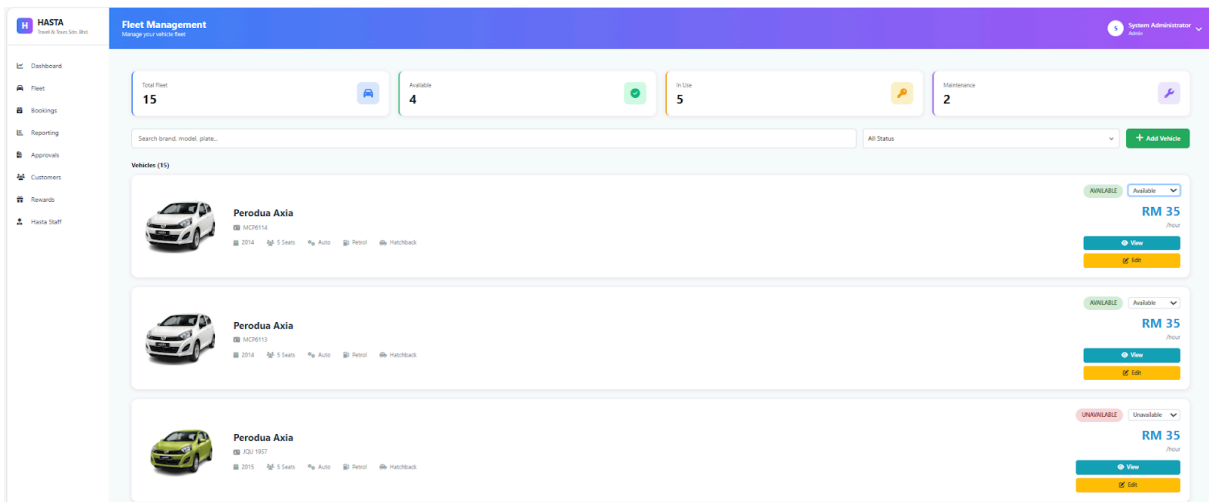


Figure 4.31: Fleet management dashboard interface

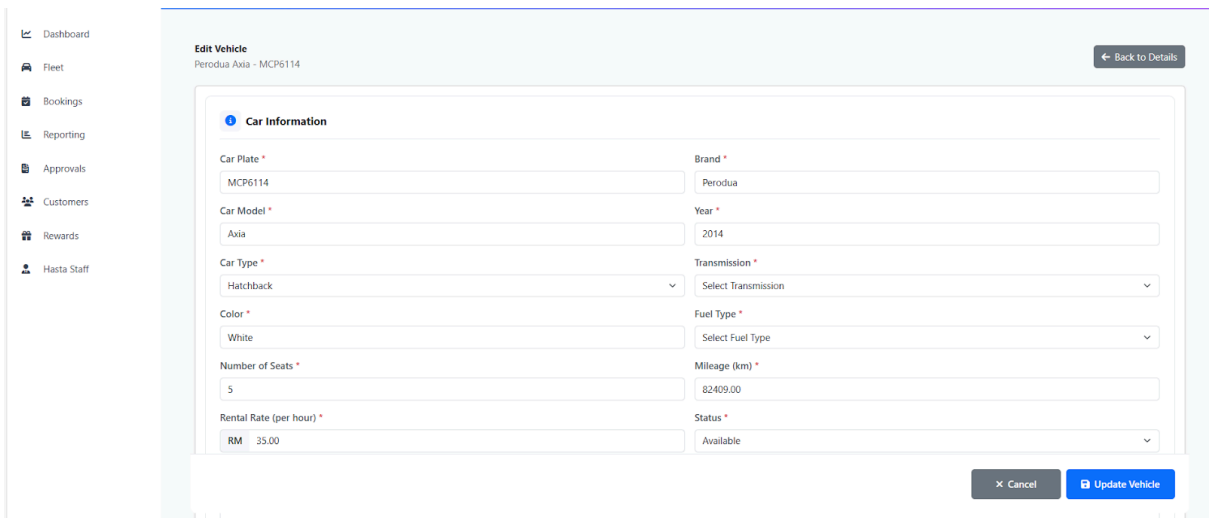


Figure 4.32: Vehicle Edit interface

Fleet Management
Vehicle Details and Information

System Administrator
Admin

Perodua Axia
MCP0114

Back to FleetAdd Vehicle



Basic Information

Status
Available

Year
2014

Transmission
Auto

Mileage
62,409.00 km

Description
Good

Brand
Perodua

Color
White

Fuel Type
Petrol

Rental Rate (per hour)
RM 35.00

Model
Axia

Car Type
Hatchback

Number of Seats
5 Seats

Pricing Table

1 Hour(s)	RM 35.00	3 Hour(s)	RM 55.00	5 Hour(s)	RM 65.00	7 Hour(s)	RM 70.00
9 Hour(s)	RM 85.00	12 Hour(s)	RM 95.00	24 Hour(s)	RM 120.00		

Service & Maintenance

Last Service Date
15 Oct 2024

Next Service Date
15 Apr 2025

Figure 4.33: Vehicle Details View

4.3.2 SQL Implementation

4.3.2.1 DDL

Uses `cars` table (already defined earlier).

4.3.2.2 DML

UC13 — Manage Vehicles

```
-- List all cars
SELECT id, brand, model, plate_number, status,
rental_rate_per_hour, created_at
FROM cars
ORDER BY created_at DESC;

-- Add car
INSERT INTO cars
(brand, model, year, color, plate_number, car_type,
transmission_type, fuel_type,
 number_of_seats, rental_rate_per_hour, mileage, description,
image, status, pricing_table,
 roadtax_path, roadtax_expiry_date, insurance_path,
insurance_expiry_date,
 last_service_date, next_service_date, created_at, updated_at)
VALUES
(?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?,
NOW(), NOW());

-- Delete car
DELETE FROM cars
WHERE id = ?;
```

UC14 — Update Vehicle Information

```
UPDATE cars
SET brand = ?,
    model = ?,
    year = ?,
```



```
color = ?,
car_type = ?,
transmission_type = ?,
fuel_type = ?,
number_of_seats = ?,
rental_rate_per_hour = ?,
mileage = ?,
description = ?,
image = ?,
status = ?,
pricing_table = ?,
roadtax_path = ?,
roadtax_expiry_date = ?,
insurance_path = ?,
insurance_expiry_date = ?,
last_service_date = ?,
next_service_date = ?,
updated_at = NOW()
WHERE id = ?;
```

4.4 Reporting Module

4.4.1 Interface Design

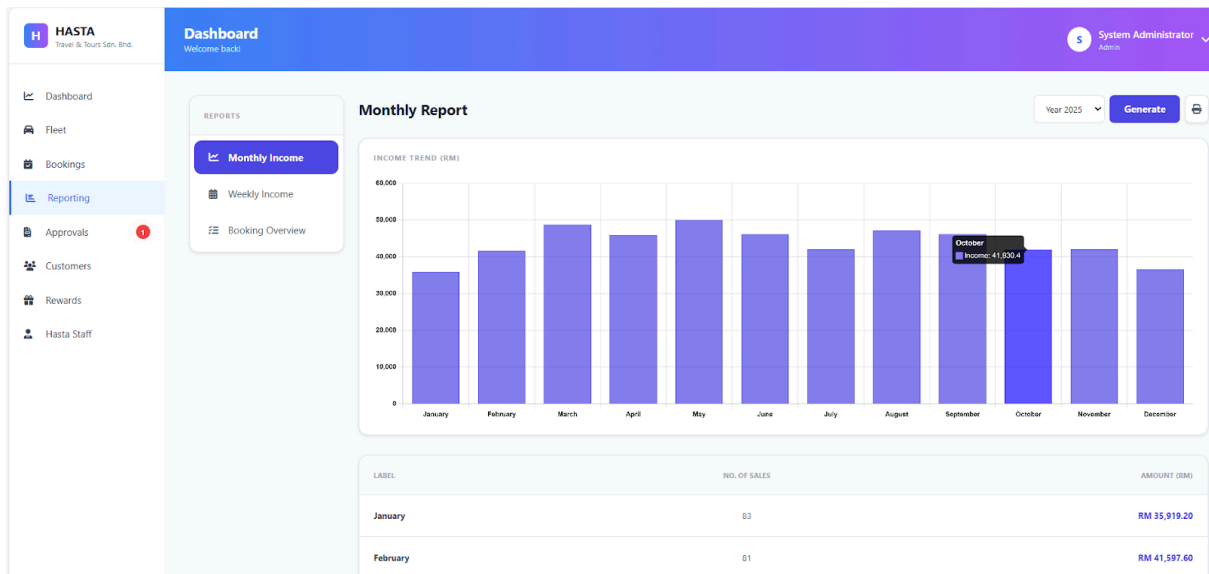


Figure 4.34: Monthly Income Report

LABEL	NO. OF SALES	AMOUNT (RM)
January	83	RM 35,919.20
February	81	RM 41,597.60
March	91	RM 48,771.50
April	89	RM 45,952.50
May	91	RM 50,030.50
June	85	RM 46,156.50
July	81	RM 42,091.00
August	85	RM 47,149.50
September	93	RM 46,158.10
October	89	RM 41,930.40
November	81	RM 42,054.00
December	69	RM 36,641.70
TOTAL SUM	1018	RM 524,452.50

Figure 4.35: Monthly report data table

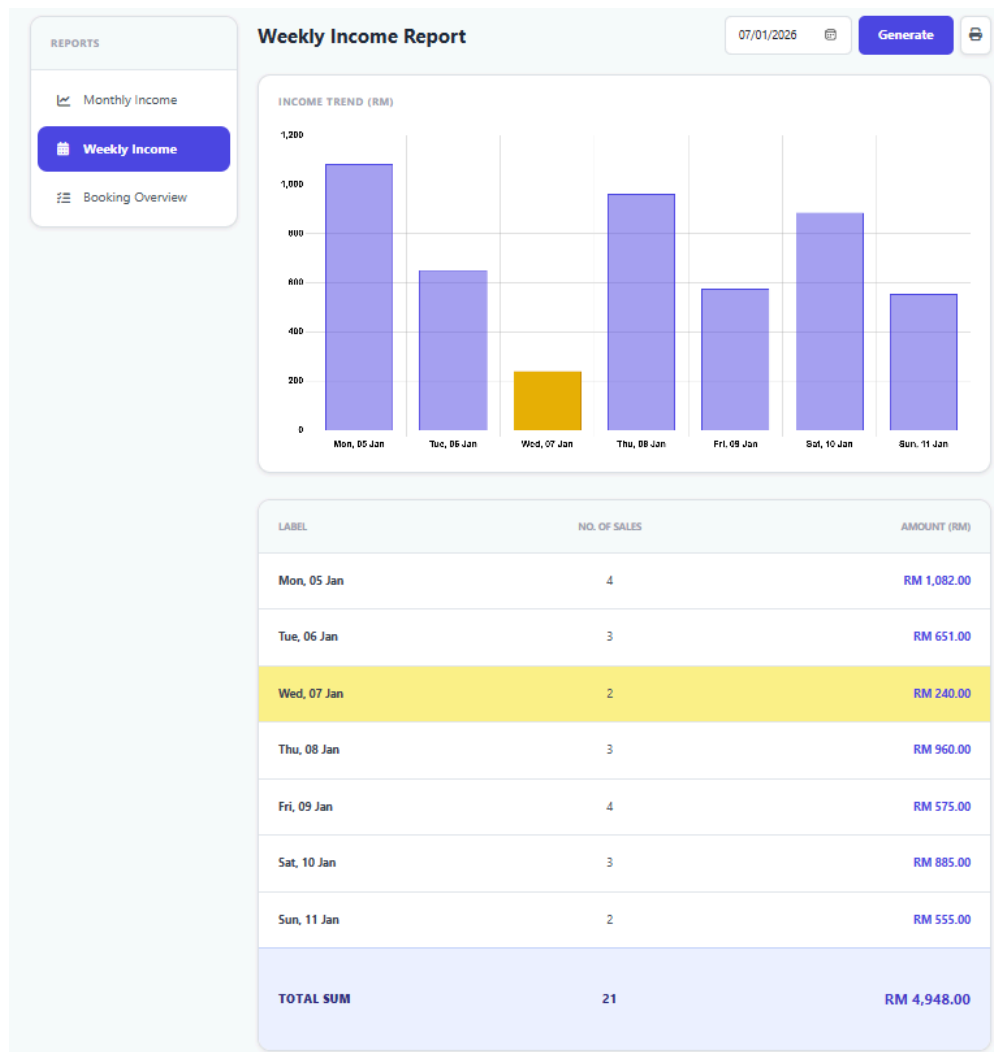


Figure 4.36: Weekly/Daily Income Report



Figure 4.37: Booking overview report

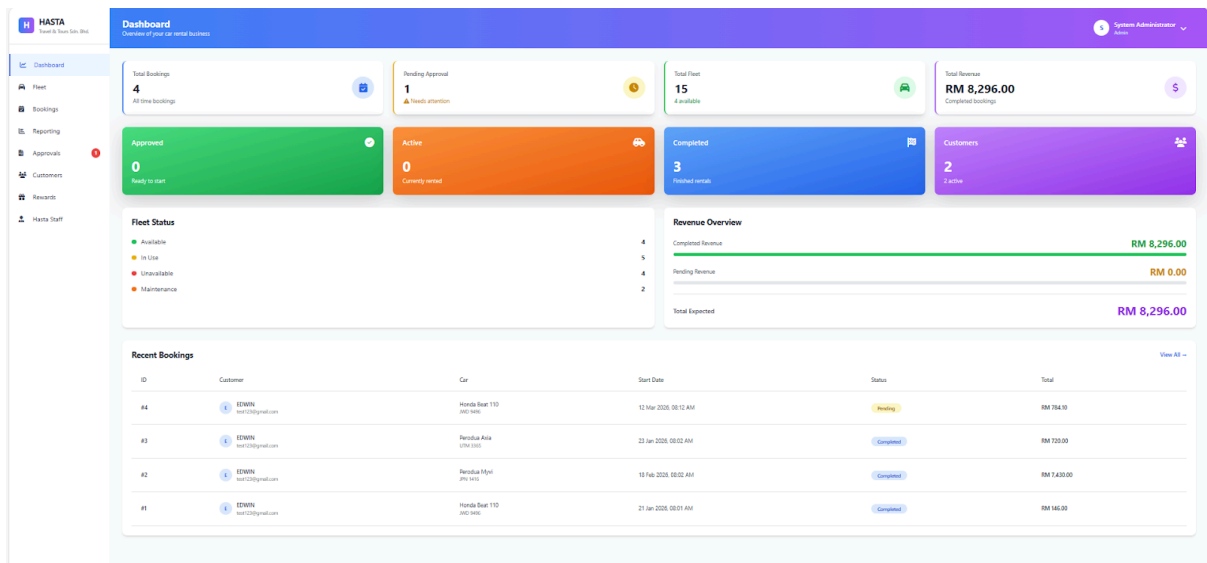


Figure 4.38: Admin dashboard interface

4.4.2 SQL Implementation

4.4.2.1 DML

Monthly Income

```
SELECT
    MONTH(created_at) AS month_no,
    SUM(total_price) AS total_revenue,
    COUNT(*) AS total_sales
FROM bookings
WHERE status IN ('approved','active','completed')
    AND YEAR(created_at) = ?
GROUP BY MONTH(created_at)
ORDER BY month_no;
```

Weekly/Daily Income

```
SELECT
    DATE(created_at) AS day_date,
    SUM(total_price) AS total_revenue,
    COUNT(*) AS total_sales
FROM bookings
WHERE status IN ('approved','active','completed')
    AND created_at >= ?
    AND created_at < ?
GROUP BY DATE(created_at)
ORDER BY day_date;
```

Booking Overview (Faculty / College breakdown)

```
-- Faculty bookings
SELECT
    u.faculty AS faculty_name,
    COUNT(*) AS total_bookings
FROM bookings b
JOIN users u ON u.id = b.user_id
WHERE b.status <> 'cancelled'
GROUP BY u.faculty
ORDER BY total_bookings DESC;
```

```
-- College bookings
SELECT
    u.college_name,
    COUNT(*) AS total_bookings
FROM bookings b
JOIN users u ON u.id = b.user_id
WHERE b.status <> 'cancelled'
GROUP BY u.college_name
ORDER BY total_bookings DESC;
```

Admin Dashboard KPIs

```
SELECT COUNT(*) AS total_bookings
FROM bookings;
```

```
SELECT COUNT(*) AS pending_approvals
FROM bookings
WHERE status = 'pending';
```

```
SELECT status, COUNT(*) AS count_by_status
FROM cars
GROUP BY status;
```

```
SELECT SUM(total_price) AS total_revenue
FROM bookings
WHERE status IN ('approved','active','completed');
```

4.5 Loyalty Card and Rewards Module

4.5.1 Interface Design

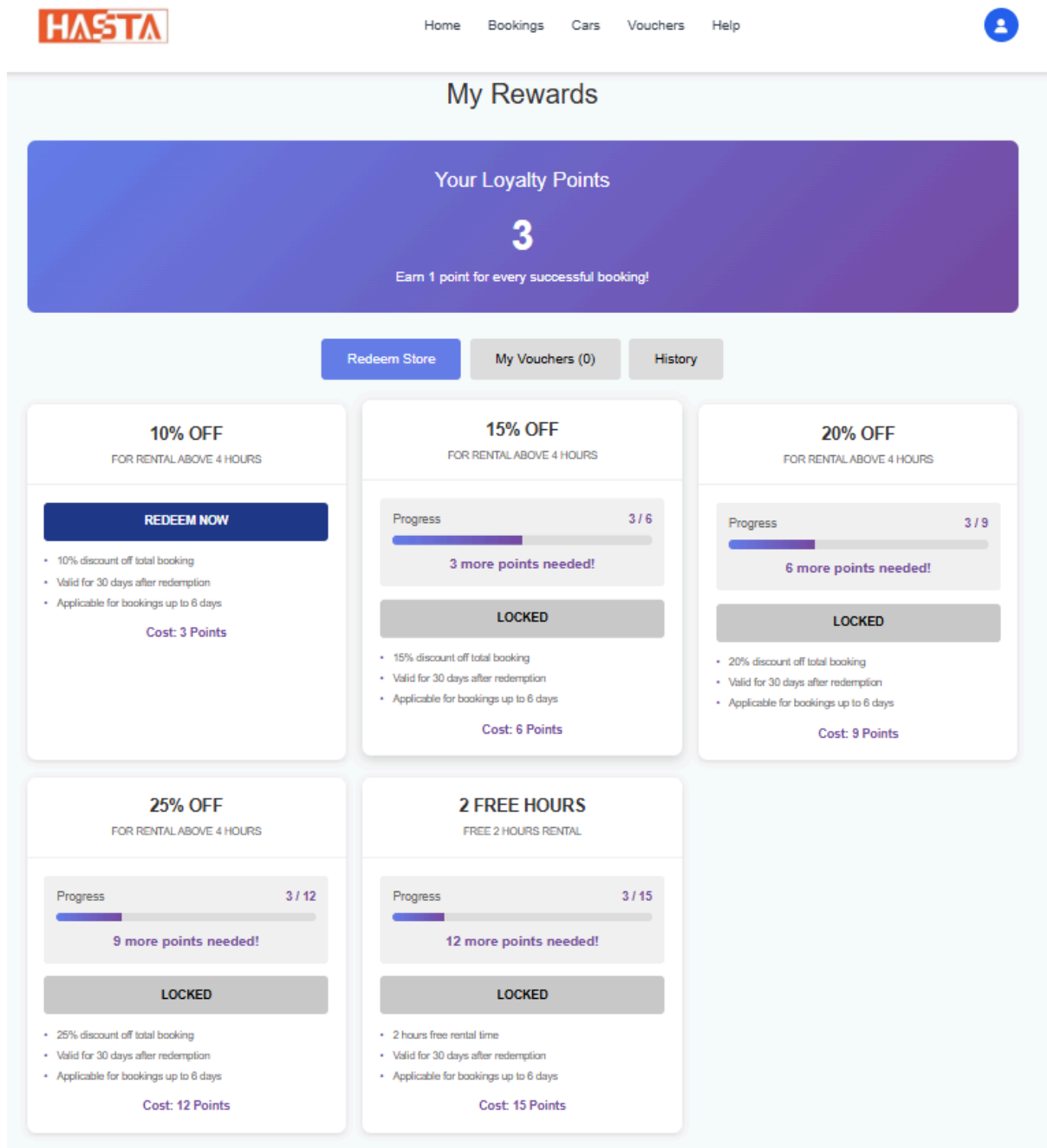


Figure 4.39: Loyalty rewards store

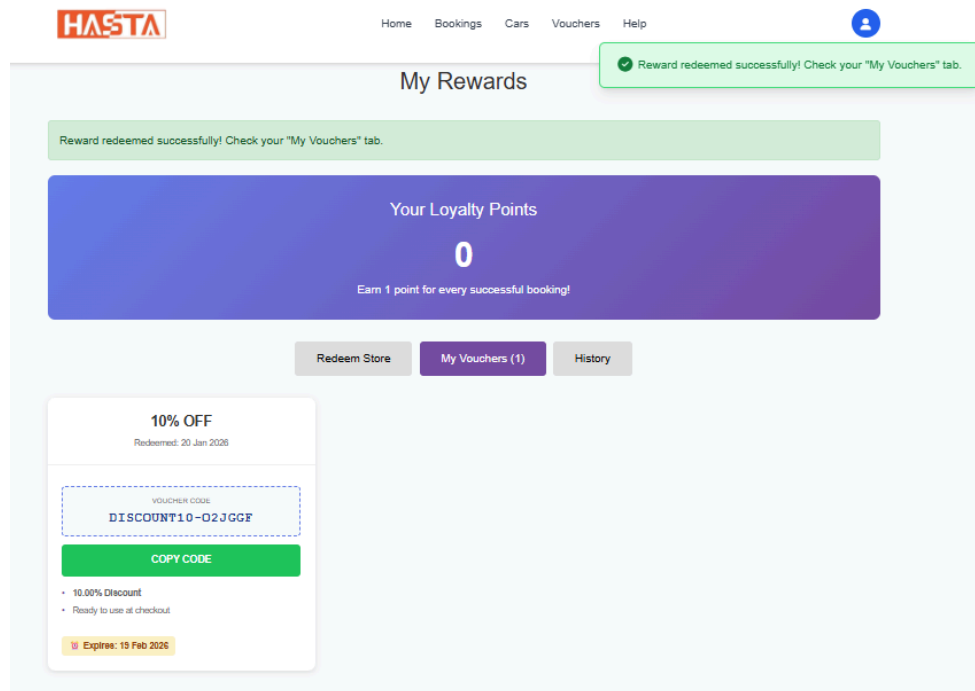


Figure 4.40: My voucher page

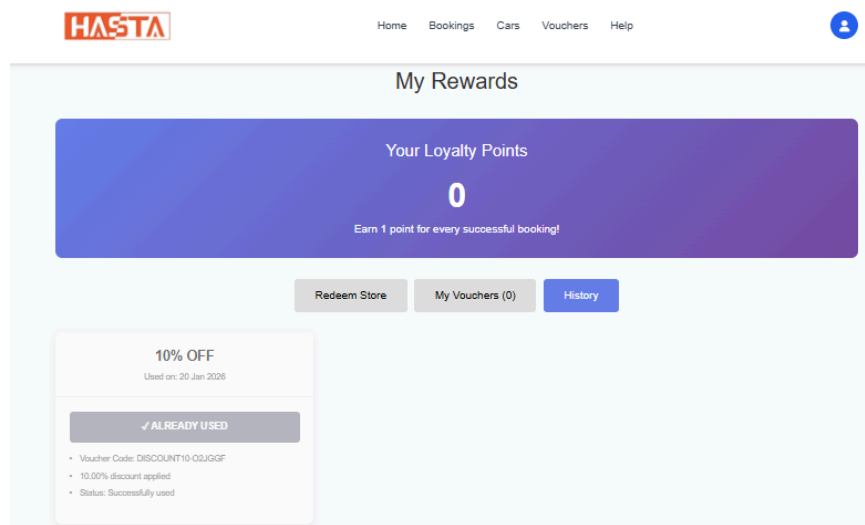


Figure 4.41: History of used voucher

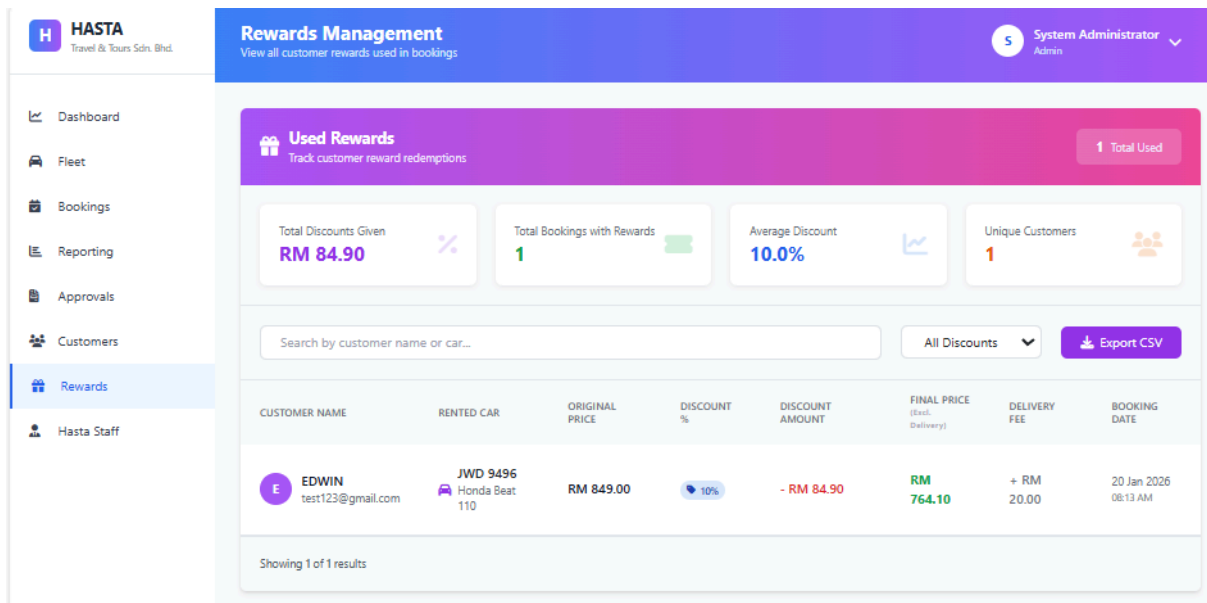


Figure 4.42: Rewards Management Dashboard

4.5.2 SQL Implementation

4.5.2.1 DDL

REWARDS

```
CREATE TABLE rewards (  
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    user_id BIGINT UNSIGNED NOT NULL,  
    reward_type VARCHAR(255) NOT NULL,  
    reward_name VARCHAR(255) NOT NULL,  
    voucher_code VARCHAR(255) NOT NULL UNIQUE,  
    points_required INT NOT NULL,  
    discount_percentage DECIMAL(5,2) NULL,  
    free_hours INT NULL,  
    status ENUM('available','used','expired') NOT NULL DEFAULT  
'available',  
    expiry_date DATE NOT NULL,  
    redeemed_at TIMESTAMP NOT NULL,  
    used_at TIMESTAMP NULL,  
    created_at TIMESTAMP NULL,  
    updated_at TIMESTAMP NULL,  
    CONSTRAINT fk_rewards_user FOREIGN KEY (user_id) REFERENCES  
users(id) ON DELETE CASCADE  
);
```

4.5.2.2 DML

UC16 — View All Vouchers (Available + Used)

```
-- Available vouchers for a user  
SELECT id, reward_name, voucher_code, points_required,  
discount_percentage, free_hours, status, expiry_date  
FROM rewards  
WHERE user_id = ? AND status = 'available'  
ORDER BY expiry_date ASC;  
  
-- Used voucher history  
SELECT id, reward_name, voucher_code, discount_percentage,  
free_hours, used_at
```

```

FROM rewards
WHERE user_id = ? AND status = 'used'
ORDER BY used_at DESC;

```

Redeem Voucher

```

-- Check points
SELECT loyalty_points
FROM users
WHERE id = ?;

-- Deduct points
UPDATE users
SET loyalty_points = loyalty_points - ?,
    updated_at = NOW()
WHERE id = ? AND loyalty_points >= ?;

-- Create voucher
INSERT INTO rewards
(user_id, reward_type, reward_name, voucher_code, points_required,
 discount_percentage, free_hours, status, expiry_date,
 redeemed_at, created_at, updated_at)
VALUES
(?, ?, ?, ?, ?, ?, ?, 'available', ?, NOW(), NOW(), NOW());

```

Apply Voucher in Booking (link reward to booking)

```

-- Validate voucher belongs to user and is available
SELECT id, discount_percentage, free_hours, status, expiry_date
FROM rewards
WHERE user_id = ? AND voucher_code = ?;

-- Apply to booking
UPDATE bookings
SET reward_id = ?,
    voucher_code = ?,
    voucher_discount = ?,
    updated_at = NOW()
WHERE id = ? AND user_id = ?;

```

Mark Voucher as Used after booking confirmed

```

UPDATE rewards
SET status = 'used',
    used_at = NOW(),
    updated_at = NOW()
WHERE id = ? AND user_id = ?;

```

5.0 Summary

This project presents a comprehensive database design for the Hasta Vehicle Rental Booking System, developed to modernize the operational infrastructure of Hasta Travel & Tours Sdn Bhd, a car rental service provider for Universiti Teknologi Malaysia (UTM) students and staff. The system addresses critical inefficiencies in the existing manual workflow, which currently relies on fragmented tools such as WhatsApp, Google Sheets, physical whiteboards, and the Drive Wahdah system.

Through systematic database modeling and normalization processes, the project delivers a fully integrated solution encompassing five core modules: User Management, Car Rental Booking, Fleet Management, Reporting & Analytics, and Loyalty Card & Rewards. The Global Conceptual ERD was refined through normalization up to Third Normal Form (3NF) and Boyce-Codd Normal Form (BCNF), resulting in a robust schema of 11 normalized tables that eliminate data redundancy and ensure referential integrity.

Complete SQL implementations including Data Definition Language (DDL) and Data Manipulation Language (DML) statements were developed to support all identified use cases, from customer registration and document verification to booking workflows, payment processing, vehicle tracking, and deposit management. The proposed system replaces manual communication channels with automated workflows, introduces real-time vehicle availability tracking, streamlines the complex booking process, and provides comprehensive reporting capabilities for revenue analysis and operational insights.

By transitioning from a file-based approach to a centralized relational database architecture, this solution significantly enhances operational efficiency, data accuracy, scalability, and overall customer experience while maintaining strict data integrity and supporting the company's long-term growth objectives.

6.0 Appendices

6.1 Use Case Diagram for the system

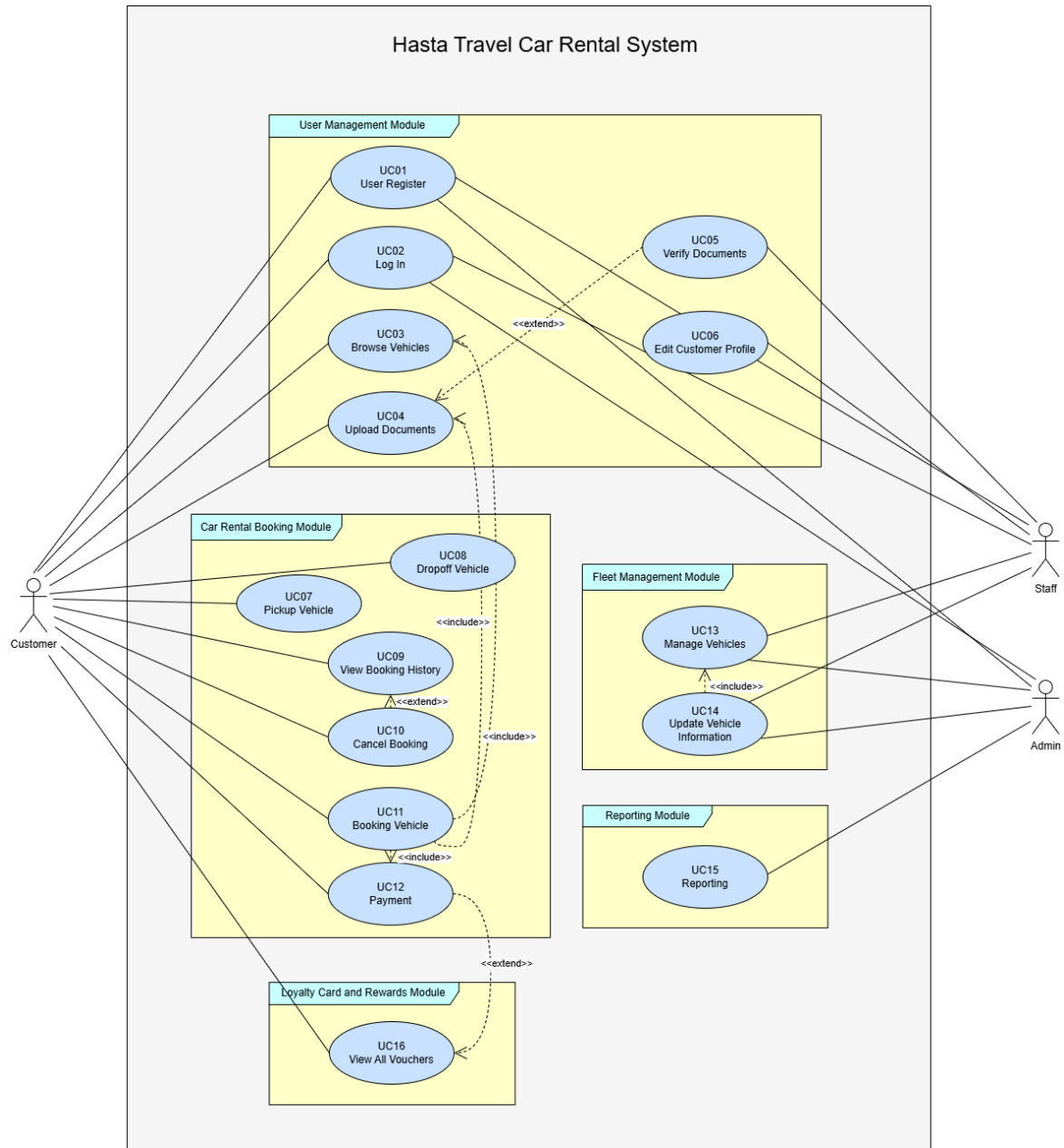


Figure 6.1: Use Case Diagram for Hasta Vehicle Rental Booking System

6.2 Activity Diagrams for every user case identified

6.2.1 UC01: User Register

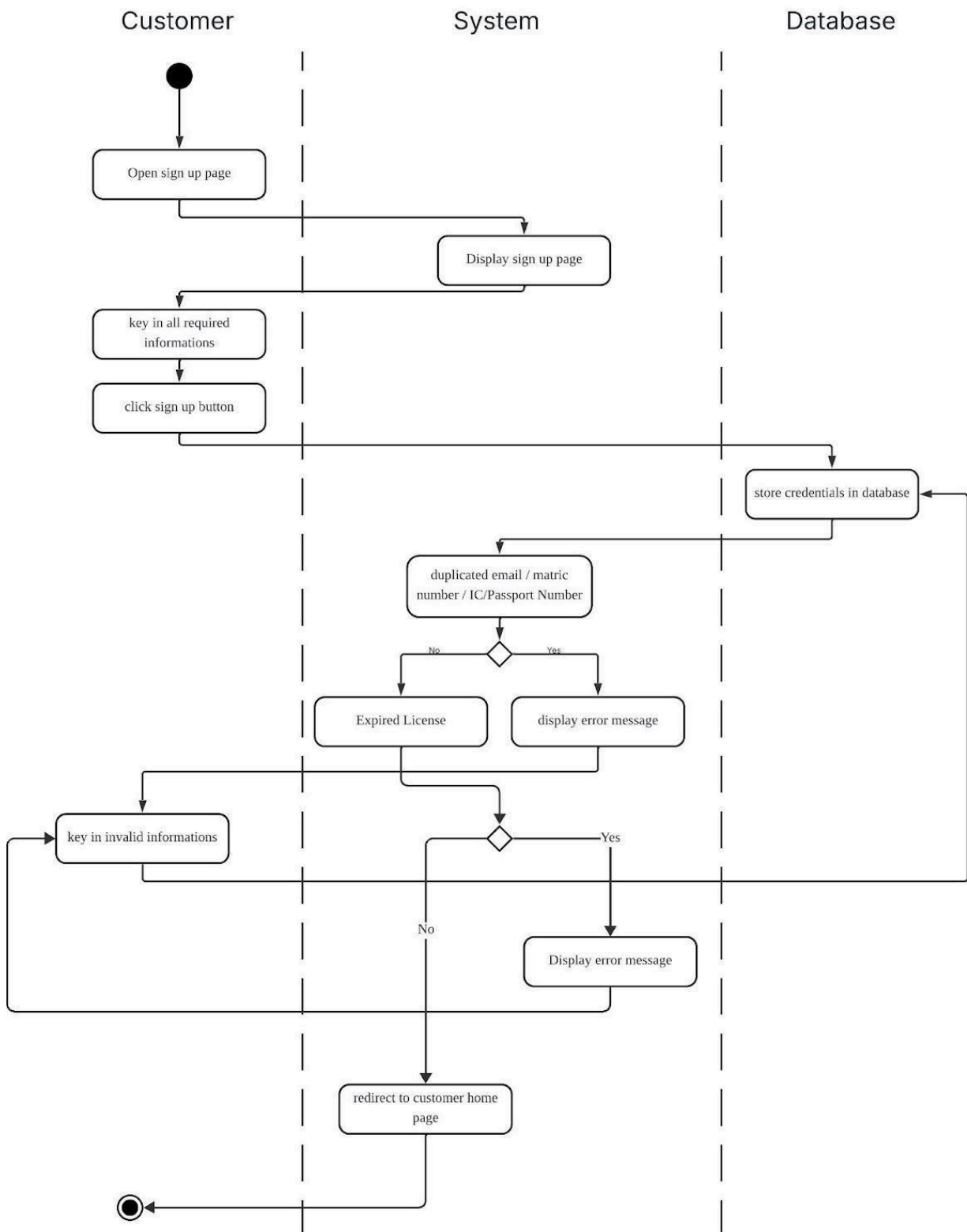


Figure 6.2: Activity Diagram for User Register

6.2.2 UC02: Log In

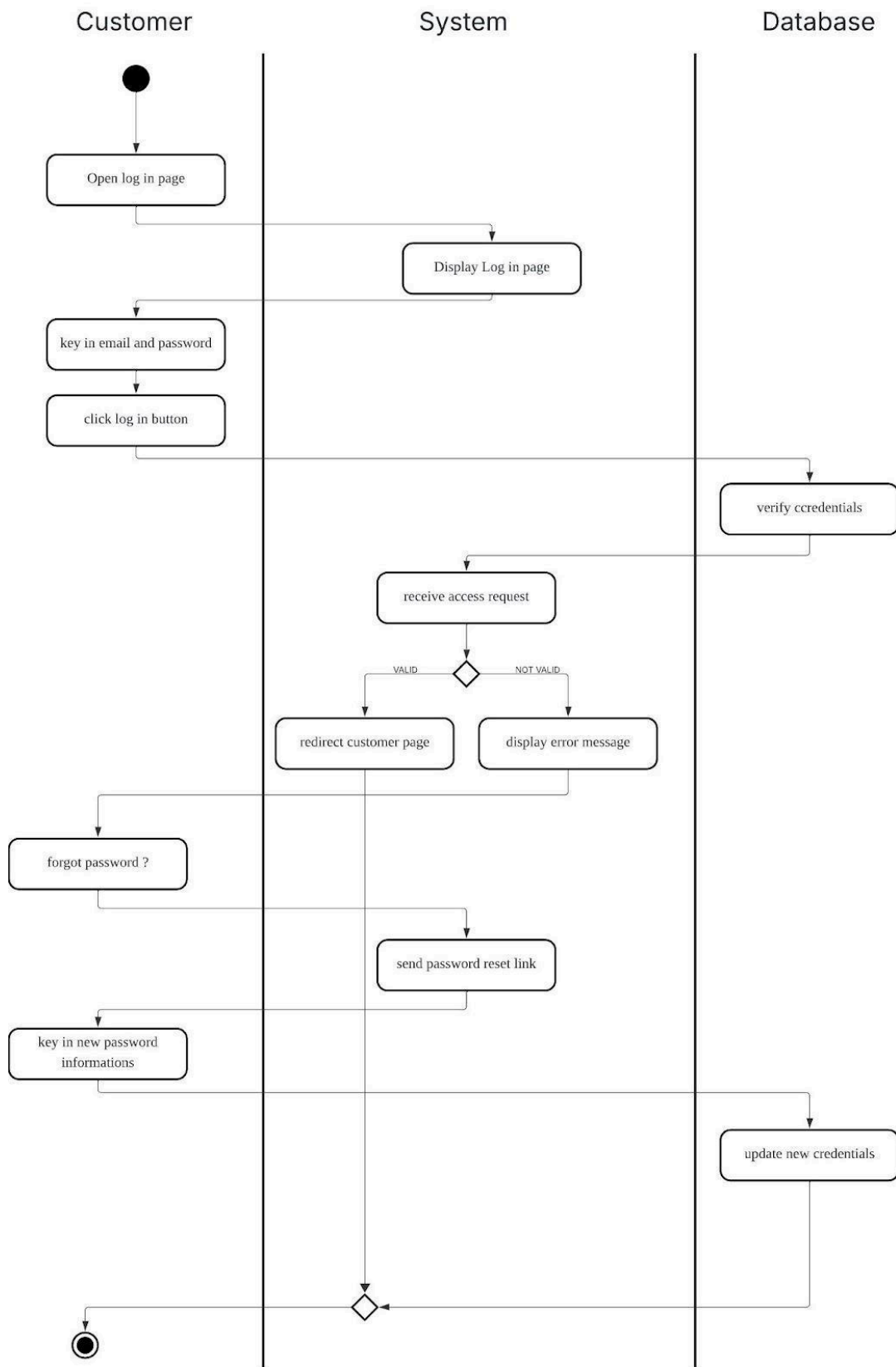


Figure 6.3: Activity Diagram for Log In

6.2.3 UC03: Browse Vehicle

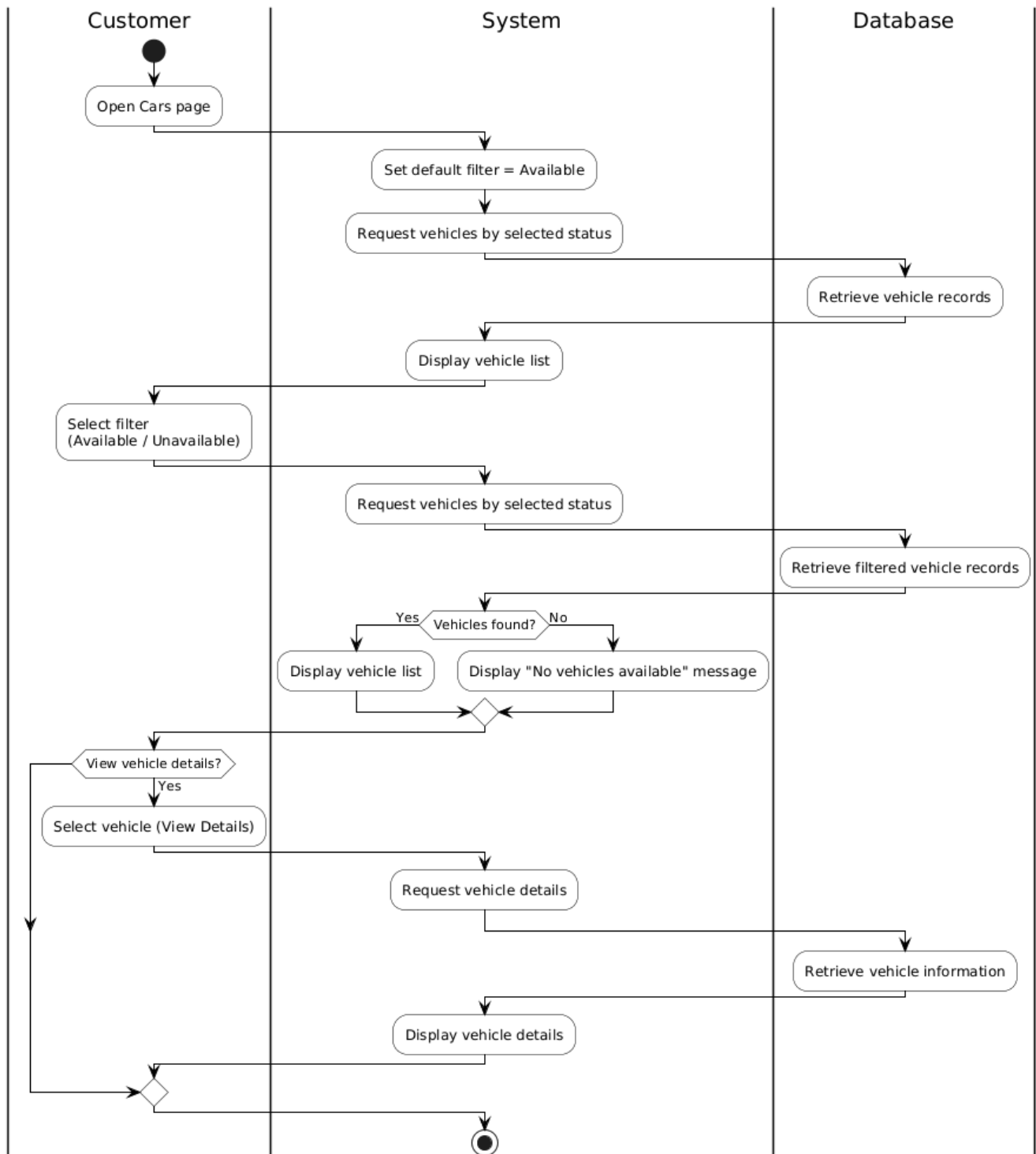


Figure 6.4: Activity Diagram for Browse Vehicle

6.2.4 UC04: Upload Documents

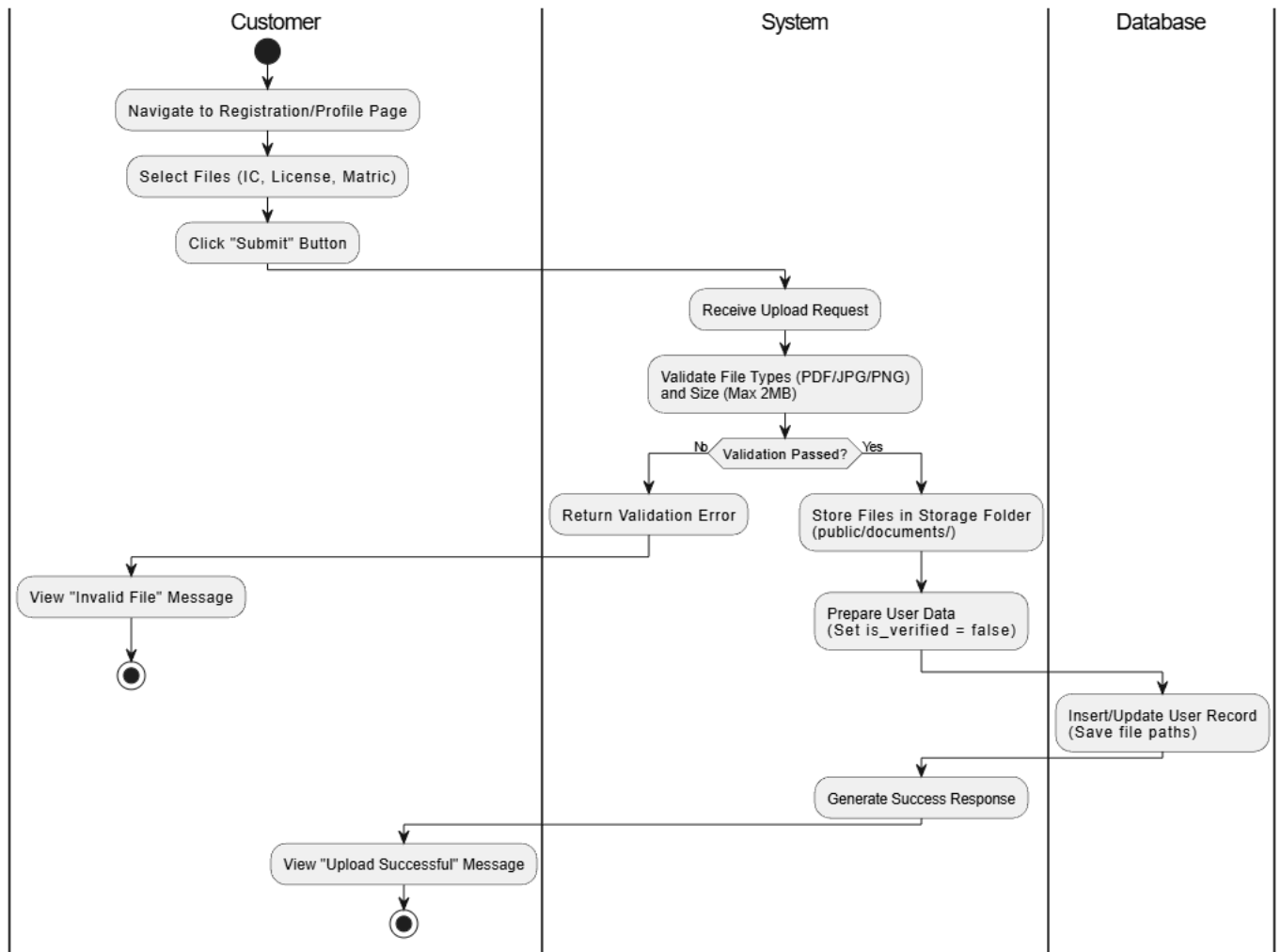


Figure 6.5: Activity Diagram for Upload Documents

6.2.5 UC05: Verify Documents

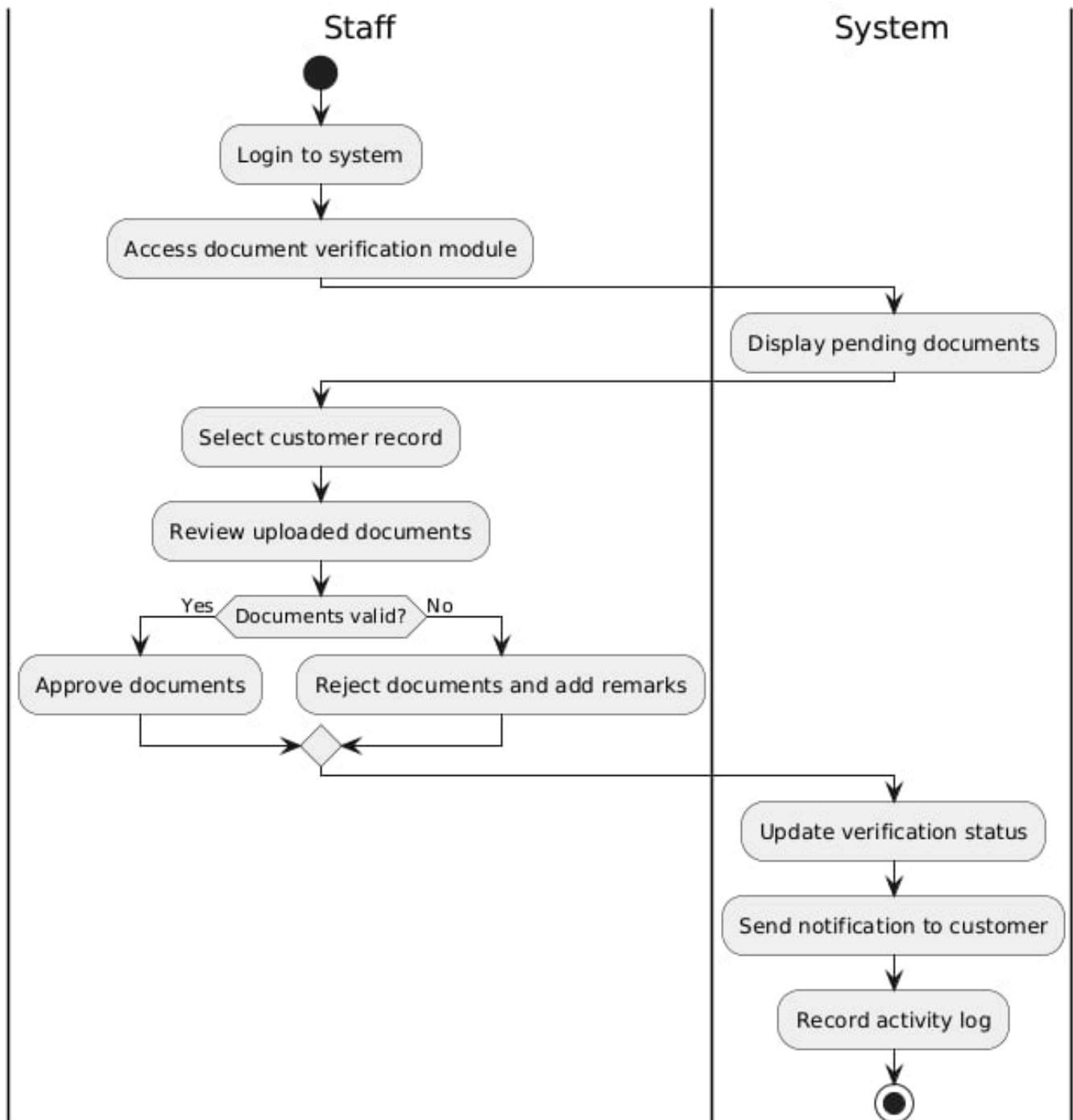


Figure 6.6: Activity Diagram for Verify Documents

6.2.6 UC06: Edit Customer Profile

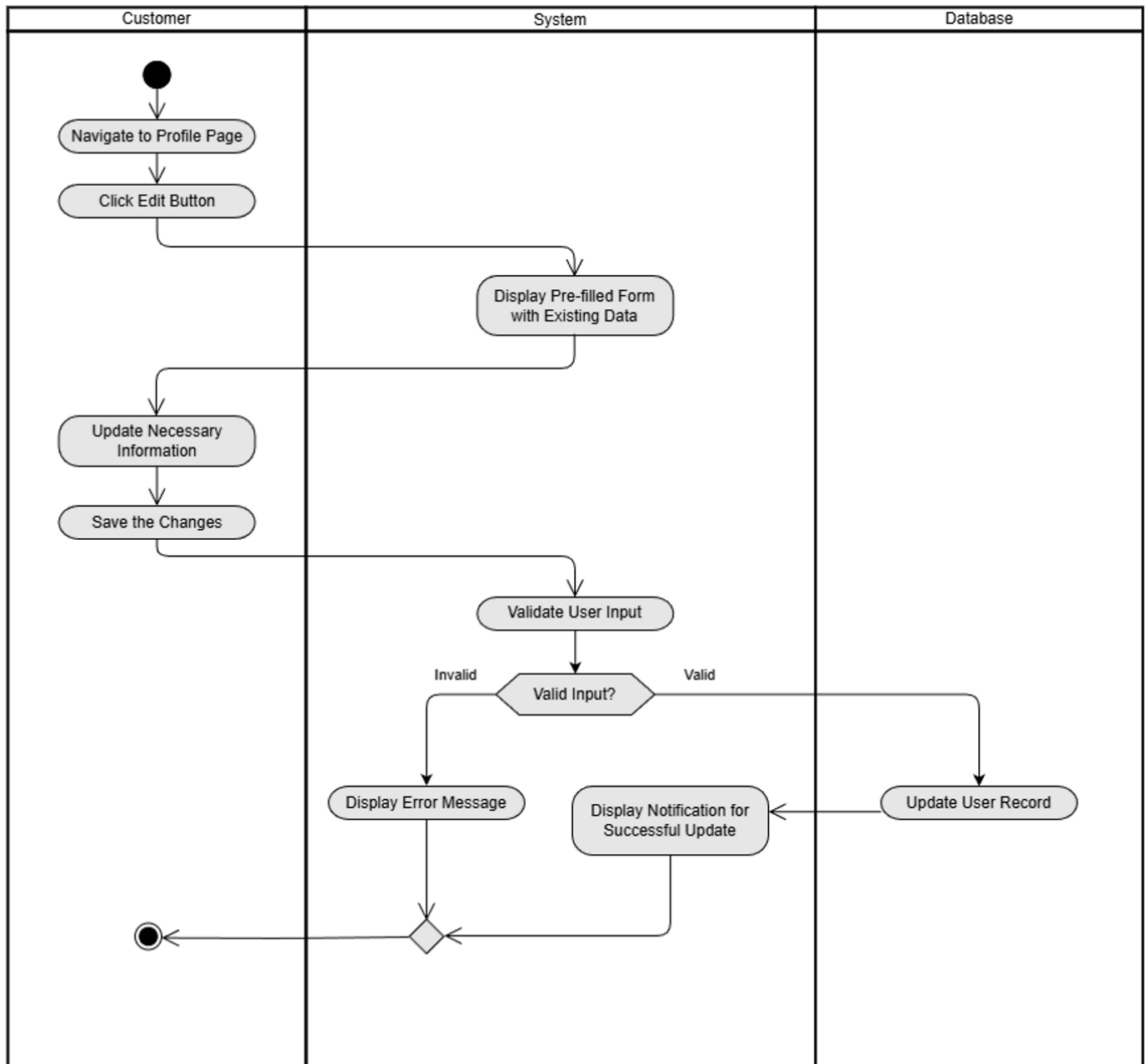


Figure 6.7: Activity Diagram for Edit Customer Profile

6.2.7 UC07: Pickup Vehicle

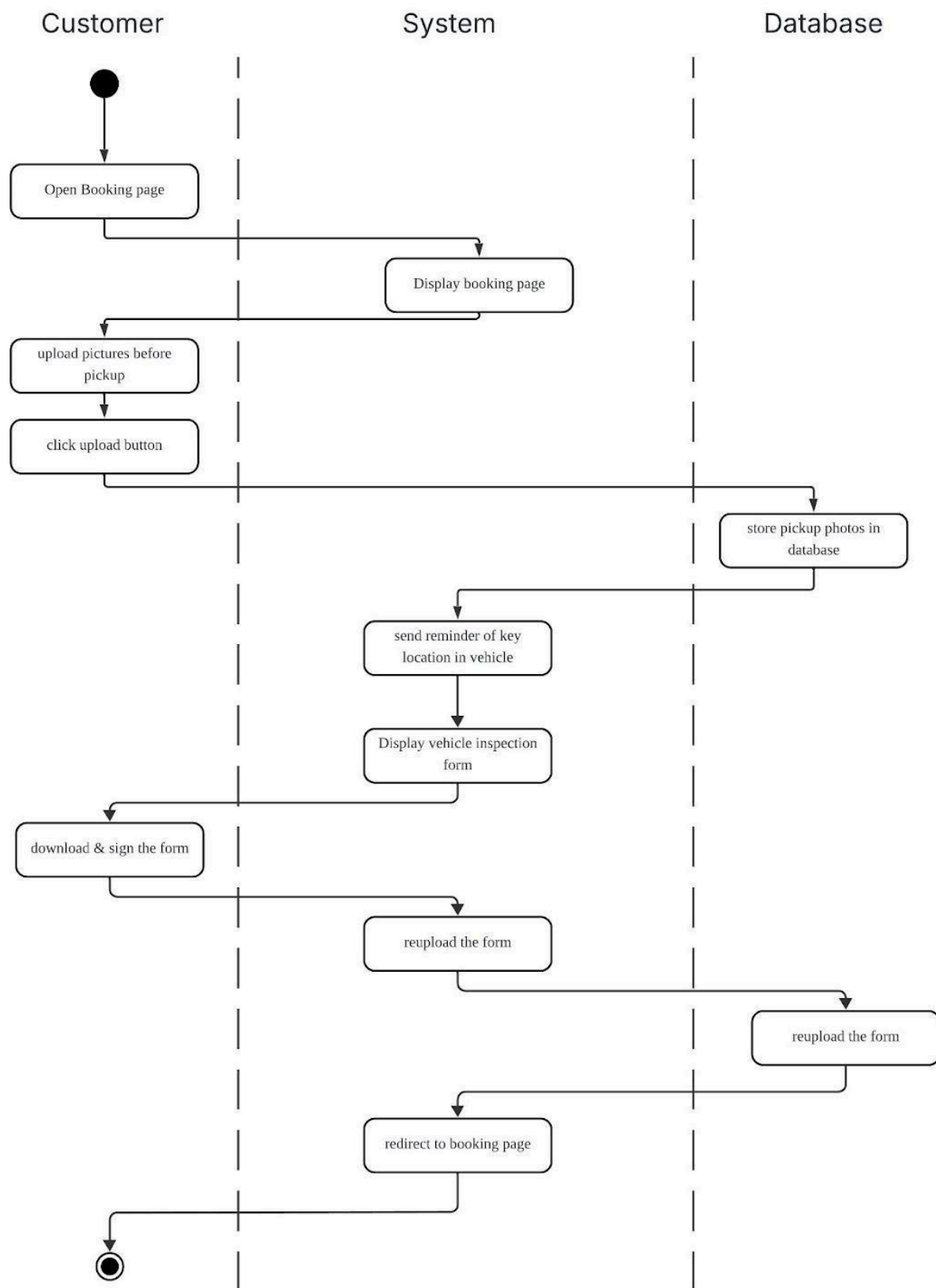


Figure 6.8: Activity Diagram for Pickup Vehicle

6.2.8 UC08: Return Vehicle

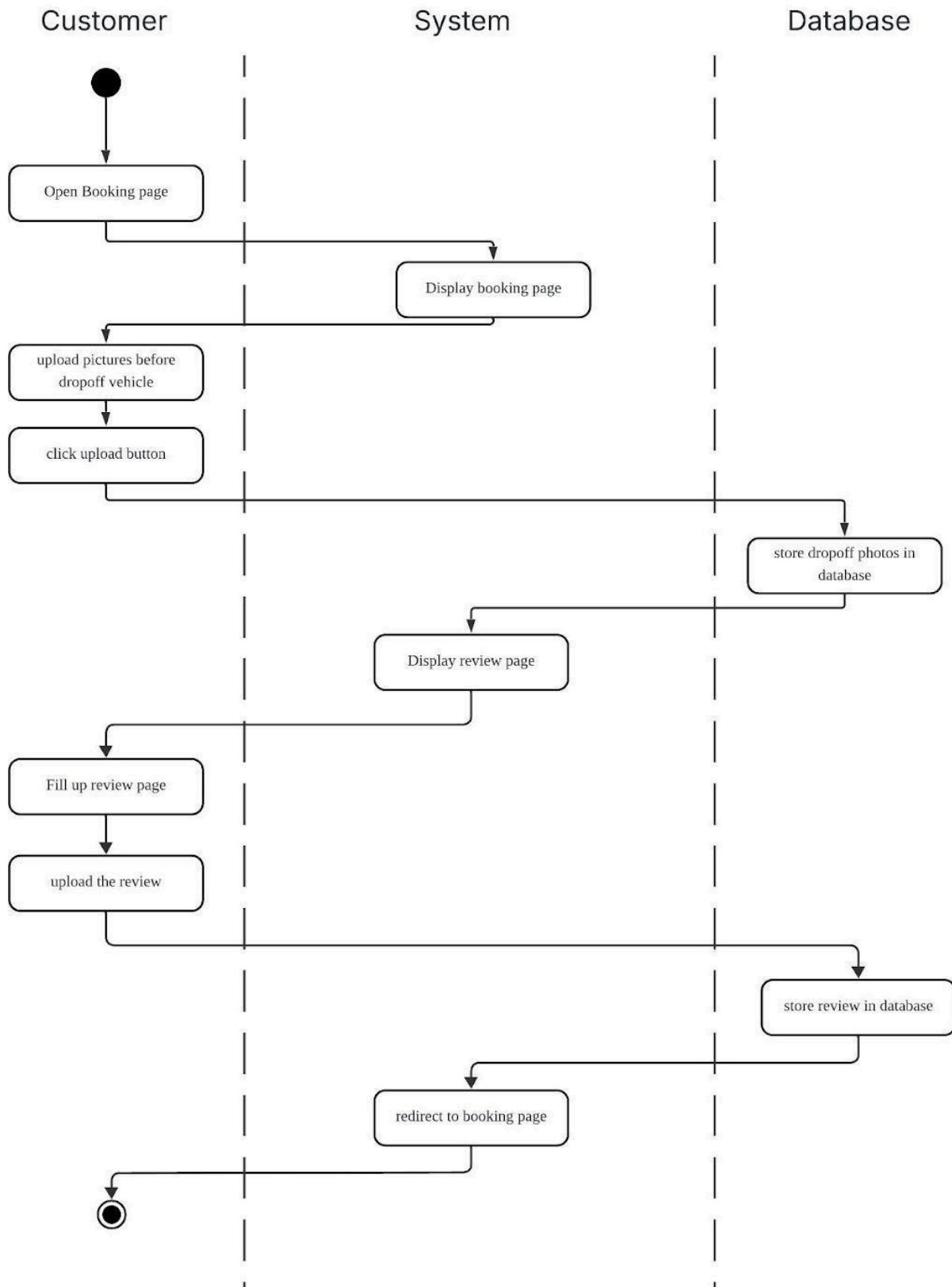


Figure 6.9: Activity Diagram for Return Vehicle

6.2.9 UC09: View Booking History

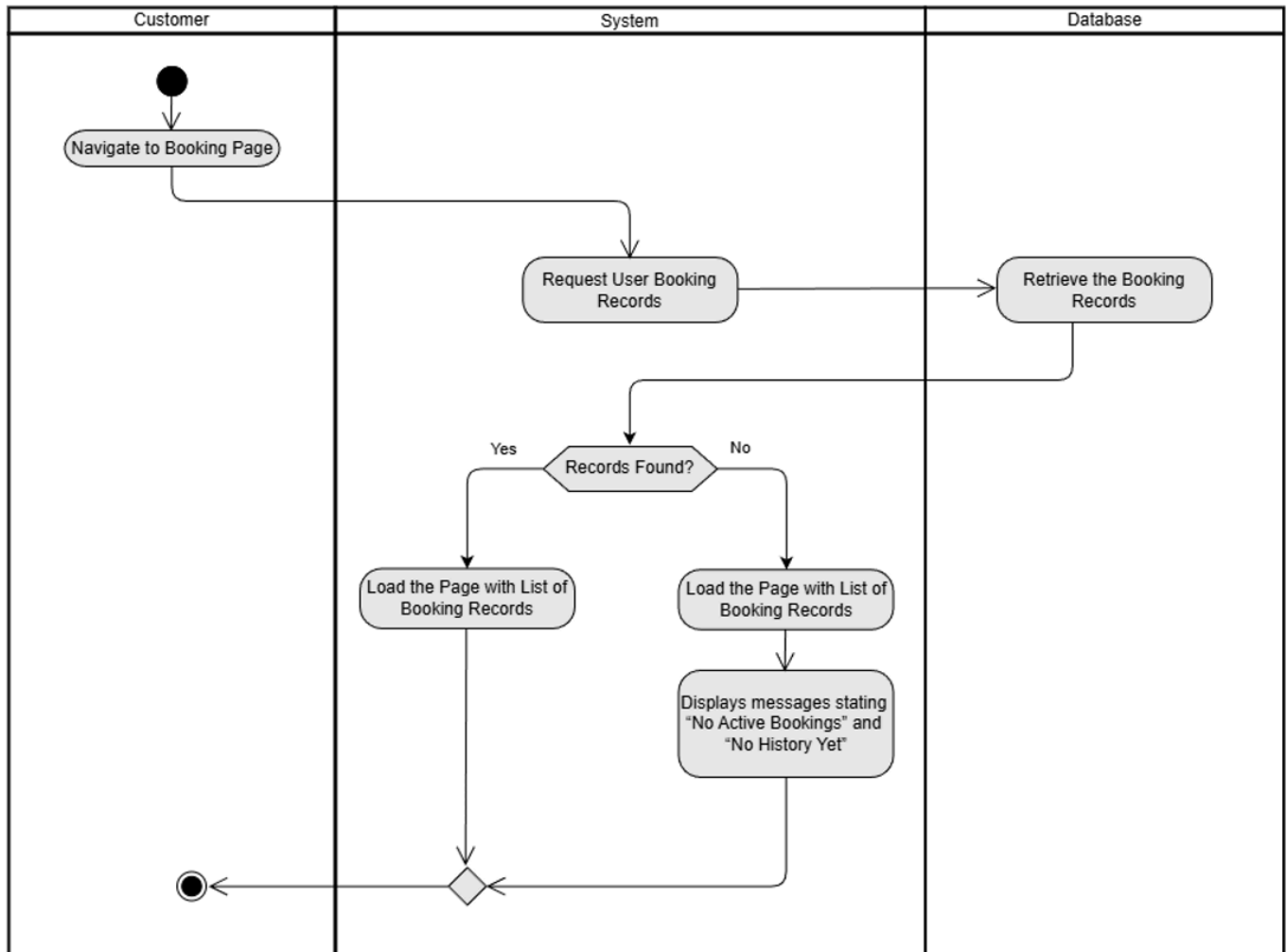


Figure 6.10: Activity Diagram for View Booking History

6.2.10 UC10: Cancel Booking

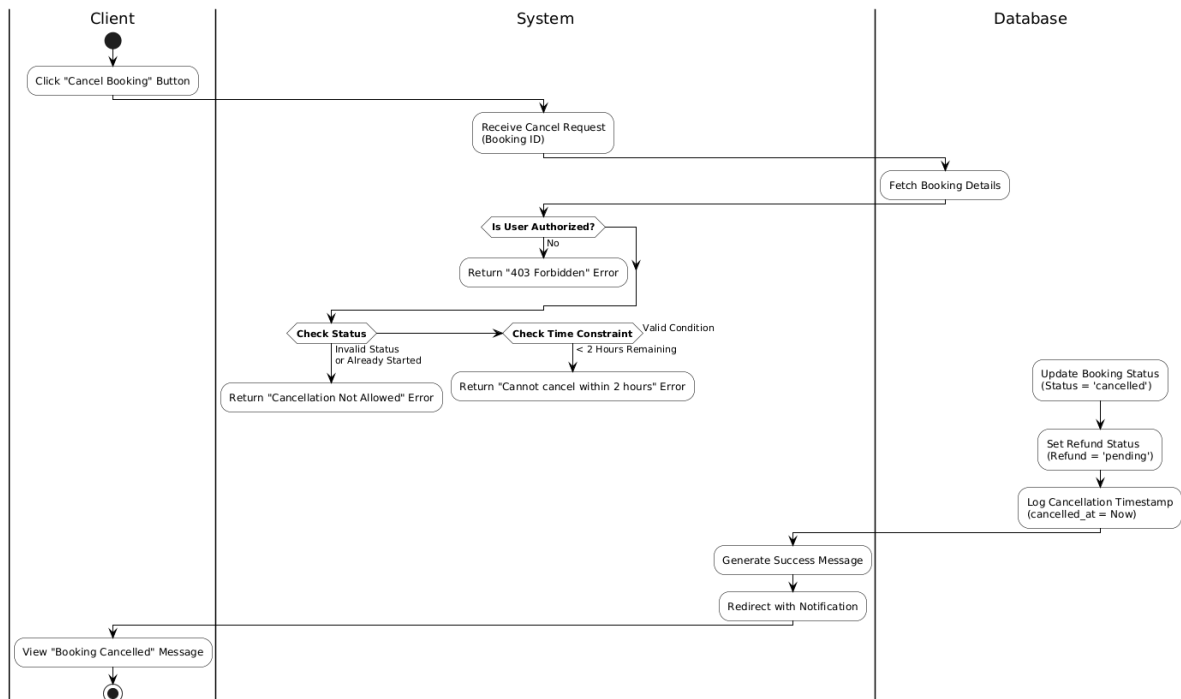


Figure 6.11: Activity Diagram for Cancel Booking

6.2.11 UC11: Booking Vehicle

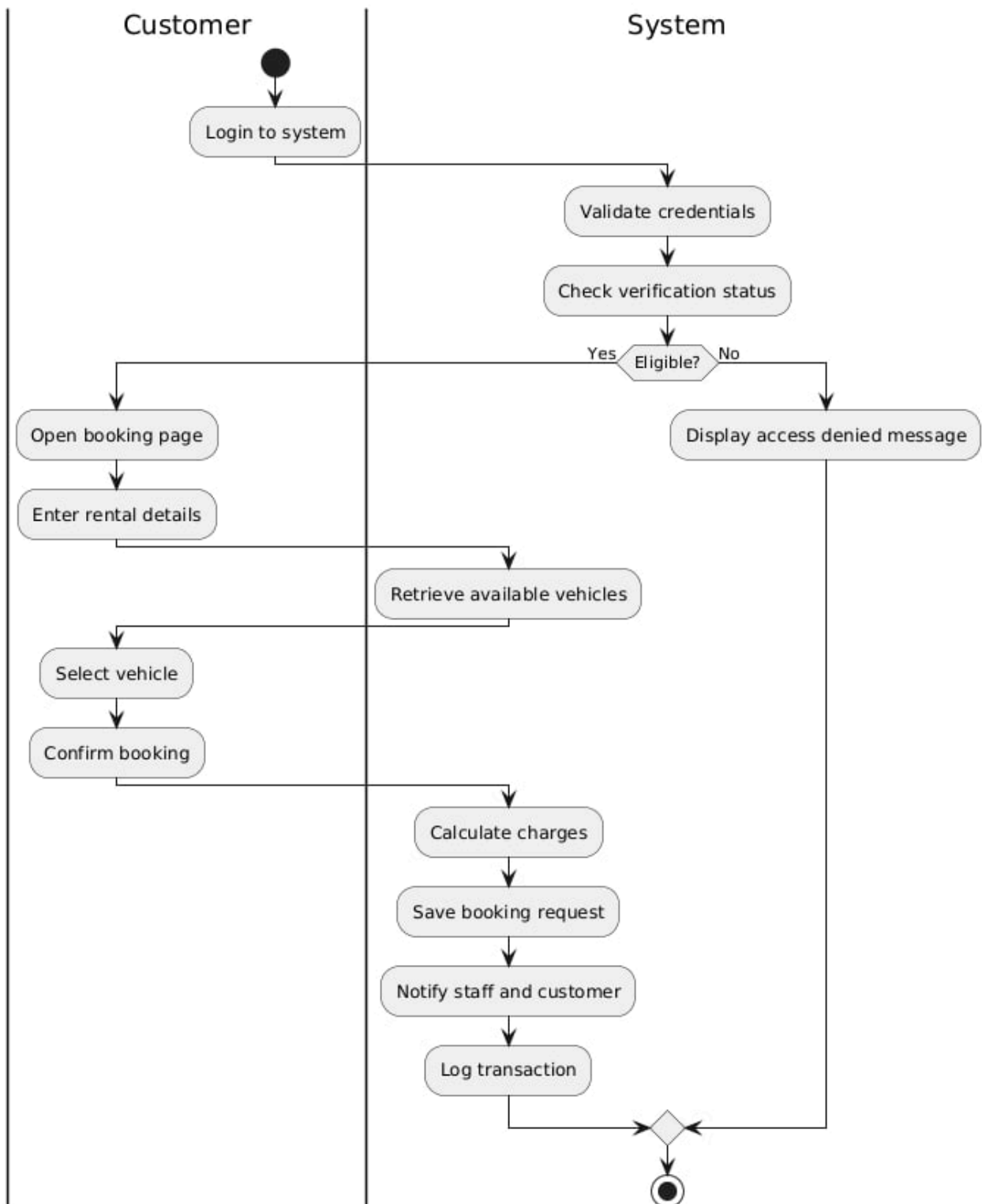


Figure 6.12: Activity Diagram for Booking Vehicle

6.2.12 UC12: Payment

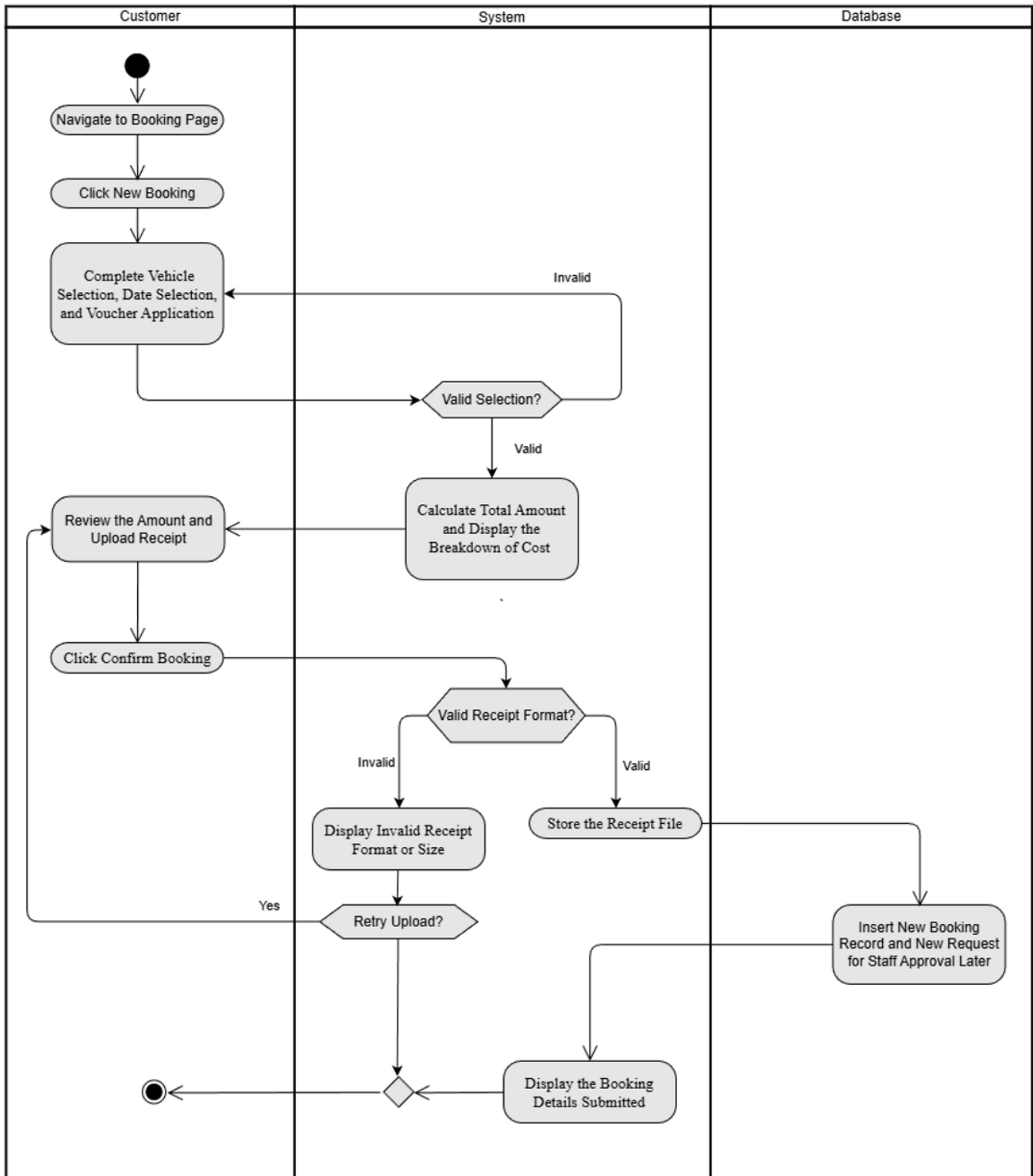


Figure 6.13: Activity Diagram for Payment

6.2.13 UC13: Manage Vehicle

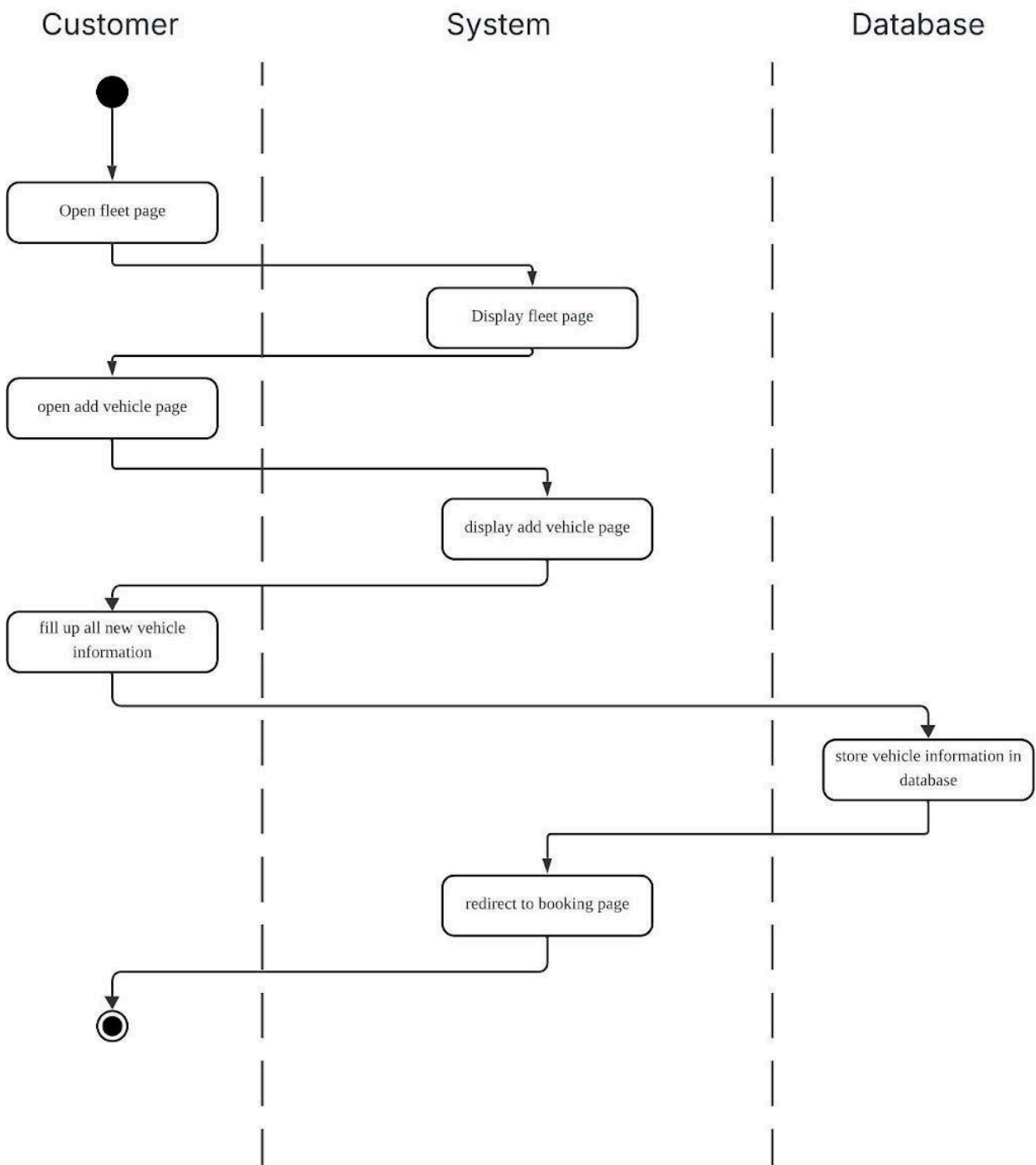


Figure 6.14: Activity Diagram for Manage Vehicle

6.2.14 UC14: Update Vehicle Information

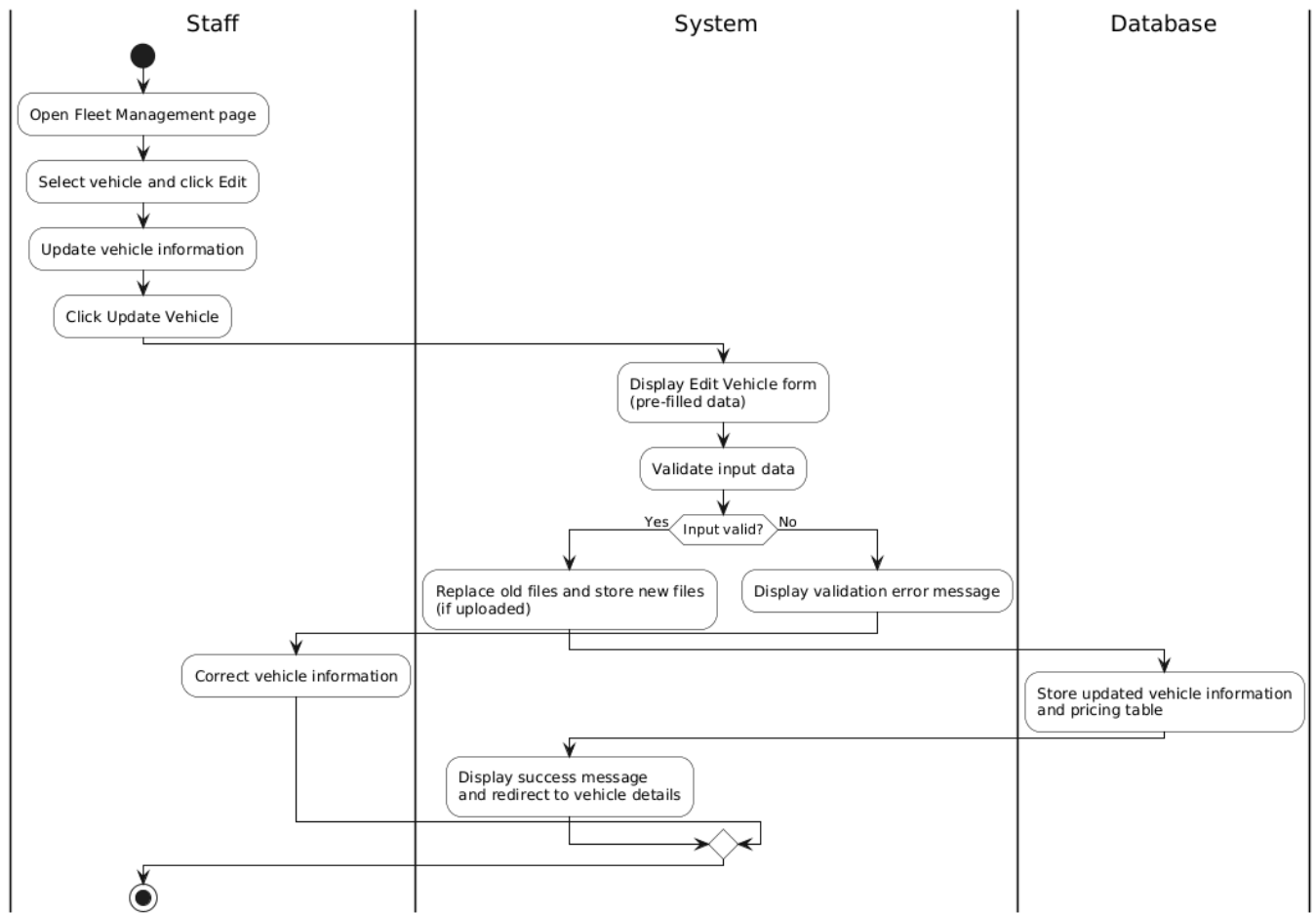


Figure 6.15: Activity Diagram for Update Vehicle Information

6.2.15 UC15: Reporting

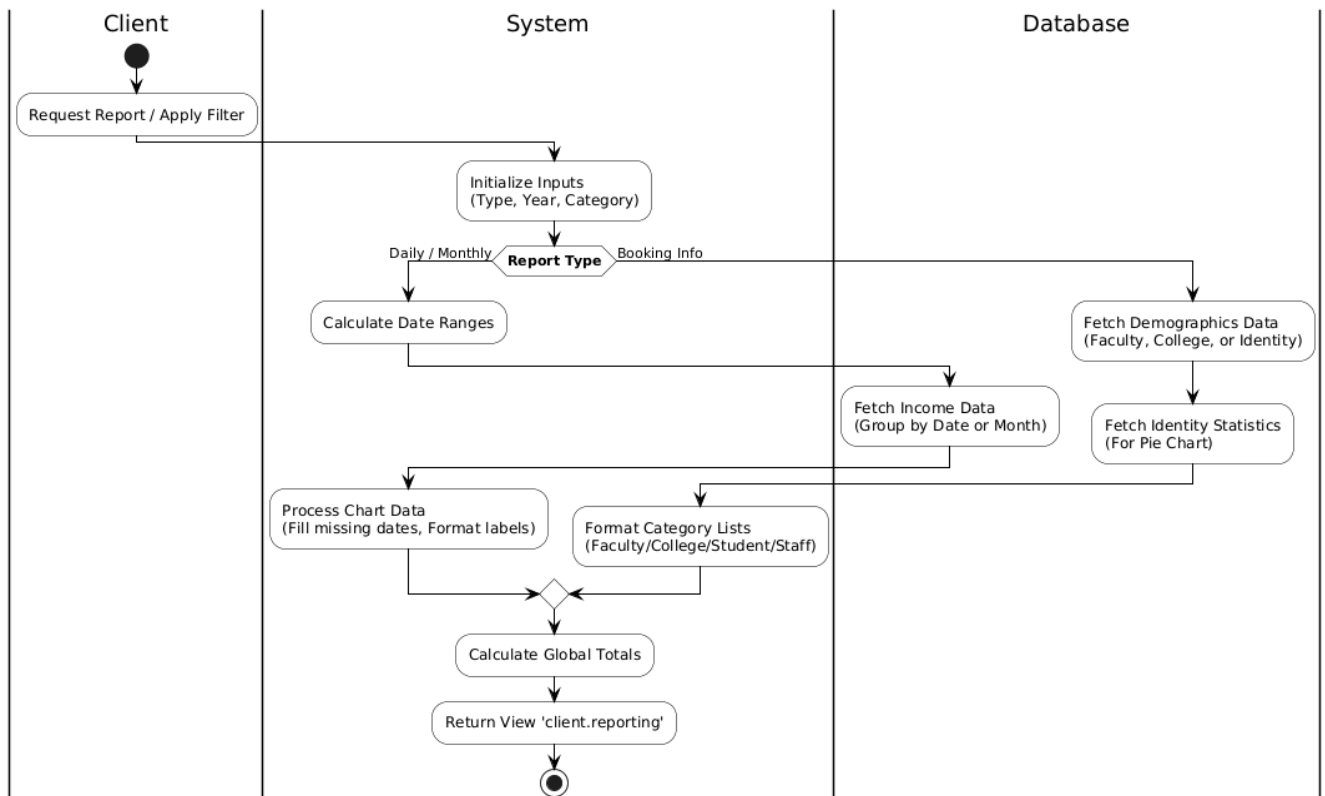


Figure 6.16: Activity Diagram for Reporting

6.2.16 UC16: View All Vouchers

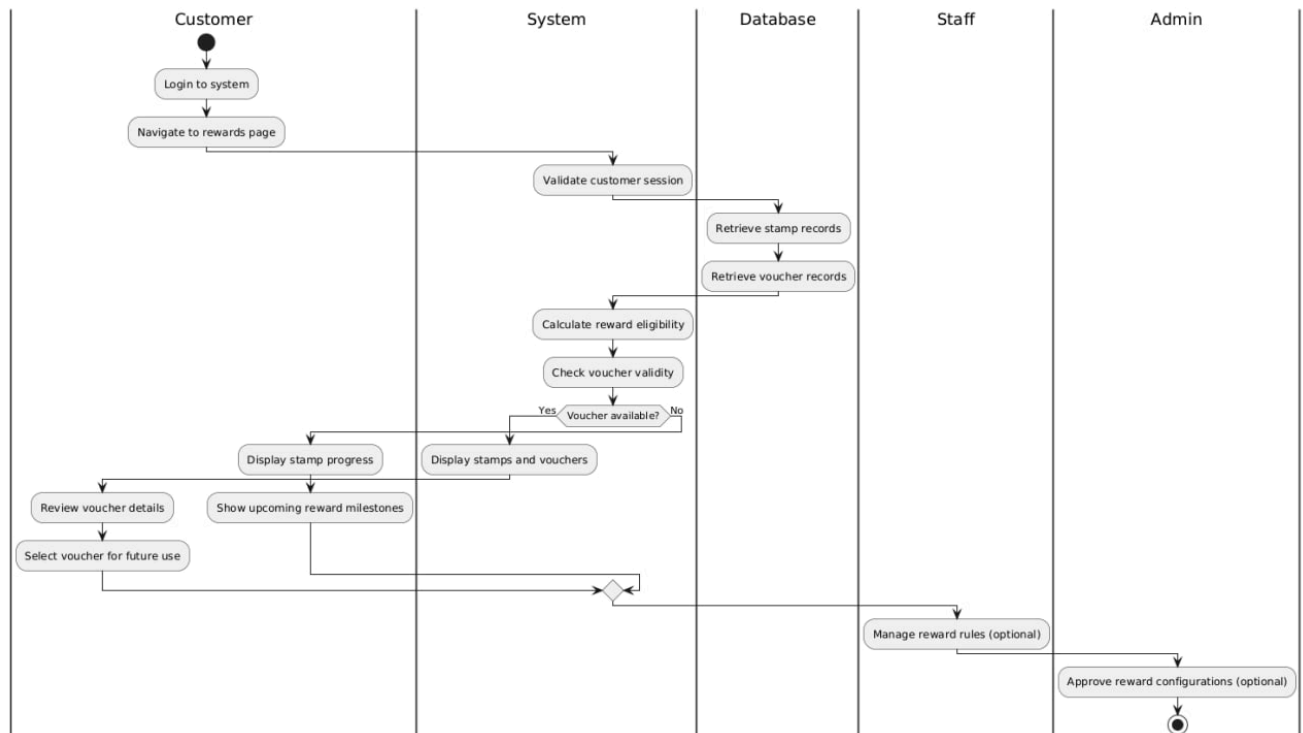


Figure 6.17: Activity Diagram for View All Vouchers

6.3 Meeting Log

Meeting #1

DATE	06 / 1 / 2026	
LOCATION	Hyflex Classroom, Faculty of Computing, N28a	
FOCUS	LOGICAL DATABASE DESIGN	
AGENDA	1. Finalizing database structure planning 2. Identify issues within ERD & EERD 3. Drafting & Drawing Logical ERD	
MEETING MC	HENG ZHI QIANG	
ATTENDANCE		
NAME	DATE	REASON FOR ABSENCE
Parthiv Gunalan	06 / 1 / 2026	-
Heng Zhi Qiang		
Leong Jia Ling		
Hong Jia Bao		
Lee Pei Yuan		
GROUP / MEETING PHOTOS		
		
Meeting was done after industrial day*		

MINUTES

Discussion:

The group analyzed the requirements of the Database Report for Phase 3. From the discussion, task division has also been aggregated. The team has also identified the issues within ERD and EERD. From this, we have identified the best way to draft for the logical ERD.

Decisions Made:

- Identifying issues within ERD & EERD
-
- Task Division for Database Report Part 3
- Drafting logical ERD
- Drawing logical ERD diagram

Action Items:

- Zhi Qiang has initialized the draft for logical ERD
- Task
- Team members to review and suggest improvements

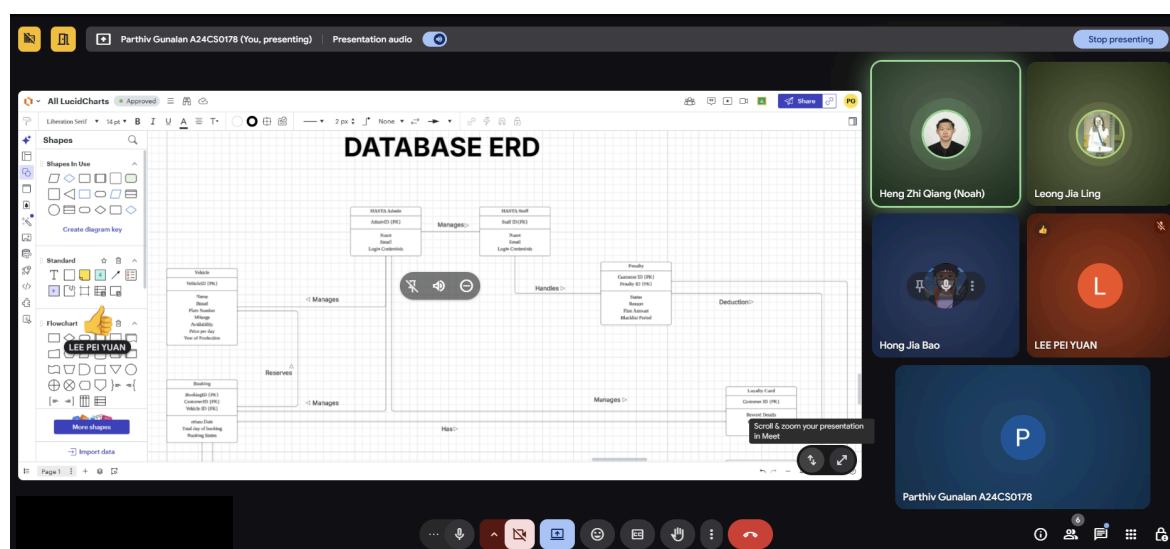
Meeting #2

DATE	16 / 1 / 2026
LOCATION	Online Google Meet
FOCUS	Logical ERD Drawing & Refinement
AGENDA	1. Review drawn logical ERD 2. Matching logical ERD with system database & old ERD 3. Revise database structure
MEETING MC	PARTHIV GUNALAN

ATTENDANCE

NAME	DATE	REASON FOR ABSENCE
Parthiv Gunalan	16 / 1 / 2026	-
Heng Zhi Qiang		
Leong Jia Ling		
Hong Jia Bao		
Lee Pei Yuan		

GROUP / MEETING PHOTOS



MINUTES

Discussion:

A team meeting was held to discuss on the drawing of logical ERD. The logical ERD also was made sure to be tally with the system database design. Logical ERD was also revised based on old ERD design.

Decisions Made:

- Tallying logical ERD with system database
- Revising old ERD
- Redraw the logical ERD

Action Items:

- Parthiv has asked to revise the drawing of logical ERD
- Team to update logical ERD accordingly

Meeting #3

DATE	01 / 2 / 2025	
LOCATION	Online Google Meet	
FOCUS	Database Report Submission	
AGENDA	1. Finishing all Database Reports Phases 2. Checking for any issues within the reports 3. Checking report quality	
MEETING MC	PARTHIV GUNALAN	
ATTENDANCE		
NAME	DATE	REASON FOR ABSENCE
Parthiv Gunalan	01 / 2 / 2025	-
Heng Zhi Qiang		
Leong Jia Ling		
Hong Jia Bao		
Lee Pei Yuan		
GROUP / MEETING PHOTOS		

Database Phase 3

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Database Phase

- Check for errors within reports
- Formatting report
- Tally information in each phase of database report

Action Items:

- Fixing errors
- Aligning report content between multiple report phases